

```

1 E:\Anaconda\envs\airsim\python.exe E:\Github\AirSim_RL\Src\
  batch_test_script.py
2 Gym has been unmaintained since 2022 and does not support NumPy 2
  .0 amongst other critical functionality.
3 Please upgrade to Gymnasium, the maintained drop-in replacement of
  Gym, or contact the authors of your software and request that they
  upgrade.
4 Users of this version of Gym should be able to simply replace 'import
  gym' with 'import gymnasium as gym' in the vast majority of cases.
5 See the migration guide at https://gymnasium.farama.org/introduction/
  migration\_guide/ for additional information.
6 Detected UE-world task coordinates in E:\Github\AirSim_RL\dataset\
  relative_coordinates_export.csv; converted to AirSim local NED using
  origin (1910.0, -458.0, 100.0) and scale /100.
7 Initializing environment...
8 ??? CSV ?? 96 ??????? 4 ????????????
9 Connected!
10 Client Ver:1 (Min Req: 1), Server Ver:1 (Min Req: 1)
11
12 检测到无人机 'Drone1'
13 [Astar_planner] Loading TD3 lower policy: E:\Github\AirSim_RL\
  Network\TD3_base\checkpoints\td3_resume_latest.zip
14 Starting batch test for 38 tasks...
15
16 --- Task 1/38 ---
17 From [101.4  29.68 -6.6 ] to [-75.1  -8.22 -16. ]
18 [Astar_planner] Top-down occupancy: 350x390, x=[-179.0, 170.0], y=[-
  200.0, 189.0], occupied=72869 cells (53.4%)
19 [Astar_planner] Occupancy base_grid=390x350, occupied=72869 cells (
  53.4%)
20 [Astar_planner] Endpoint nearest-free cells are disconnected; using
  reachable cells start=(69, 230), goal=(245, 192), search_radius=36.
21 [Astar_planner] Occupancy A*: grid=390x350, occupied=53.4%,
  safety_margin=1.5
22 [Astar_planner] Endpoint nearest-free cells are disconnected; using
  reachable cells start=(69, 230), goal=(245, 192), search_radius=36.
23 [Astar_planner] Occupancy A*: grid=390x350, occupied=53.4%,
  safety_margin=1.0
24 [Astar_planner] Endpoint nearest-free cells are disconnected; using
  reachable cells start=(69, 230), goal=(245, 192), search_radius=36.
25 [Astar_planner] Occupancy A*: grid=390x350, occupied=53.4%,
  safety_margin=0.6
26 [Astar_planner] Endpoint nearest-free cells are disconnected; using
  reachable cells start=(69, 230), goal=(245, 192), search_radius=36.
27 [Astar_planner] Occupancy A*: grid=390x350, occupied=53.4%,
  safety_margin=0.3
28 [Astar_planner] Safe occupancy A* failed; retrying legacy A* without
  clearance penalties.

```

```

29 [Astar_planner] Legacy occupancy A*: grid=390x350, occupied=53.4%,
    safety_margin=1.5
30 [Astar_planner] WARNING: using legacy occupancy fallback; local
    targets may be closer to obstacles.
31 Path planned via occupancy_legacy (Points: 18, Dist: 208.6m). Local
    target z/alt: 0:-6.63(6.63m), 1:-20.97(20.97m), 2:-21.09(21.09m), 3:-20.
    29(20.29m), 4:-23.89(23.89m), 5:-20.75(20.75m), 6:-24.69(24.69m), 7:-
    22.93(22.93m), 8:-22.31(22.31m), 9:-20.94(20.94m), 10:-23.49(23.49m
    ), 11:-23.98(23.98m), 12:-23.35(23.35m), 13:-24.78(24.78m), 14:-20.45(
    20.45m), 15:-20.17(20.17m), 16:-20.26(20.26m), 17:-15.99(15.99m)
32 [Astar_planner] Segment 0: [101.    30.    -6.626673] -> [ 92.    21
    .    -20.97309]
33 [Astar_planner] Placing first segment start at [101.    30.    -6.
    626673]
34 [Astar_planner] Initial climb to 20.0m.
35 [Step 18 | Total    0] Target: ( 92.0, 21.0, -21.0) | Pos: ( 94.2, 21.6, -
    20.6) | Vel: ( -1.4, 0.1, 0.1) | Acc: ( -0.8, 1.6, 2.0) | Rwd: 2.12
36 [Astar_planner] Segment 0 reached in 18 steps, dist=2.31, axis_error=[
    2.216072 0.5506916 0.3675518], reason=running
37 [Astar_planner] Segment 1 starts from actual pos [ 94.21607 21.
    550692 -20.605537] (planned waypoint was [ 92.    21.    -20.97309])
38 [Astar_planner] Segment 1: [ 94.21607 21.550692 -20.605537] -> [ 92
    .    10.    -21.085625]
39 [Step 17 | Total    0] Target: ( 92.0, 10.0, -21.1) | Pos: ( 90.7, 12.7, -
    20.5) | Vel: ( -0.1, -1.3, 0.0) | Acc: ( -0.1, -1.3, -0.3) | Rwd: 1.24
40 [Astar_planner] Segment 1 reached in 17 steps, dist=3.08, axis_error=[
    1.2615433 2.745205 0.5737686], reason=running
41 [Astar_planner] Segment 2 starts from actual pos [ 90.738304 12.
    741265 -20.511837] (planned waypoint was [ 92.    10.    -21.
    085625])
42 [Astar_planner] Segment 2: [ 90.738304 12.741265 -20.511837] -> [ 90
    .    7.    -20.290905]
43 [Step 3 | Total    0] Target: ( 90.0, 7.0, -20.3) | Pos: ( 89.9, 9.7, -19
    .2) | Vel: ( -0.6, -1.5, 0.0) | Acc: ( -0.1, -0.4, -0.8) | Rwd: 2.50
44 [Astar_planner] Segment 2 reached in 3 steps, dist=2.94, axis_error=[0.
    11402893 2.7153015 1.112072 ], reason=running
45 [Astar_planner] Segment 3 starts from actual pos [ 89.882416 9.
    706337 -19.178633] (planned waypoint was [ 90.    7.    -20.290905
    ])
46 [Astar_planner] Segment 3: [ 89.882416 9.706337 -19.178633] -> [ 77
    .    7.    -23.88771]
47 [Step 45 | Total    0] Target: ( 77.0, 7.0, -23.9) | Pos: ( 75.0, 7.3, -
    21.4) | Vel: ( -0.1, 1.5, -2.2) | Acc: ( -2.0, 1.5, 2.0) | Rwd: 6.29
48 [Astar_planner] Segment 3 reached in 45 steps, dist=3.20, axis_error=[
    2.029091 0.25325727 2.4661808 ], reason=running
49 [Astar_planner] Segment 4 starts from actual pos [ 74.970535 7.
    2579007 -21.428038 ] (planned waypoint was [ 77.    7.    -23.88771
    ])

```

```

50 [Astar_planner] Segment 4: [ 74.970535  7.2579007 -21.428038 ] -> [
    62.      7.      -20.750288]
51 [Step 12 | Total    0] Target: ( 62.0,  7.0, -20.8) | Pos: ( 64.1,  6.1, -
    18.6) | Vel: ( -1.7, -0.6, -0.1) | Acc: ( -0.9, -0.6,  0.5) | Rwd: 1.34
52 [Astar_planner] Segment 4 reached in 12 steps, dist=3.12, axis_error=[
    2.0815887 0.9354234 2.1317043], reason=running
53 [Astar_planner] Segment 5 starts from actual pos [ 64.07656  6.
    062745 -18.618893] (planned waypoint was [ 62.      7.      -20.750288
    ])
54 [Astar_planner] Segment 5: [ 64.07656  6.062745 -18.618893] -> [ 55
    .      7.      -24.687864]
55 [Step 23 | Total    0] Target: ( 55.0,  7.0, -24.7) | Pos: ( 55.8,  5.3, -
    22.7) | Vel: ( -0.5,  0.8, -1.9) | Acc: ( -1.7, -0.4,  1.0) | Rwd: 6.57
56 [Astar_planner] Segment 5 reached in 23 steps, dist=2.69, axis_error=[
    0.7512932 1.6760087 1.9689732], reason=running
57 [Astar_planner] Segment 6 starts from actual pos [ 55.749664  5.
    326364 -22.724712] (planned waypoint was [ 55.      7.      -24.687864
    ])
58 [Astar_planner] Segment 6: [ 55.749664  5.326364 -22.724712] -> [ 40
    .      6.      -22.925047]
59 [Step 23 | Total    0] Target: ( 40.0,  6.0, -22.9) | Pos: ( 38.5,  3.0, -
    20.3) | Vel: ( -0.3,  1.2,  0.0) | Acc: ( -0.9, -0.7,  0.3) | Rwd: 0.69
60 [Astar_planner] Segment 6 reached in 23 steps, dist=4.20, axis_error=[
    1.460659 2.9745233 2.5754662], reason=running
61 [Astar_planner] Segment 7 starts from actual pos [ 38.53834  3.
    0289428 -20.349491 ] (planned waypoint was [ 40.      6.      -22.
    925047])
62 [Astar_planner] Segment 7: [ 38.53834  3.0289428 -20.349491 ] -> [
    25.      6.      -22.31489]
63 [Step 20 | Total    0] Target: ( 25.0,  6.0, -22.3) | Pos: ( 24.5,  3.0, -
    20.1) | Vel: ( -0.3,  1.2,  0.0) | Acc: ( -0.2, -0.6, -0.8) | Rwd: 2.01
64 [Astar_planner] Segment 7 reached in 20 steps, dist=3.76, axis_error=[
    0.509367 2.9877353 2.232748 ], reason=running
65 [Astar_planner] Segment 8 starts from actual pos [ 24.48856  3.
    0194814 -20.081879 ] (planned waypoint was [ 25.      6.      -22.31489
    ])
66 [Astar_planner] Segment 8: [ 24.48856  3.0194814 -20.081879 ] -> [
    10.      6.      -20.94346]
67 [Step 20 | Total    0] Target: ( 10.0,  6.0, -20.9) | Pos: (  9.1,  3.1, -18
    .8) | Vel: ( -0.6,  1.0, -0.0) | Acc: ( -1.3, -0.3,  0.0) | Rwd: 0.91
68 [Astar_planner] Segment 8 reached in 20 steps, dist=3.72, axis_error=[
    0.86298466 2.9280136 2.1303787 ], reason=running
69 [Astar_planner] Segment 9 starts from actual pos [  9.133626  3.
    077769 -18.813263] (planned waypoint was [ 10.      6.      -20.94346])
70 [Astar_planner] Segment 9: [  9.133626  3.077769 -18.813263] -> [ 0
    .      6.      -23.485023]
71 [Step 18 | Total    0] Target: (  0.0,  6.0, -23.5) | Pos: (  1.2,  5.6, -21.
    2) | Vel: ( -0.3,  0.6, -1.1) | Acc: ( -0.1,  2.0,  1.6) | Rwd: 5.46

```

```

72 [Astar_planner] Segment 9 reached in 18 steps, dist=2.61, axis_error
   =[1.161689  0.38315964 2.3059902 ], reason=running
73 [Astar_planner] Segment 10 starts from actual pos [ 1.1607579  5.
   618623 -21.182196 ] (planned waypoint was [ 0.      6.      -23.
   485023])
74 [Astar_planner] Segment 10: [ 1.1607579  5.618623 -21.182196
   ] -> [ -8.      -2.      -23.979246]
75 [Step 10 | Total    0] Target: ( -8.0, -2.0, -24.0) | Pos: ( -5.4, -1.9, -
   21.7) | Vel: ( -0.8, -1.0, -0.0) | Acc: ( 0.3, -0.8, -0.3) | Rwd: 1.78
76 [Astar_planner] Segment 10 reached in 10 steps, dist=3.41, axis_error
   =[2.5790524 0.1472584 2.230795 ], reason=running
77 [Astar_planner] Segment 11 starts from actual pos [ -5.4232154 -1.
   8556228 -21.748528 ] (planned waypoint was [ -8.      -2.      -23.
   979246])
78 [Astar_planner] Segment 11: [ -5.4232154 -1.8556228 -21.748528
   ] -> [ -23.      -2.      -23.349474]
79 [Step 29 | Total    0] Target: (-23.0, -2.0, -23.3) | Pos: (-21.1, -1.6, -
   20.4) | Vel: ( -0.8, 0.3, -0.5) | Acc: ( -1.1, 2.0, 0.4) | Rwd: 3.42
80 [Astar_planner] Segment 11 reached in 29 steps, dist=3.53, axis_error
   =[1.9067974 0.39410496 2.9494286 ], reason=running
81 [Astar_planner] Segment 12 starts from actual pos [-21.0955  -1.
   6050694 -20.40139 ] (planned waypoint was [-23.      -2.      -23.
   349474])
82 [Astar_planner] Segment 12: [-21.0955  -1.6050694 -20.40139 ] -> [-
   38.      -2.      -24.778013]
83 [Step 53 | Total    0] Target: (-38.0, -2.0, -24.8) | Pos: (-35.0, -2.4, -
   22.3) | Vel: ( -0.8, 0.4, 0.1) | Acc: ( 0.7, -0.5, -0.7) | Rwd: 1.92
84 [Astar_planner] Segment 12 reached in 53 steps, dist=3.91, axis_error
   =[2.9957771 0.3984096 2.4783306], reason=running
85 [Astar_planner] Segment 13 starts from actual pos [-35.00648  -2.
   3971157 -22.299534 ] (planned waypoint was [-38.      -2.      -24.
   778013])
86 [Astar_planner] Segment 13: [-35.00648  -2.3971157 -22.299534
   ] -> [-53.      -2.      -20.445219]
87 [Step 49 | Total    0] Target: (-53.0, -2.0, -20.4) | Pos: (-50.5, -0.5, -
   21.2) | Vel: ( -0.6, 0.7, 0.0) | Acc: ( 0.5, 1.5, -0.7) | Rwd: 0.23
88 [Astar_planner] Segment 13 reached in 49 steps, dist=3.06, axis_error
   =[2.52594  1.5259349 0.7944622], reason=running
89 [Astar_planner] Segment 14 starts from actual pos [-50.475864  -0.
   47209376 -21.239656 ] (planned waypoint was [-53.      -2.      -20.
   445219])
90 [Astar_planner] Segment 14: [-50.475864  -0.47209376 -21.239656
   ] -> [-64.      -2.      -20.167496]
91 [Step 39 | Total    0] Target: (-64.0, -2.0, -20.2) | Pos: (-61.3, -0.8, -
   20.0) | Vel: ( -1.2, 0.0, 0.1) | Acc: ( -1.9, 0.3, -1.0) | Rwd: 3.92
92 [Astar_planner] Segment 14 reached in 39 steps, dist=2.93, axis_error
   =[2.6825333 1.1687969 0.19020462], reason=running
93 [Astar_planner] Segment 15 starts from actual pos [-61.32105  -0.

```

```

93 83114547 -19.977037 ] (planned waypoint was [-64.    -2.    -20.
    167496])
94 [Astar_planner] Segment 15: [-61.32105    -0.83114547 -19.977037
    ] -> [-70.    -3.    -20.262274]
95 [Step   7 | Total    0] Target: (-70.0, -3.0, -20.3) | Pos: (-67.5, -2.3, -
    18.4) | Vel: ( -0.9, -0.4,  0.0) | Acc: (  0.2, -1.1, -0.6) | Rwd:  2.54
96 [Astar_planner] Segment 15 reached in 7 steps, dist=3.16, axis_error
    =[2.4524765 0.7283542 1.8480434], reason=running
97 [Astar_planner] Segment 16 starts from actual pos [-67.5529    -2.
    273897 -18.414019] (planned waypoint was [-70.    -3.    -20.
    262274])
98 [Astar_planner] Segment 16: [-67.5529    -2.273897 -18.414019] -> [-
    75.    -8.    -15.992509]
99 [Step   9 | Total    0] Target: (-75.0, -8.0, -16.0) | Pos: (-74.1, -9.5, -
    15.3) | Vel: ( -0.9, -0.6,  0.1) | Acc: (  1.5,  1.7, -0.7) | Rwd:  1.79
100 [Astar_planner] Segment 16 reached in 9 steps, dist=1.84, axis_error
    =[0.9011841 1.4630136 0.65520954], reason=running
101 Task 1 result: success (Success: Reached destination (17/17
    segments). Shield interventions=96, emergencies=2, recoveries=17,
    recovery_steps=96.)
102
103 --- Task 2/38 ---
104 From [ -61.1 -101.02  -3.9 ] to [-10.97211182 -13.19131592 -14.
    84776611]
105 [Astar_planner] Top-down occupancy: 350x390, x=[-179.0, 170.0], y=[-
    200.0, 189.0], occupied=72869 cells (53.4%)
106 [Astar_planner] Occupancy base_grid=390x350, occupied=72869 cells
    (53.4%)
107 [Astar_planner] Occupancy A*: grid=390x350, occupied=53.4%,
    safety_margin=1.5
108 Path planned via occupancy (Points: 11, Dist: 143.5m). Local target z/
    alt: 0:-3.91(3.91m), 1:-20.47(20.47m), 2:-24.15(24.15m), 3:-23.79(23.
    79m), 4:-24.31(24.31m), 5:-24.28(24.28m), 6:-21.60(21.60m), 7:-24.07
    (24.07m), 8:-20.31(20.31m), 9:-24.14(24.14m), 10:-14.85(14.85m)
109 [Astar_planner] Segment 0: [ -61.    -101.    -3.91104] -> [-59.    -
    91.    -20.470758]
110 [Astar_planner] Placing first segment start at [ -61.    -101.    -3.
    91104]
111 [Astar_planner] Initial climb to 20.0m.
112 [Step   11 | Total    0] Target: (-59.0, -91.0, -20.5) | Pos: (-60.1, -93.9
    , -18.2) | Vel: ( -0.6,  1.3, -0.1) | Acc: ( -1.2,  1.8, -0.0) | Rwd:  1.65
113 [Astar_planner] Segment 0 reached in 11 steps, dist=3.87, axis_error
    =[1.0990334 2.8955994 2.3155117], reason=running
114 [Astar_planner] Segment 1 starts from actual pos [-60.099033 -93.
    8956  -18.155247] (planned waypoint was [-59.    -91.    -20.
    470758])
115 [Astar_planner] Segment 1: [-60.099033 -93.8956  -18.155247] -> [-46
    .    -88.    -24.14949]

```

```

116 [Step 48 | Total 0] Target: (-46.0, -88.0, -24.1) | Pos: (-52.1, -87.0
, -21.1) | Vel: ( 0.1, -0.1, 0.1) | Acc: ( 0.3, -0.8, -0.8) | Rwd: -0.10
117 [Astar_planner] Segment 1 reached in 48 steps, dist=6.94, axis_error
=[6.145447 0.96889496 3.070118 ], reason=
stuck_near_intermediate
118 [Astar_planner] Segment 2 starts from actual pos [-52.14519 -87.
03144 -21.079056] (planned waypoint was [-46. -88. -24.14949
])
119 [Astar_planner] Segment 2: [-52.14519 -87.03144 -21.079056] -> [-32
. -88. -23.7898]
120 [Step 24 | Total 0] Target: (-32.0, -88.0, -23.8) | Pos: (-32.1, -90.4
, -21.9) | Vel: ( 0.5, -0.1, -1.3) | Acc: ( -0.6, 0.1, -0.7) | Rwd: 5.02
121 [Astar_planner] Segment 2 reached in 24 steps, dist=3.08, axis_error
=[0.06802368 2.4313812 1.8902912 ], reason=running
122 [Astar_planner] Segment 3 starts from actual pos [-32.066486 -90.
43171 -21.90343 ] (planned waypoint was [-32. -88. -23.7898])
123 [Astar_planner] Segment 3: [-32.066486 -90.43171 -21.90343 ] -> [-19
. -88. -24.312586]
124 [Step 121 | Total 0] Target: (-19.0, -88.0, -24.3) | Pos: (-21.9, -87.2
, -21.9) | Vel: ( 0.6, -0.1, 0.1) | Acc: ( -0.4, 0.3, -0.7) | Rwd: 1.17
125 [Astar_planner] Segment 3 reached in 121 steps, dist=3.84, axis_error
=[2.8746662 0.7813034 2.427349 ], reason=running
126 [Astar_planner] Segment 4 starts from actual pos [-21.872833 -87.
218956 -21.885025] (planned waypoint was [-19. -88. -24.
312586])
127 [Astar_planner] Segment 4: [-21.872833 -87.218956 -21.885025] -> [-
11. -80. -24.275568]
128 [Step 22 | Total 0] Target: (-11.0, -80.0, -24.3) | Pos: (-13.1, -80.1
, -22.0) | Vel: ( 1.0, -0.6, -0.1) | Acc: ( 0.4, -0.2, 0.2) | Rwd: 1.17
129 [Astar_planner] Segment 4 reached in 22 steps, dist=3.11, axis_error
=[2.1386633 0.05480957 2.2582111 ], reason=running
130 [Astar_planner] Segment 5 starts from actual pos [-13.135783 -80.
05652 -22.017593] (planned waypoint was [-11. -80. -24.
275568])
131 [Astar_planner] Segment 5: [-13.135783 -80.05652 -22.017593] -> [-
10. -66. -21.5978]
132 [Step 15 | Total 0] Target: (-10.0, -66.0, -21.6) | Pos: ( -7.7, -68.1
, -18.9) | Vel: ( -0.2, 0.4, 0.4) | Acc: ( -1.9, 1.4, -1.9) | Rwd: 1.04
133 [Astar_planner] Segment 5 reached in 15 steps, dist=4.08, axis_error
=[2.2997556 2.0528488 2.6756802], reason=running
134 [Astar_planner] Segment 6 starts from actual pos [ -7.7015433 -68.
050285 -18.919905 ] (planned waypoint was [-10. -66. -21.5978])
135 [Astar_planner] Segment 6: [ -7.7015433 -68.050285 -18.919905
] -> [-10. -52. -24.066788]
136 [Step 18 | Total 0] Target: (-10.0, -52.0, -24.1) | Pos: ( -9.2, -52.1
, -22.6) | Vel: ( -0.4, 0.4, -0.7) | Acc: ( -1.6, 1.7, 1.6) | Rwd: 607.30
-> Episode End: ARRIVED!
137

```

```

138 [Astar_planner] Segment 6 reached in 18 steps, dist=1.67, axis_error
    =[0.808548 0.12846375 1.4501915 ], reason=arrived
139 [Astar_planner] Segment 7 starts from actual pos [ -9.192794 -52.
    12728 -22.618778] (planned waypoint was [-10.    -52.    -24.
    066788])
140 [Astar_planner] Segment 7: [ -9.192794 -52.12728 -22.618778] -> [ -9
    .    -39.    -20.314104]
141 [Step 10 | Total 0] Target: ( -9.0, -39.0, -20.3) | Pos: ( -9.5, -41.9, -
    17.7) | Vel: ( 0.1, 1.7, -0.2) | Acc: ( 1.7, 2.0, 0.6) | Rwd: 2.40
142 [Astar_planner] Segment 7 reached in 10 steps, dist=3.96, axis_error
    =[0.4603157 2.917572 2.6372356], reason=running
143 [Astar_planner] Segment 8 starts from actual pos [ -9.45948 -41.
    90719 -17.677792] (planned waypoint was [ -9.    -39.    -20.
    314104])
144 [Astar_planner] Segment 8: [ -9.45948 -41.90719 -17.677792] -> [ -9
    .    -25.    -24.136427]
145 [Step 63 | Total 0] Target: ( -9.0, -25.0, -24.1) | Pos: ( -9.9, -27.4, -
    21.3) | Vel: ( -0.7, 0.8, -0.2) | Acc: ( 1.2, -1.2, 1.9) | Rwd: -0.71
146 [Astar_planner] Segment 8 reached in 63 steps, dist=3.85, axis_error
    =[0.9382067 2.3712444 2.8842316], reason=running
147 [Astar_planner] Segment 9 starts from actual pos [ -9.940186 -27.
    368721 -21.252747] (planned waypoint was [ -9.    -25.    -24.
    136427])
148 [Astar_planner] Segment 9: [ -9.940186 -27.368721 -21.252747] -> [ -
    11.    -13.    -14.847765]
149 [Step 15 | Total 0] Target: (-11.0, -13.0, -14.8) | Pos: (-10.6, -13.8
    , -14.6) | Vel: ( -0.2, 0.8, 0.1) | Acc: ( 0.6, -0.7, -0.9) | Rwd: 603.50
    -> Episode End: ARRIVED!
150
151 [Astar_planner] Segment 9 reached in 15 steps, dist=0.92, axis_error
    =[0.4117546 0.7826233 0.26399803], reason=arrived
152 Task 2 result: success (Success: Reached destination (10/10
    segments). Shield interventions=121, emergencies=3, recoveries=21,
    recovery_steps=121.)
153
154 --- Task 3/38 ---
155 From [119.7 78.28 -7.4 ] to [-106.2 125.18 -4.9 ]
156 [Astar_planner] Top-down occupancy: 350x390, x=[-179.0, 170.0], y=[-
    200.0, 189.0], occupied=72869 cells (53.4%)
157 [Astar_planner] Occupancy base_grid=390x350, occupied=72869 cells
    (53.4%)
158 [Astar_planner] Occupancy A*: grid=390x350, occupied=53.4%,
    safety_margin=1.5
159 Path planned via occupancy (Points: 25, Dist: 319.2m). Local target z/
    alt: 0:-7.40(7.40m), 1:-24.09(24.09m), 2:-21.41(21.41m), 3:-23.82(23.
    82m), 4:-20.50(20.50m), 5:-21.09(21.09m), 6:-22.48(22.48m), 7:-21.89
    (21.89m), 8:-24.75(24.75m), 9:-22.89(22.89m), 10:-23.59(23.59m), 11
    :-21.65(21.65m), 12:-20.79(20.79m), 13:-21.69(21.69m), 14:-24.16(24.

```

```

159 16m), 15:-21.26(21.26m), 16:-21.79(21.79m), 17:-20.42(20.42m), 18:-
    24.74(24.74m), 19:-22.32(22.32m), 20:-22.44(22.44m), 21:-22.51(22.
    51m), 22:-23.31(23.31m), 23:-22.74(22.74m), 24:-4.90(4.90m)
160 [Astar_planner] Segment 0: [120.      78.      -7.3955536] -> [110
    .      84.      -24.088959]
161 [Astar_planner] Placing first segment start at [120.      78.      -7.
    3955536]
162 [Astar_planner] Initial climb to 20.0m.
163 [Step 14 | Total 0] Target: (110.0, 84.0, -24.1) | Pos: (112.6, 83.5
    , -21.6) | Vel: ( -0.9, 0.1, 0.0) | Acc: ( 1.1, -0.6, -0.2) | Rwd: 0.61
164 [Astar_planner] Segment 0 reached in 14 steps, dist=3.66, axis_error
    =[2.5858078 0.53331757 2.5374546 ], reason=running
165 [Astar_planner] Segment 1 starts from actual pos [112.58581 83.
    46668 -21.551504] (planned waypoint was [110.      84.      -24.
    088959])
166 [Astar_planner] Segment 1: [112.58581 83.46668 -21.551504] -> [
    105.      95.      -21.405779]
167 [Step 14 | Total 0] Target: (105.0, 95.0, -21.4) | Pos: (106.9, 92.4
    , -20.3) | Vel: ( 0.1, 0.8, -1.5) | Acc: ( 0.9, -1.7, -0.7) | Rwd: 0.04
168 [Astar_planner] Segment 1 reached in 14 steps, dist=3.38, axis_error
    =[1.8615723 2.586998 1.1139603], reason=running
169 [Astar_planner] Segment 2 starts from actual pos [106.861755 92.
    41539 -20.296185] (planned waypoint was [105.      95.      -21.
    405779])
170 [Astar_planner] Segment 2: [106.861755 92.41539 -20.296185] -> [
    105.      108.      -23.824038]
171 [Step 17 | Total 0] Target: (105.0, 108.0, -23.8) | Pos: (107.4, 94.
    5, -22.3) | Vel: ( -0.7, 2.2, 2.7) | Acc: ( 0.8, 0.5, -0.7) | Rwd: -908.
    90 -> Episode End: CRASHED!
172
173 [Astar_planner] Segment 2 ended: collision
174 [Astar_planner] Stop after failed segment 2.
175 Task 3 result: failed (Failed at segment 2: collision (2/24 reached).
    Shield interventions=17, emergencies=2, recoveries=3,
    recovery_steps=17.)
176
177 --- Task 4/38 ---
178 From [16. 3.18 -7.2 ] to [-106.2 125.18 -4.9 ]
179 [Astar_planner] Top-down occupancy: 350x390, x=[-179.0, 170.0], y=[-
    200.0, 189.0], occupied=72869 cells (53.4%)
180 [Astar_planner] Occupancy base_grid=390x350, occupied=72869 cells
    (53.4%)
181 [Astar_planner] Occupancy A*: grid=390x350, occupied=53.4%,
    safety_margin=1.5
182 Path planned via occupancy (Points: 19, Dist: 251.9m). Local target z/
    alt: 0:-7.20(7.20m), 1:-22.62(22.62m), 2:-24.93(24.93m), 3:-23.72(23.
    72m), 4:-24.76(24.76m), 5:-20.49(20.49m), 6:-20.54(20.54m), 7:-22.77
    (22.77m), 8:-24.37(24.37m), 9:-24.19(24.19m), 10:-22.61(22.61m), 11

```



```

182 :-23.29(23.29m), 12:-23.13(23.13m), 13:-20.10(20.10m), 14:-20.23(20.
    23m), 15:-20.73(20.73m), 16:-23.90(23.90m), 17:-23.55(23.55m), 18:-
    4.90(4.90m)
183 [Astar_planner] Segment 0: [16.      3.      -7.1976023] -> [ 3.      3
    .      -22.616127]
184 [Astar_planner] Placing first segment start at [16.      3.      -7.
    1976023]
185 [Astar_planner] Initial climb to 20.0m.
186 [Step 15 | Total 0] Target: ( 3.0, 3.0, -22.6) | Pos: ( 3.4, 0.3, -
    20.4) | Vel: ( -0.2, 0.9, 0.1) | Acc: ( -0.1, 0.9, -0.7) | Rwd: 1.69
187 [Astar_planner] Segment 0 reached in 15 steps, dist=3.48, axis_error
    =[0.4014945 2.6573808 2.2137547], reason=running
188 [Astar_planner] Segment 1 starts from actual pos [ 3.4008646 0.
    34545437 -20.402142 ] (planned waypoint was [ 3.      3.      -22.
    616127])
189 [Astar_planner] Segment 1: [ 3.4008646 0.34545437 -20.402142
    ] -> [ -9.      -2.      -24.92505]
190 [Step 29 | Total 0] Target: ( -9.0, -2.0, -24.9) | Pos: ( -6.5, -2.8, -
    23.0) | Vel: ( 0.0, 0.1, -1.5) | Acc: ( 0.5, -0.2, 1.6) | Rwd: 6.42
191 [Astar_planner] Segment 1 reached in 29 steps, dist=3.25, axis_error
    =[2.5246606 0.81855917 1.8785286 ], reason=running
192 [Astar_planner] Segment 2 starts from actual pos [ -6.475207 -2.
    8181655 -23.050983 ] (planned waypoint was [ -9.      -2.      -24.
    92505])
193 [Astar_planner] Segment 2: [ -6.475207 -2.8181655 -23.050983 ] -> [ -
    21.      0.      -23.717371]
194 [Step 26 | Total 0] Target: (-21.0, 0.0, -23.7) | Pos: (-16.1, -1.1, -
    20.5) | Vel: ( -0.0, -0.2, 0.1) | Acc: ( 0.2, -0.8, -0.6) | Rwd: -1.10
195 [Astar_planner] Segment 2 reached in 26 steps, dist=5.92, axis_error
    =[4.8758564 1.0899311 3.1826286], reason=stuck_near_intermediate
196 [Astar_planner] Segment 3 starts from actual pos [-16.124372 -1.
    0909101 -20.534275 ] (planned waypoint was [-21.      0.      -23.
    717371])
197 [Astar_planner] Segment 3: [-16.124372 -1.0909101 -20.534275
    ] -> [-31.      8.      -24.761395]
198 [Step 43 | Total 0] Target: (-31.0, 8.0, -24.8) | Pos: (-27.1, 5.9, -
    20.8) | Vel: ( -0.0, -0.2, 0.1) | Acc: ( -0.3, -0.5, -0.9) | Rwd: -1.31
199 [Astar_planner] Segment 3 reached in 43 steps, dist=5.95, axis_error
    =[3.859846 2.1424742 3.995514 ], reason=stuck_near_intermediate
200 [Astar_planner] Segment 4 starts from actual pos [-27.140276 5.
    856783 -20.765585] (planned waypoint was [-31.      8.      -24.
    761395])
201 [Astar_planner] Segment 4: [-27.140276 5.856783 -20.765585] -> [ -
    41.      18.      -20.494864]
202 [Step 17 | Total 0] Target: (-41.0, 18.0, -20.5) | Pos: (-38.4, 19.8
    , -19.2) | Vel: ( -0.8, 0.1, 0.1) | Acc: ( -0.1, 0.0, -1.0) | Rwd: 1.82
203 [Astar_planner] Segment 4 reached in 17 steps, dist=3.42, axis_error
    =[2.6097221 1.8042831 1.265358 ], reason=running

```

```

204 [Astar_planner] Segment 5 starts from actual pos [-38.392624 19.804495 -19.229094] (planned waypoint was [-41.    18.    -20.494864])
205 [Astar_planner] Segment 5: [-38.392624 19.804495 -19.229094] -> [-50.    27.    -20.537632]
206 [Step 14 | Total    0] Target: (-50.0, 27.0, -20.5) | Pos: (-47.1, 28.9, -19.2) | Vel: ( -0.9, 0.1, 0.1) | Acc: ( -0.5, 0.3, -0.7) | Rwd: 1.73
207 [Astar_planner] Segment 5 reached in 14 steps, dist=3.69, axis_error=[2.890049 1.8806572 1.3071995], reason=running
208 [Astar_planner] Segment 6 starts from actual pos [-47.112606 28.881027 -19.230211] (planned waypoint was [-50.    27.    -20.537632])
209 [Astar_planner] Segment 6: [-47.112606 28.881027 -19.230211] -> [-59.    37.    -22.769846]
210 [Step 20 | Total    0] Target: (-59.0, 37.0, -22.8) | Pos: (-56.3, 36.3, -20.1) | Vel: ( -0.9, 0.1, -0.1) | Acc: ( -0.9, -0.1, 0.6) | Rwd: -0.05
211 [Astar_planner] Segment 6 reached in 20 steps, dist=3.89, axis_error=[2.73423 0.71118164 2.6706429 ], reason=running
212 [Astar_planner] Segment 7 starts from actual pos [-56.26577 36.28882 -20.099203] (planned waypoint was [-59.    37.    -22.769846])
213 [Astar_planner] Segment 7: [-56.26577 36.28882 -20.099203] -> [-60.    50.    -24.372286]
214 [Step 41 | Total    0] Target: (-60.0, 50.0, -24.4) | Pos: (-57.1, 50.2, -21.4) | Vel: ( -0.2, -0.1, 0.1) | Acc: ( -0.3, 0.1, -0.8) | Rwd: -0.33
215 [Astar_planner] Segment 7 reached in 41 steps, dist=4.13, axis_error=[2.8550987 0.17864609 2.9720383 ], reason=running
216 [Astar_planner] Segment 8 starts from actual pos [-57.145466 50.178486 -21.40002 ] (planned waypoint was [-60.    50.    -24.372286])
217 [Astar_planner] Segment 8: [-57.145466 50.178486 -21.40002 ] -> [-53.    61.    -24.185177]
218 [Step 12 | Total    0] Target: (-53.0, 61.0, -24.2) | Pos: (-54.0, 60.5, -21.7) | Vel: ( 0.6, 1.4, -1.1) | Acc: ( -0.1, 1.2, -1.1) | Rwd: 4.62
219 [Astar_planner] Segment 8 reached in 12 steps, dist=2.67, axis_error=[0.97237396 0.49482346 2.43861 ], reason=running
220 [Astar_planner] Segment 9 starts from actual pos [-53.97052 60.509396 -21.749886] (planned waypoint was [-53.    61.    -24.185177])
221 [Astar_planner] Segment 9: [-53.97052 60.509396 -21.749886] -> [-43.    68.    -22.611385]
222 [Step 147 | Total    0] Target: (-43.0, 68.0, -22.6) | Pos: (-45.8, 68.4, -22.0) | Vel: ( 1.9, 0.4, -0.0) | Acc: ( 1.8, 1.1, 1.1) | Rwd: 2.53
223 [Astar_planner] Segment 9 reached in 147 steps, dist=2.91, axis_error=[2.815403 0.36681366 0.6228409 ], reason=running
224 [Astar_planner] Segment 10 starts from actual pos [-45.80979 68.36798 -21.988605] (planned waypoint was [-43.    68.    -22.611385])

```

```

225 [Astar_planner] Segment 10: [-45.80979  68.36798 -21.988605] -> [-
    44.      80.      -23.294916]
226 [Step  14 | Total    0] Target: (-44.0, 80.0, -23.3) | Pos: (-46.1, 78.5
    , -20.5) | Vel: ( -0.6,  1.2, -1.5) | Acc: ( -1.6,  1.1, -0.0) | Rwd: 2.07
227 [Astar_planner] Segment 10 reached in 14 steps, dist=3.79, axis_error
    =[2.0556717 1.4934921 2.805811 ], reason=running
228 [Astar_planner] Segment 11 starts from actual pos [-46.059494 78.
    51399 -20.498257] (planned waypoint was [-44.      80.      -23.
    294916])
229 [Astar_planner] Segment 11: [-46.059494 78.51399 -20.498257] -> [-
    53.      89.      -23.129202]
230 [Step  15 | Total    0] Target: (-53.0, 89.0, -23.1) | Pos: (-51.4, 87.4
    , -20.8) | Vel: ( 0.0,  0.1, -1.8) | Acc: ( -0.3, -0.4,  2.0) | Rwd: 5.30
231 [Astar_planner] Segment 11 reached in 15 steps, dist=3.23, axis_error
    =[1.6292496 1.5575867 2.3159657], reason=running
232 [Astar_planner] Segment 12 starts from actual pos [-51.3707  87.
    442665 -20.818737] (planned waypoint was [-53.      89.      -23.
    129202])
233 [Astar_planner] Segment 12: [-51.3707  87.442665 -20.818737] -> [-
    67.      89.      -20.09961]
234 [Step  12 | Total    0] Target: (-67.0, 89.0, -20.1) | Pos: (-64.3, 90.0
    , -17.5) | Vel: ( -2.5,  0.4,  0.1) | Acc: ( 0.5, -0.1,  0.4) | Rwd: 1.52
235 [Astar_planner] Segment 12 reached in 12 steps, dist=3.91, axis_error
    =[2.6791916 1.0376663 2.6465797], reason=running
236 [Astar_planner] Segment 13 starts from actual pos [-64.328156 90.
    03893 -17.452662] (planned waypoint was [-67.      89.      -20.09961
    ])
237 [Astar_planner] Segment 13: [-64.328156 90.03893 -17.452662] -> [-
    80.      89.      -20.225142]
238 [Step  12 | Total    0] Target: (-80.0, 89.0, -20.2) | Pos: (-78.0, 89.8
    , -17.5) | Vel: ( -1.8,  0.6,  0.2) | Acc: ( -0.5,  1.2,  0.8) | Rwd: -0.34
239 [Astar_planner] Segment 13 reached in 12 steps, dist=3.52, axis_error
    =[2.038147 0.8136368 2.7477493], reason=running
240 [Astar_planner] Segment 14 starts from actual pos [-77.96732  89.
    81547 -17.476751] (planned waypoint was [-80.      89.      -20.
    225142])
241 [Astar_planner] Segment 14: [-77.96732  89.81547 -17.476751] -> [-
    93.      90.      -20.726328]
242 [Step  15 | Total    0] Target: (-93.0, 90.0, -20.7) | Pos: (-90.4, 87.1
    , -18.2) | Vel: ( -1.8,  0.0,  0.0) | Acc: ( -0.3, -0.1, -0.3) | Rwd: 0.44
243 [Astar_planner] Segment 14 reached in 15 steps, dist=4.63, axis_error
    =[2.603531 2.892212 2.51552 ], reason=running
244 [Astar_planner] Segment 15 starts from actual pos [-90.401726 87.
    10782 -18.210701] (planned waypoint was [-93.      90.      -20.
    726328])
245 [Astar_planner] Segment 15: [-90.401726 87.10782 -18.210701] -> [-
    100.     101.     -23.900085]
246 [Step  20 | Total    0] Target: (-100.0, 101.0, -23.9) | Pos: (-99.0, 104

```

```

246 .0, -22.7) | Vel: ( -1.3, -1.0, -0.6) | Acc: ( 1.1, 0.9, 0.3) | Rwd: 3.12

247 [Astar_planner] Segment 15 reached in 20 steps, dist=3.38, axis_error
    =[1.0468674 2.9853973 1.1982498], reason=running
248 [Astar_planner] Segment 16 starts from actual pos [-98.96072 103.
    97925 -22.70542] (planned waypoint was [-100.    101.    -23.
    900085])
249 [Astar_planner] Segment 16: [-98.96072 103.97925 -22.70542] -> [-
    101.    114.    -23.545528]
250 [Step 36 | Total    0] Target: (-101.0, 114.0, -23.5) | Pos: (-99.0, 111
    .8, -22.1) | Vel: ( -0.4, 1.7, -0.8) | Acc: ( -0.3, 0.7, -0.0) | Rwd: 2.20

251 [Astar_planner] Segment 16 reached in 36 steps, dist=3.30, axis_error
    =[2.002449 2.1705017 1.4791965], reason=running
252 [Astar_planner] Segment 17 starts from actual pos [-98.999954 111.
    839714 -22.07086 ] (planned waypoint was [-101.    114.    -23.
    545528])
253 [Astar_planner] Segment 17: [-98.999954 111.839714 -22.07086 ] -> [-
    106.    125.    -4.903579]
254 [Step 33 | Total    0] Target: (-106.0, 125.0, -4.9) | Pos: (-107.3,
    125.8, -5.7) | Vel: ( 0.2, -0.5, 1.4) | Acc: ( -1.9, -1.3, 0.7) | Rwd: 1
    .28
255 [Astar_planner] Segment 17 reached in 33 steps, dist=1.76, axis_error
    =[1.3298874 0.81627655 0.82445955], reason=running
256 Task 4 result: success (Success: Reached destination (18/18
    segments). Shield interventions=126, emergencies=0, recoveries=21,
    recovery_steps=126.)
257
258 --- Task 5/38 ---
259 From [-106.2 125.18 -4.9 ] to [-10.1 -76.12 -9.8 ]
260 [Astar_planner] Top-down occupancy: 350x390, x=[-179.0, 170.0], y=[-
    200.0, 189.0], occupied=72869 cells (53.4%)
261 [Astar_planner] Occupancy base_grid=390x350, occupied=72869 cells
    (53.4%)
262 [Astar_planner] Occupancy A*: grid=390x350, occupied=53.4%,
    safety_margin=1.5
263 Path planned via occupancy (Points: 24, Dist: 295.2m). Local target z/
    alt: 0:-4.91(4.91m), 1:-21.38(21.38m), 2:-21.35(21.35m), 3:-21.87(21.
    87m), 4:-24.77(24.77m), 5:-21.82(21.82m), 6:-20.76(20.76m), 7:-20.93
    (20.93m), 8:-21.57(21.57m), 9:-20.75(20.75m), 10:-20.36(20.36m), 11
    :-22.99(22.99m), 12:-24.52(24.52m), 13:-20.56(20.56m), 14:-22.08(22.
    08m), 15:-20.55(20.55m), 16:-21.63(21.63m), 17:-23.32(23.32m), 18:-
    23.88(23.88m), 19:-22.94(22.94m), 20:-20.95(20.95m), 21:-24.97(24.
    97m), 22:-24.47(24.47m), 23:-9.80(9.80m)
264 [Astar_planner] Segment 0: [-106.    125.    -4.905911] -> [-100
    .    115.    -21.380838]
265 [Astar_planner] Placing first segment start at [-106.    125.    -4.
    905911]

```

```

266 [Astar_planner] Initial climb to 20.0m.
267 [Step 12 | Total 0] Target: (-100.0, 115.0, -21.4) | Pos: (-100.7,
    117.4, -20.1) | Vel: ( 0.8, -1.3, -0.0) | Acc: ( 0.7, 1.2, -0.3) | Rwd:
    2.71
268 [Astar_planner] Segment 0 reached in 12 steps, dist=2.84, axis_error
    =[0.709198 2.439186 1.2657642], reason=running
269 [Astar_planner] Segment 1 starts from actual pos [-100.70417 117.
    43156 -20.11512] (planned waypoint was [-100. 115. -21.
    380838])
270 [Astar_planner] Segment 1: [-100.70417 117.43156 -20.11512] -> [-
    100. 103. -21.34862]
271 [Step 14 | Total 0] Target: (-100.0, 103.0, -21.3) | Pos: (-100.1,
    105.2, -20.1) | Vel: ( 0.4, -1.4, -0.0) | Acc: ( 0.4, 1.6, -0.7) | Rwd:
    3.12
272 [Astar_planner] Segment 1 reached in 14 steps, dist=2.52, axis_error
    =[0.08891296 2.186966 1.2444115], reason=running
273 [Astar_planner] Segment 2 starts from actual pos [-100.086624 105.
    1787 -20.10426] (planned waypoint was [-100. 103. -21.
    34862])
274 [Astar_planner] Segment 2: [-100.086624 105.1787 -20.10426] -> [-
    95. 92. -21.86944]
275 [Step 15 | Total 0] Target: (-95.0, 92.0, -21.9) | Pos: (-97.8, 94.2
    , -19.1) | Vel: ( 0.4, -0.8, 0.1) | Acc: ( -1.3, 0.5, -0.4) | Rwd: 0.83
276 [Astar_planner] Segment 2 reached in 15 steps, dist=4.45, axis_error
    =[2.7586594 2.1546402 2.7398167], reason=running
277 [Astar_planner] Segment 3 starts from actual pos [-97.75755 94.1523
    -19.129442] (planned waypoint was [-95. 92. -21.86944])
278 [Astar_planner] Segment 3: [-97.75755 94.1523 -19.129442] -> [-84
    . 89. -24.770988]
279 [Step 49 | Total 0] Target: (-84.0, 89.0, -24.8) | Pos: (-85.4, 87.8
    , -22.7) | Vel: ( 0.4, 0.5, -1.5) | Acc: ( 1.9, -0.8, 2.0) | Rwd: 5.65
280 [Astar_planner] Segment 3 reached in 49 steps, dist=2.79, axis_error
    =[1.4169388 1.1886292 2.085865], reason=running
281 [Astar_planner] Segment 4 starts from actual pos [-85.41428 87.8146
    -22.694172] (planned waypoint was [-84. 89. -24.770988])
282 [Astar_planner] Segment 4: [-85.41428 87.8146 -22.694172] -> [-71
    . 89. -21.821756]
283 [Step 11 | Total 0] Target: (-71.0, 89.0, -21.8) | Pos: (-73.9, 87.8
    , -18.9) | Vel: ( 1.8, 0.1, -0.1) | Acc: ( -2.0, -2.0, -1.4) | Rwd: 2.18
284 [Astar_planner] Segment 4 reached in 11 steps, dist=4.27, axis_error
    =[2.9291992 1.1554794 2.8782902], reason=running
285 [Astar_planner] Segment 5 starts from actual pos [-73.923836 87.
    8447 -18.943705] (planned waypoint was [-71. 89. -21.
    821756])
286 [Astar_planner] Segment 5: [-73.923836 87.8447 -18.943705] -> [-59
    . 89. -20.763355]
287 [Step 11 | Total 0] Target: (-59.0, 89.0, -20.8) | Pos: (-59.9, 88.7
    , -18.0) | Vel: ( 1.8, -0.1, -1.3) | Acc: ( -2.0, -2.0, -1.4) | Rwd: 2.82

```

```

288 [Astar_planner] Segment 5 reached in 11 steps, dist=2.91, axis_error
    =[0.92325974 0.31586456 2.7463474 ], reason=running
289 [Astar_planner] Segment 6 starts from actual pos [-59.91797  88.
    68375 -18.020735] (planned waypoint was [-59.    89.    -20.
    763355])
290 [Astar_planner] Segment 6: [-59.91797  88.68375 -18.020735] -> [-48
    .    84.    -20.929054]
291 [Step  9 | Total    0] Target: (-48.0, 84.0, -20.9) | Pos: (-50.2, 84.2
    , -18.8) | Vel: ( 1.4, -0.3, -0.0) | Acc: ( -1.9, 1.6, -0.2) | Rwd: 2.14
292 [Astar_planner] Segment 6 reached in 9 steps, dist=3.09, axis_error=[
    2.2420654 0.18126678 2.115219 ], reason=running
293 [Astar_planner] Segment 7 starts from actual pos [-50.237812 84.
    18046 -18.813911] (planned waypoint was [-48.    84.    -20.
    929054])
294 [Astar_planner] Segment 7: [-50.237812 84.18046 -18.813911] -> [-
    41.    75.    -21.57088]
295 [Step 12 | Total    0] Target: (-41.0, 75.0, -21.6) | Pos: (-43.6, 75.6
    , -21.1) | Vel: ( 1.0, -0.7, 0.1) | Acc: ( 0.7, -0.5, -0.9) | Rwd: 2.53
296 [Astar_planner] Segment 7 reached in 12 steps, dist=2.70, axis_error
    =[2.5970726 0.61920166 0.43187904], reason=running
297 [Astar_planner] Segment 8 starts from actual pos [-43.594044 75.
    6171 -21.138775] (planned waypoint was [-41.    75.    -21.57088])
298 [Astar_planner] Segment 8: [-43.594044 75.6171 -21.138775] -> [-47
    .    66.    -20.747387]
299 [Step 11 | Total    0] Target: (-47.0, 66.0, -20.7) | Pos: (-45.9, 68.3
    , -19.6) | Vel: ( -1.3, -0.9, 0.0) | Acc: ( -1.4, 0.5, -0.2) | Rwd: 2.53
300 [Astar_planner] Segment 8 reached in 11 steps, dist=2.80, axis_error
    =[1.1275826 2.28199 1.1662922], reason=running
301 [Astar_planner] Segment 9 starts from actual pos [-45.876347 68.
    27916 -19.581095] (planned waypoint was [-47.    66.    -20.
    747387])
302 [Astar_planner] Segment 9: [-45.876347 68.27916 -19.581095] -> [-
    56.    58.    -20.363777]
303 [Step 11 | Total    0] Target: (-56.0, 58.0, -20.4) | Pos: (-54.7, 60.2
    , -20.4) | Vel: ( -0.9, -1.0, -0.0) | Acc: ( 0.7, -0.6, -0.2) | Rwd: 3.28
304 [Astar_planner] Segment 9 reached in 11 steps, dist=2.58, axis_error
    =[1.3193398 2.214756 0.08350945], reason=running
305 [Astar_planner] Segment 10 starts from actual pos [-54.683224 60.
    21183 -20.447409] (planned waypoint was [-56.    58.    -20.
    363777])
306 [Astar_planner] Segment 10: [-54.683224 60.21183 -20.447409] -> [-
    60.    47.    -22.988493]
307 [Step 11 | Total    0] Target: (-60.0, 47.0, -23.0) | Pos: (-59.5, 49.7
    , -21.7) | Vel: ( -0.3, -0.9, 0.0) | Acc: ( -0.1, 0.1, -0.6) | Rwd: 2.16
308 [Astar_planner] Segment 10 reached in 11 steps, dist=3.05, axis_error
    =[0.49835968 2.7266998 1.2705345 ], reason=running
309 [Astar_planner] Segment 11 starts from actual pos [-59.50164 49.
    7267 -21.717958] (planned waypoint was [-60.    47.    -22.

```

```

309 988493])
310 [Astar_planner] Segment 11: [-59.50164  49.7267 -21.717958] -> [-
58.    36.    -24.522413]
311 [Step  13 | Total    0] Target: (-58.0,  36.0, -24.5) | Pos: (-57.0,  37.8
, -23.4) | Vel: ( -0.3, -1.6, -0.1) | Acc: ( -0.4, -1.9, -0.1) | Rwd:  4.03
312 [Astar_planner] Segment 11 reached in 13 steps, dist=2.38, axis_error
=[1.0167084 1.8152771 1.1461468], reason=running
313 [Astar_planner] Segment 12 starts from actual pos [-56.98429  37.
810387 -23.376677] (planned waypoint was [-58.    36.    -24.
522413])
314 [Astar_planner] Segment 12: [-56.98429  37.810387 -23.376677] -> [-
49.    26.    -20.563896]
315 [Step  12 | Total    0] Target: (-49.0,  26.0, -20.6) | Pos: (-51.1,  28.5
, -21.0) | Vel: (  0.4, -0.9, -0.1) | Acc: (  0.4, -1.5, -0.1) | Rwd:  2.98
316 [Astar_planner] Segment 12 reached in 12 steps, dist=3.27, axis_error
=[2.0834465 2.4900131 0.39681435], reason=running
317 [Astar_planner] Segment 13 starts from actual pos [-51.082188  28.
48717 -20.960938] (planned waypoint was [-49.    26.    -20.
563896])
318 [Astar_planner] Segment 13: [-51.082188  28.48717 -20.960938] -> [-
41.    18.    -22.077385]
319 [Step  19 | Total    0] Target: (-41.0,  18.0, -22.1) | Pos: (-42.4,  15.3
, -20.6) | Vel: (  0.7,  0.3, -0.0) | Acc: (  0.1, -0.7,  0.6) | Rwd:  0.57
320 [Astar_planner] Segment 13 reached in 19 steps, dist=3.40, axis_error
=[1.4393082 2.7264175 1.4315071], reason=running
321 [Astar_planner] Segment 14 starts from actual pos [-42.43709  15.
274424 -20.646  ] (planned waypoint was [-41.    18.    -22.
077385])
322 [Astar_planner] Segment 14: [-42.43709  15.274424 -20.646  ] -> [-32
.    9.    -20.54739]
323 [Step  16 | Total    0] Target: (-32.0,  9.0, -20.5) | Pos: (-34.9,  8.5, -
19.9) | Vel: (  1.1, -0.6, -0.1) | Acc: (  0.9, -1.5,  0.6) | Rwd:  1.89
324 [Astar_planner] Segment 14 reached in 16 steps, dist=2.97, axis_error
=[2.8522186 0.5004425 0.64577675], reason=running
325 [Astar_planner] Segment 15 starts from actual pos [-34.84902  8.
497761 -19.901901] (planned waypoint was [-32.    9.    -20.54739
])
326 [Astar_planner] Segment 15: [-34.84902  8.497761 -19.901901] -> [-
23.    1.    -21.62932]
327 [Step  36 | Total    0] Target: (-23.0,  1.0, -21.6) | Pos: (-23.8,  3.1, -
18.9) | Vel: (  0.6, -0.3, -0.8) | Acc: (  1.2,  0.1,  0.3) | Rwd:  3.85
328 [Astar_planner] Segment 15 reached in 36 steps, dist=3.50, axis_error
=[0.7635231 2.0992694 2.6952019], reason=running
329 [Astar_planner] Segment 16 starts from actual pos [-23.761858  3.
0984764 -18.93654  ] (planned waypoint was [-23.    1.    -21.
62932])
330 [Astar_planner] Segment 16: [-23.761858  3.0984764 -18.93654
] -> [-11.    -3.    -23.318838]

```

```

331 [Step 42 | Total 0] Target: (-11.0, -3.0, -23.3) | Pos: (-13.3, -3.3, -
    20.9) | Vel: ( -0.0, 0.5, -0.7) | Acc: ( -1.5, 0.4, 2.0) | Rwd: 4.70
332 [Astar_planner] Segment 16 reached in 42 steps, dist=3.33, axis_error
    =[2.263301 0.26672816 2.433895 ], reason=running
333 [Astar_planner] Segment 17 starts from actual pos [-13.263359 -3.
    2652986 -20.887165 ] (planned waypoint was [-11. -3. -23.
    318838])
334 [Astar_planner] Segment 17: [-13.263359 -3.2652986 -20.887165
    ] -> [ -9. -14. -23.87793]
335 [Step 63 | Total 0] Target: ( -9.0, -14.0, -23.9) | Pos: (-11.7, -13.2
    , -21.6) | Vel: ( 0.8, -0.8, -0.1) | Acc: ( 1.4, -1.4, 0.3) | Rwd: 0.38
336 [Astar_planner] Segment 17 reached in 63 steps, dist=3.63, axis_error
    =[2.6719303 0.82018757 2.3172646 ], reason=running
337 [Astar_planner] Segment 18 starts from actual pos [-11.67193 -13.
    179812 -21.560665] (planned waypoint was [ -9. -14. -23.87793
    ])
338 [Astar_planner] Segment 18: [-11.67193 -13.179812 -21.560665] -> [ -
    9. -26. -22.944597]
339 [Step 12 | Total 0] Target: ( -9.0, -26.0, -22.9) | Pos: (-11.5, -23.6
    , -20.1) | Vel: ( 0.7, -0.7, -0.0) | Acc: ( 1.8, 0.4, 0.3) | Rwd: 1.09
340 [Astar_planner] Segment 18 reached in 12 steps, dist=4.48, axis_error
    =[2.5171814 2.4280872 2.8032207], reason=running
341 [Astar_planner] Segment 19 starts from actual pos [-11.515003 -23.
    574024 -20.141506] (planned waypoint was [ -9. -26. -22.
    944597])
342 [Astar_planner] Segment 19: [-11.515003 -23.574024 -20.141506
    ] -> [ -9. -39. -20.95142]
343 [Step 12 | Total 0] Target: ( -9.0, -39.0, -21.0) | Pos: (-11.7, -36.3
    , -20.3) | Vel: ( 0.2, -1.1, 0.1) | Acc: ( 0.3, 0.5, -0.9) | Rwd: 2.87
344 [Astar_planner] Segment 19 reached in 12 steps, dist=3.88, axis_error
    =[2.7307901 2.678421 0.6598606], reason=running
345 [Astar_planner] Segment 20 starts from actual pos [-11.730151 -36.
    32475 -20.29138 ] (planned waypoint was [ -9. -39. -20.95142])
346 [Astar_planner] Segment 20: [-11.730151 -36.32475 -20.29138 ] -> [ -
    9. -51. -24.969835]
347 [Step 14 | Total 0] Target: ( -9.0, -51.0, -25.0) | Pos: (-11.8, -52.9
    , -22.7) | Vel: ( 0.5, -0.9, -0.2) | Acc: ( 0.7, -1.8, 1.4) | Rwd: -2.71
348 [Astar_planner] Segment 20 reached in 14 steps, dist=4.02, axis_error
    =[2.7528028 1.8768654 2.243639 ], reason=running
349 [Astar_planner] Segment 21 starts from actual pos [-11.751419 -52.
    879467 -22.7267 ] (planned waypoint was [ -9. -51. -24.
    969835])
350 [Astar_planner] Segment 21: [-11.751419 -52.879467 -22.7267 ] -> [ -
    9. -64. -24.471504]
351 [Step 10 | Total 0] Target: ( -9.0, -64.0, -24.5) | Pos: (-11.4, -63.6
    , -22.3) | Vel: ( -0.2, -1.6, -1.6) | Acc: ( -0.4, -1.4, 0.9) | Rwd: 5.65
352 [Astar_planner] Segment 21 reached in 10 steps, dist=3.24, axis_error
    =[2.3509798 0.38845444 2.2000027 ], reason=running

```



```

353 [Astar_planner] Segment 22 starts from actual pos [-11.351435 -63.
616364 -22.27628 ] (planned waypoint was [ -9.    -64.    -24.
471504])
354 [Astar_planner] Segment 22: [-11.351435 -63.616364 -22.27628 ] -> [-
10.    -76.    -9.798569]
355 [Step 16 | Total    0] Target: (-10.0, -76.0, -9.8) | Pos: ( -9.1, -74.5, -
10.2) | Vel: ( 0.2, -0.6, 0.0) | Acc: ( 0.1, 0.2, 0.5) | Rwd: 1.23
356 [Astar_planner] Segment 22 reached in 16 steps, dist=1.75, axis_error
=[0.8941374 1.455864 0.35592556], reason=running
357 Task 5 result: success (Success: Reached destination (23/23
segments). Shield interventions=6, emergencies=0, recoveries=1,
recovery_steps=6.)
358
359 --- Task 6/38 ---
360 From [64.2  3.18 -7.4 ] to [-60.9 -95.62 -8.8 ]
361 [Astar_planner] Top-down occupancy: 350x390, x=[-179.0, 170.0], y=[-
200.0, 189.0], occupied=72869 cells (53.4%)
362 [Astar_planner] Occupancy base_grid=390x350, occupied=72869 cells
(53.4%)
363 [Astar_planner] Occupancy A*: grid=390x350, occupied=53.4%,
safety_margin=1.5
364 Path planned via occupancy (Points: 17, Dist: 221.9m). Local target z/
alt: 0:-7.40(7.40m), 1:-21.27(21.27m), 2:-23.57(23.57m), 3:-22.74(22.
74m), 4:-21.97(21.97m), 5:-22.48(22.48m), 6:-24.43(24.43m), 7:-21.35
(21.35m), 8:-24.95(24.95m), 9:-22.18(22.18m), 10:-22.06(22.06m), 11
:-23.97(23.97m), 12:-24.07(24.07m), 13:-20.95(20.95m), 14:-23.90(23.
90m), 15:-22.23(22.23m), 16:-8.80(8.80m)
365 [Astar_planner] Segment 0: [64.    3.    -7.4023633] -> [ 51.
2.    -21.266174]
366 [Astar_planner] Placing first segment start at [64.    3.    -7.
4023633]
367 [Astar_planner] Initial climb to 20.0m.
368 [Step 10 | Total    0] Target: ( 51.0,  2.0, -21.3) | Pos: ( 53.9,  1.4, -
19.1) | Vel: ( -2.0, -0.7, -0.1) | Acc: ( -1.7, -0.8, -0.3) | Rwd: 3.38
369 [Astar_planner] Segment 0 reached in 10 steps, dist=3.70, axis_error
=[2.9486122 0.5586451 2.1635723], reason=running
370 [Astar_planner] Segment 1 starts from actual pos [ 53.94247  1.
4392439 -19.102823 ] (planned waypoint was [ 51.    2.    -21.
266174])
371 [Astar_planner] Segment 1: [ 53.94247  1.4392439 -19.102823 ] -> [
38.    2.    -23.56636]
372 [Step 113 | Total    0] Target: ( 38.0,  2.0, -23.6) | Pos: ( 40.5, -0.2
, -21.4) | Vel: ( 0.2,  1.9,  0.0) | Acc: ( 0.6,  1.1, -0.4) | Rwd: 1.25
373 [Astar_planner] Segment 1 reached in 113 steps, dist=3.96, axis_error
=[2.4745636 2.1784408 2.1956863], reason=running
374 [Astar_planner] Segment 2 starts from actual pos [ 40.47574  -0.
16733591 -21.370497 ] (planned waypoint was [ 38.    2.    -23.
56636])

```

```

375 [Astar_planner] Segment 2: [ 40.47574   -0.16733591 -21.370497
    ] -> [ 25.       1.       -22.738485]
376 [Step 12 | Total    0] Target: ( 25.0,   1.0, -22.7) | Pos: ( 27.5, -1.6, -
    20.6) | Vel: ( -1.7, -0.5, -0.1) | Acc: ( -1.4, -0.6, -0.0) | Rwd: 1.51
377 [Astar_planner] Segment 2 reached in 12 steps, dist=4.21, axis_error
    =[2.4712257 2.6198997 2.1766853], reason=running
378 [Astar_planner] Segment 3 starts from actual pos [ 27.4662   -1.
    6214888 -20.562216 ] (planned waypoint was [ 25.       1.       -22.
    738485])
379 [Astar_planner] Segment 3: [ 27.4662   -1.6214888 -20.562216 ] -> [
    12.       1.       -21.96954]
380 [Step 77 | Total    0] Target: ( 12.0,   1.0, -22.0) | Pos: ( 12.3, -1.6, -
    21.9) | Vel: ( -0.4,  0.8,  0.0) | Acc: (  1.3,  1.7,  0.1) | Rwd: 1.46
381 [Astar_planner] Segment 3 reached in 77 steps, dist=2.66, axis_error
    =[0.28123856 2.6387024 0.11662483], reason=running
382 [Astar_planner] Segment 4 starts from actual pos [ 12.280082  -1.
    6362985 -21.852798 ] (planned waypoint was [ 12.       1.       -21.
    96954])
383 [Astar_planner] Segment 4: [ 12.280082  -1.6362985 -21.852798
    ] -> [ -2.       1.       -22.475687]
384 [Step 15 | Total    0] Target: ( -2.0,   1.0, -22.5) | Pos: ( -0.9, -1.8, -
    20.2) | Vel: ( -0.8,  0.5, -0.1) | Acc: ( -0.3,  0.0,  1.3) | Rwd: -0.33
385 [Astar_planner] Segment 4 reached in 15 steps, dist=3.77, axis_error
    =[1.0895648 2.7761865 2.3098621], reason=running
386 [Astar_planner] Segment 5 starts from actual pos [ -0.91281223 -1.
    7748318 -20.166092 ] (planned waypoint was [ -2.       1.       -22.
    475687])
387 [Astar_planner] Segment 5: [ -0.91281223 -1.7748318 -20.166092
    ] -> [ -9.       -9.       -24.431667]
388 [Step 12 | Total    0] Target: ( -9.0,  -9.0, -24.4) | Pos: ( -6.2, -9.5, -
    21.5) | Vel: ( -0.4, -0.3,  0.0) | Acc: (  0.6, -0.9, -0.1) | Rwd: -0.99
389 [Astar_planner] Segment 5 reached in 12 steps, dist=4.07, axis_error
    =[2.8240771 0.4655447 2.889145 ], reason=running
390 [Astar_planner] Segment 6 starts from actual pos [ -6.1780696 -9.
    467523  -21.542437 ] (planned waypoint was [ -9.       -9.       -24.
    431667])
391 [Astar_planner] Segment 6: [ -6.1780696 -9.467523  -21.542437
    ] -> [ -9.       -22.       -21.352524]
392 [Step 10 | Total    0] Target: ( -9.0, -22.0, -21.4) | Pos: ( -7.7, -19.3, -
    21.6) | Vel: ( -0.2, -1.0,  0.0) | Acc: ( -0.8, -0.3, -1.0) | Rwd: 2.30
393 [Astar_planner] Segment 6 reached in 10 steps, dist=2.99, axis_error
    =[1.3256459 2.6762924 0.22298622], reason=running
394 [Astar_planner] Segment 7 starts from actual pos [ -7.674824 -19.
    326693 -21.575365] (planned waypoint was [ -9.       -22.       -21.
    352524])
395 [Astar_planner] Segment 7: [ -7.674824 -19.326693 -21.575365] -> [ -9
    .       -35.       -24.953146]
396 [Step 13 | Total    0] Target: ( -9.0, -35.0, -25.0) | Pos: ( -9.2, -33.7, -

```

```

396 23.2) | Vel: ( 0.3, -1.2, -1.5) | Acc: ( 0.3, -1.2, -0.3) | Rwd: 608.02
    -> Episode End: ARRIVED!
397
398 [Astar_planner] Segment 7 reached in 13 steps, dist=2.18, axis_error
    =[0.20545101 1.3268127 1.7165356 ], reason=arrived
399 [Astar_planner] Segment 8 starts from actual pos [ -9.204651 -33.
    676647 -23.24118 ] (planned waypoint was [ -9.    -35.    -24.
    953146])
400 [Astar_planner] Segment 8: [ -9.204651 -33.676647 -23.24118 ] -> [ -9
    .    -48.    -22.18418]
401 [Step 12 | Total    0] Target: ( -9.0, -48.0, -22.2) | Pos: ( -8.7, -45.1, -
    21.0) | Vel: ( 0.8, -0.9, -0.0) | Acc: ( 1.2, 0.5, 0.3) | Rwd: 1.68
402 [Astar_planner] Segment 8 reached in 12 steps, dist=3.16, axis_error
    =[0.31726933 2.9095955 1.1893349 ], reason=running
403 [Astar_planner] Segment 9 starts from actual pos [ -8.682731 -45.
    090405 -20.994844] (planned waypoint was [ -9.    -48.    -22.18418
    ])
404 [Astar_planner] Segment 9: [ -8.682731 -45.090405 -20.994844] -> [ -9
    .    -62.    -22.060314]
405 [Step 14 | Total    0] Target: ( -9.0, -62.0, -22.1) | Pos: ( -8.1, -59.3, -
    21.3) | Vel: ( 0.3, -1.3, -0.0) | Acc: ( 0.5, -1.0, -0.1) | Rwd: 2.59
406 [Astar_planner] Segment 9 reached in 14 steps, dist=2.89, axis_error
    =[0.8568249 2.666397 0.7178936], reason=running
407 [Astar_planner] Segment 10 starts from actual pos [ -8.143175 -59.
    333603 -21.34242 ] (planned waypoint was [ -9.    -62.    -22.
    060314])
408 [Astar_planner] Segment 10: [ -8.143175 -59.333603 -21.34242 ] -> [-
    10.    -75.    -23.966547]
409 [Step 11 | Total    0] Target: (-10.0, -75.0, -24.0) | Pos: ( -9.8, -72.2
    , -22.1) | Vel: ( -0.2, -1.7, 0.0) | Acc: ( -0.2, -1.9, -0.9) | Rwd: 4.27
410 [Astar_planner] Segment 10 reached in 11 steps, dist=3.43, axis_error
    =[0.18479347 2.8392563 1.9078636 ], reason=running
411 [Astar_planner] Segment 11 starts from actual pos [ -9.815793 -72.
    16586 -22.058605] (planned waypoint was [-10.    -75.    -23.
    966547])
412 [Astar_planner] Segment 11: [ -9.815793 -72.16586 -22.058605] -> [-
    13.    -86.    -24.065504]
413 [Step 10 | Total    0] Target: (-13.0, -86.0, -24.1) | Pos: (-12.2, -83.0
    , -22.1) | Vel: ( -0.3, -1.4, -0.1) | Acc: ( -0.3, -1.2, 0.5) | Rwd: 2.28
414 [Astar_planner] Segment 11 reached in 10 steps, dist=3.65, axis_error
    =[0.8132696 2.960907 1.9814739], reason=running
415 [Astar_planner] Segment 12 starts from actual pos [-12.187768 -83.
    04323 -22.084387] (planned waypoint was [-13.    -86.    -24.
    065504])
416 [Astar_planner] Segment 12: [-12.187768 -83.04323 -22.084387] -> [-
    26.    -88.    -20.949385]
417 [Step 12 | Total    0] Target: (-26.0, -88.0, -20.9) | Pos: (-23.6, -88.2
    , -19.9) | Vel: ( -1.6, -0.6, -0.1) | Acc: ( -1.4, -0.7, -0.1) | Rwd: 3.28

```

```

418 [Astar_planner] Segment 12 reached in 12 steps, dist=2.60, axis_error
    =[2.3541832 0.19203186 1.0920544 ], reason=running
419 [Astar_planner] Segment 13 starts from actual pos [-23.650667 -88.
    19369 -19.857645] (planned waypoint was [-26.    -88.    -20.
    949385])
420 [Astar_planner] Segment 13: [-23.650667 -88.19369 -19.857645] -> [-
    39.    -88.    -23.899075]
421 [Step 12 | Total    0] Target: (-39.0, -88.0, -23.9) | Pos: (-36.4, -90.0
    , -20.9) | Vel: ( -1.0, 0.2, 0.2) | Acc: ( -0.3, -0.3, -1.8) | Rwd: 2.40
422 [Astar_planner] Segment 13 reached in 12 steps, dist=4.43, axis_error
    =[2.5969467 2.0404816 2.9566574], reason=running
423 [Astar_planner] Segment 14 starts from actual pos [-36.406025 -90.
    0399 -20.941874] (planned waypoint was [-39.    -88.    -23.
    899075])
424 [Astar_planner] Segment 14: [-36.406025 -90.0399 -20.941874] -> [-
    53.    -88.    -22.226208]
425 [Step 124 | Total    0] Target: (-53.0, -88.0, -22.2) | Pos: (-50.1, -90.6
    , -22.8) | Vel: ( -1.1, 0.2, -0.1) | Acc: ( -0.2, 0.5, 0.2) | Rwd: 2.35
426 [Astar_planner] Segment 14 reached in 124 steps, dist=4.00,
    axis_error=[2.948742 2.638115 0.5609627], reason=running
427 [Astar_planner] Segment 15 starts from actual pos [-50.054695 -90.
    63756 -22.787373] (planned waypoint was [-53.    -88.    -22.
    226208])
428 [Astar_planner] Segment 15: [-50.054695 -90.63756 -22.787373] -> [-
    61. -96. -8.8]
429 [Step 51 | Total    0] Target: (-61.0, -96.0, -8.8) | Pos: (-59.7, -95.1
    , -9.4) | Vel: ( -0.2, 0.1, -0.1) | Acc: ( 1.1, 0.9, 1.0) | Rwd: 1.44
430 [Astar_planner] Segment 15 reached in 51 steps, dist=1.70, axis_error
    =[1.3184929 0.9008484 0.5786829], reason=running
431 Task 6 result: success (Success: Reached destination (16/16
    segments). Shield interventions=282, emergencies=18, recoveries=47
    , recovery_steps=282.)
432
433 --- Task 7/38 ---
434 From [108. -87.82 -5.4 ] to [ -61. -139.82 -11.6 ]
435 [Astar_planner] Top-down occupancy: 350x390, x=[-179.0, 170.0], y=[-
    200.0, 189.0], occupied=72869 cells (53.4%)
436 [Astar_planner] Occupancy base_grid=390x350, occupied=72869 cells
    (53.4%)
437 [Astar_planner] Occupancy A*: grid=390x350, occupied=53.4%,
    safety_margin=1.5
438 Path planned via occupancy (Points: 18, Dist: 240.7m). Local target z/
    alt: 0:-5.41(5.41m), 1:-21.07(21.07m), 2:-22.29(22.29m), 3:-23.09(23.
    09m), 4:-21.82(21.82m), 5:-21.90(21.90m), 6:-24.53(24.53m), 7:-21.51
    (21.51m), 8:-20.89(20.89m), 9:-21.90(21.90m), 10:-21.74(21.74m), 11
    :-21.74(21.74m), 12:-23.62(23.62m), 13:-21.90(21.90m), 14:-23.15(23.
    15m), 15:-22.50(22.50m), 16:-22.30(22.30m), 17:-11.60(11.60m)
439 [Astar_planner] Segment 0: [108.    -88.    -5.4063115] -> [ 98

```

```

439 .      -98.      -21.073708]
440 [Astar_planner] Placing first segment start at [108.      -88.      -5.
4063115]
441 [Astar_planner] Initial climb to 20.0m.
442 [Step 16 | Total      0] Target: ( 98.0, -98.0, -21.1) | Pos: (100.5, -95.4
, -20.6) | Vel: ( -0.4, -0.9, -0.2) | Acc: ( 1.0, 1.4, -0.4) | Rwd: 1.87
443 [Astar_planner] Segment 0 reached in 16 steps, dist=3.64, axis_error
=[2.5297928 2.5689774 0.5005989], reason=running
444 [Astar_planner] Segment 1 starts from actual pos [100.52856 -95.
433655 -20.573711] (planned waypoint was [ 98.      -98.      -21.
073708])
445 [Astar_planner] Segment 1: [100.52856 -95.433655 -20.573711] -> [
97.      -111.      -22.292807]
446 [Step 33 | Total      0] Target: ( 97.0, -111.0, -22.3) | Pos: ( 97.4, -108
.4, -22.2) | Vel: ( -0.3, -1.0, -0.4) | Acc: ( 0.0, 0.4, 0.5) | Rwd: 1.71

447 [Astar_planner] Segment 1 reached in 33 steps, dist=2.62, axis_error
=[0.39816284 2.5886765 0.05935287], reason=running
448 [Astar_planner] Segment 2 starts from actual pos [ 97.397385 -108.
41424 -22.234737] (planned waypoint was [ 97.      -111.      -22.
292807])
449 [Astar_planner] Segment 2: [ 97.397385 -108.41424 -22.234737
] -> [ 90.      -122.      -23.093605]
450 [Step 22 | Total      0] Target: ( 90.0, -122.0, -23.1) | Pos: ( 90.8, -119
.3, -22.6) | Vel: ( -0.1, -0.8, -0.3) | Acc: ( 1.5, 0.8, -0.3) | Rwd: 1.15

451 [Astar_planner] Segment 2 reached in 22 steps, dist=2.84, axis_error
=[0.7693863 2.6759949 0.54292107], reason=running
452 [Astar_planner] Segment 3 starts from actual pos [ 90.76908 -119.
32651 -22.551651] (planned waypoint was [ 90.      -122.      -23.
093605])
453 [Astar_planner] Segment 3: [ 90.76908 -119.32651 -22.551651
] -> [ 82.      -133.      -21.815144]
454 [Step 20 | Total      0] Target: ( 82.0, -133.0, -21.8) | Pos: ( 84.7, -130
.2, -21.7) | Vel: ( 0.1, -1.2, -0.5) | Acc: ( 0.9, 0.6, 0.2) | Rwd: 1.44
455 [Astar_planner] Segment 3 reached in 20 steps, dist=3.86, axis_error
=[2.6798935 2.7760468 0.07432365], reason=running
456 [Astar_planner] Segment 4 starts from actual pos [ 84.68006 -130.
22751 -21.742363] (planned waypoint was [ 82.      -133.      -21.
815144])
457 [Astar_planner] Segment 4: [ 84.68006 -130.22751 -21.742363
] -> [ 81.      -147.      -21.896803]
458 [Step 26 | Total      0] Target: ( 81.0, -147.0, -21.9) | Pos: ( 82.9, -144
.2, -21.9) | Vel: ( -0.2, -0.7, -0.4) | Acc: ( 1.2, 0.9, -0.5) | Rwd: 1.29

459 [Astar_planner] Segment 4 reached in 26 steps, dist=3.40, axis_error
=[1.8912125 2.8292542 0.02624512], reason=running
460 [Astar_planner] Segment 5 starts from actual pos [ 82.89056 -144.

```

```

460 1729 -21.871698] (planned waypoint was [ 81.    -147.    -21.
      896803])
461 [Astar_planner] Segment 5: [ 82.89056 -144.1729 -21.871698] -> [
      73.    -156.    -24.53044]
462 [Step 16 | Total    0] Target: ( 73.0, -156.0, -24.5) | Pos: ( 75.9, -153
      .8, -24.4) | Vel: ( -0.6, -1.0, -0.1) | Acc: ( -0.2,  0.6,  0.0) | Rwd:  1.48

463 [Astar_planner] Segment 5 reached in 16 steps, dist=3.69, axis_error
      =[2.9385605  2.2194977  0.16545296], reason=running
464 [Astar_planner] Segment 6 starts from actual pos [ 75.936646 -153.
      78358 -24.365177] (planned waypoint was [ 73.    -156.    -24.
      53044])
465 [Astar_planner] Segment 6: [ 75.936646 -153.78358 -24.365177
      ] -> [ 59.    -156.    -21.510908]
466 [Step 47 | Total    0] Target: ( 59.0, -156.0, -21.5) | Pos: ( 56.7, -157
      .5, -19.0) | Vel: (  0.3, -0.5, -1.9) | Acc: (  0.3, -1.9, -0.2) | Rwd:  1.47

467 [Astar_planner] Segment 6 reached in 47 steps, dist=3.74, axis_error
      =[2.340294  1.4658661  2.5227127], reason=running
468 [Astar_planner] Segment 7 starts from actual pos [ 56.661407 -157.
      46889 -18.999578] (planned waypoint was [ 59.    -156.    -21.
      510908])
469 [Astar_planner] Segment 7: [ 56.661407 -157.46889 -18.999578
      ] -> [ 45.    -156.    -20.893023]
470 [Step 13 | Total    0] Target: ( 45.0, -156.0, -20.9) | Pos: ( 47.2, -157
      .5, -18.7) | Vel: ( -2.1, -0.7,  0.2) | Acc: ( -0.8, -1.0,  0.3) | Rwd:  0.72

471 [Astar_planner] Segment 7 reached in 13 steps, dist=3.47, axis_error
      =[2.2284584  1.4912415  2.197258 ], reason=running
472 [Astar_planner] Segment 8 starts from actual pos [ 47.215878 -157.
      49573 -18.694853] (planned waypoint was [ 45.    -156.    -20.
      893023])
473 [Astar_planner] Segment 8: [ 47.215878 -157.49573 -18.694853
      ] -> [ 32.    -159.    -21.90152]
474 [Step 18 | Total    0] Target: ( 32.0, -159.0, -21.9) | Pos: ( 29.8, -161
      .2, -19.1) | Vel: ( -0.4,  0.4,  0.0) | Acc: ( -0.3, -1.6, -1.9) | Rwd:  0.93

475 [Astar_planner] Segment 8 reached in 18 steps, dist=4.22, axis_error
      =[2.212286  2.223755  2.8284016], reason=running
476 [Astar_planner] Segment 9 starts from actual pos [ 29.785595 -161.
      22131 -19.073017] (planned waypoint was [ 32.    -159.    -21.
      90152])
477 [Astar_planner] Segment 9: [ 29.785595 -161.22131 -19.073017
      ] -> [ 19.    -160.    -21.741156]
478 [Step 29 | Total    0] Target: ( 19.0, -160.0, -21.7) | Pos: ( 21.6, -161
      .4, -22.2) | Vel: ( -0.5,  0.2,  0.0) | Acc: (  0.9, -0.2, -0.0) | Rwd:  1.54

479 [Astar_planner] Segment 9 reached in 29 steps, dist=3.01, axis_error

```

```

479 =[2.5971298 1.4282379 0.5033684], reason=running
480 [Astar_planner] Segment 10 starts from actual pos [ 21.59713 -161.
    42824 -22.244524] (planned waypoint was [ 19.    -160.    -21.
    741156])
481 [Astar_planner] Segment 10: [ 21.59713 -161.42824 -22.244524
    ] -> [ 5.    -161.    -21.739557]
482 [Step 37 | Total    0] Target: ( 5.0, -161.0, -21.7) | Pos: ( 7.6, -162.
    5, -20.6) | Vel: ( -1.6, -0.2, -0.0) | Acc: ( 1.4, 0.3, -0.7) | Rwd: 2.91
483 [Astar_planner] Segment 10 reached in 37 steps, dist=3.24, axis_error
    =[2.6293902 1.500946 1.1410866], reason=running
484 [Astar_planner] Segment 11 starts from actual pos [ 7.6247244 -162.
    50154 -20.598566 ] (planned waypoint was [ 5.    -161.    -21.
    739557])
485 [Astar_planner] Segment 11: [ 7.6247244 -162.50154 -20.598566
    ] -> [ -9.    -161.    -23.62334]
486 [Step 93 | Total    0] Target: ( -9.0, -161.0, -23.6) | Pos: ( 1.6, -162.
    9, -23.3) | Vel: ( 0.7, -0.1, 0.1) | Acc: ( 1.5, 0.7, -0.7) | Rwd: -911.
    29 -> Episode End: CRASHED!
487
488 [Astar_planner] Segment 11 ended: collision
489 [Astar_planner] Stop after failed segment 11.
490 Task 7 result: failed (Failed at segment 11: collision (11/17 reached).
    Shield interventions=132, emergencies=8, recoveries=23,
    recovery_steps=132.)
491
492 --- Task 8/38 ---
493 From [-73.2 -85.22 -5.6 ] to [ 57.9 19.78 -11.3 ]
494 [Astar_planner] Top-down occupancy: 350x390, x=[-179.0, 170.0], y=[-
    200.0, 189.0], occupied=72869 cells (53.4%)
495 [Astar_planner] Occupancy base_grid=390x350, occupied=72869 cells
    (53.4%)
496 [Astar_planner] Occupancy A*: grid=390x350, occupied=53.4%,
    safety_margin=1.5
497 Path planned via occupancy (Points: 20, Dist: 233.7m). Local target z/
    alt: 0:-5.61(5.61m), 1:-24.16(24.16m), 2:-24.42(24.42m), 3:-20.43(20.
    43m), 4:-24.99(24.99m), 5:-24.13(24.13m), 6:-21.43(21.43m), 7:-23.35
    (23.35m), 8:-24.64(24.64m), 9:-24.79(24.79m), 10:-21.67(21.67m), 11
    :-21.94(21.94m), 12:-21.70(21.70m), 13:-22.89(22.89m), 14:-22.11(22.
    11m), 15:-20.33(20.33m), 16:-22.39(22.39m), 17:-24.02(24.02m), 18:-
    24.54(24.54m), 19:-11.30(11.30m)
498 [Astar_planner] Segment 0: [-73.    -85.    -5.61009] -> [-62.    -87
    .    -24.164154]
499 [Astar_planner] Placing first segment start at [-73.    -85.    -5.61009
    ]
500 [Astar_planner] Initial climb to 20.0m.
501 [Step 18 | Total    0] Target: (-62.0, -87.0, -24.2) | Pos: (-64.2, -88.5
    , -22.4) | Vel: ( 0.5, -0.5, -1.5) | Acc: ( 0.4, -1.0, 0.6) | Rwd: 5.19
502 [Astar_planner] Segment 0 reached in 18 steps, dist=3.20, axis_error

```

```

502 =[2.2121658 1.4805679 1.7698956], reason=running
503 [Astar_planner] Segment 1 starts from actual pos [-64.2091 -88.4837
    -22.40306] (planned waypoint was [-62.    -87.    -24.164154])
504 [Astar_planner] Segment 1: [-64.2091 -88.4837 -22.40306] -> [-50
    -87.    -24.422922]
505 [Step  50 | Total    0] Target: (-50.0, -87.0, -24.4) | Pos: (-52.1, -86.5
    , -22.3) | Vel: ( -0.2,  0.2, -0.8) | Acc: ( -1.9,  1.2,  0.7) | Rwd:  4.24
506 [Astar_planner] Segment 1 reached in 50 steps, dist=3.03, axis_error
    =[2.1018257 0.52472687 2.1165047 ], reason=running
507 [Astar_planner] Segment 2 starts from actual pos [-52.102825 -86.
    47404 -22.311062] (planned waypoint was [-50.    -87.    -24.
    422922])
508 [Astar_planner] Segment 2: [-52.102825 -86.47404 -22.311062] -> [-
    39.    -88.    -20.433897]
509 [Step  16 | Total    0] Target: (-39.0, -88.0, -20.4) | Pos: (-41.9, -88.6
    , -19.0) | Vel: (  0.6, -0.5,  0.1) | Acc: ( -1.2, -2.0, -0.9) | Rwd:  1.23
510 [Astar_planner] Segment 2 reached in 16 steps, dist=3.26, axis_error
    =[2.8595238 0.59008026 1.4517746 ], reason=running
511 [Astar_planner] Segment 3 starts from actual pos [-41.85774 -88.
    59152 -18.981909] (planned waypoint was [-39.    -88.    -20.
    433897])
512 [Astar_planner] Segment 3: [-41.85774 -88.59152 -18.981909] -> [-27
    -88.    -24.991615]
513 [Step  41 | Total    0] Target: (-27.0, -88.0, -25.0) | Pos: (-29.7, -89.3
    , -22.1) | Vel: (  0.4, -0.1,  0.2) | Acc: (  0.8, -0.7, -1.5) | Rwd:  0.08
514 [Astar_planner] Segment 3 reached in 41 steps, dist=4.19, axis_error
    =[2.7391453 1.3386078 2.8774052], reason=running
515 [Astar_planner] Segment 4 starts from actual pos [-29.737934 -89.
    338806 -22.113735] (planned waypoint was [-27.    -88.    -24.
    991615])
516 [Astar_planner] Segment 4: [-29.737934 -89.338806 -22.113735] -> [-
    16.    -88.    -24.130348]
517 [Step  44 | Total    0] Target: (-16.0, -88.0, -24.1) | Pos: (-18.7, -88.0
    , -22.2) | Vel: (  0.6, -0.1,  0.1) | Acc: (  0.7, -0.5, -0.7) | Rwd:  0.83
518 [Astar_planner] Segment 4 reached in 44 steps, dist=3.33, axis_error
    =[2.7199364e+00 1.5640259e-03 1.9279251e+00], reason=running
519 [Astar_planner] Segment 5 starts from actual pos [-18.718199 -88.
    00187 -22.202133] (planned waypoint was [-16.    -88.    -24.
    130348])
520 [Astar_planner] Segment 5: [-18.718199 -88.00187 -22.202133] -> [-
    11.    -79.    -21.431599]
521 [Step  25 | Total    0] Target: (-11.0, -79.0, -21.4) | Pos: (-13.7, -80.0
    , -18.8) | Vel: (  0.5,  0.8, -0.1) | Acc: (  1.0,  1.9,  1.3) | Rwd: -0.85
522 [Astar_planner] Segment 5 reached in 25 steps, dist=3.90, axis_error
    =[2.7129993 1.0327682 2.608923 ], reason=running
523 [Astar_planner] Segment 6 starts from actual pos [-13.709894 -80.
    027725 -18.823198] (planned waypoint was [-11.    -79.    -21.
    431599])

```



```

524 [Astar_planner] Segment 6: [-13.709894 -80.027725 -18.823198] -> [-
    10.    -67.    -23.352242]
525 [Step  18 | Total    0] Target: (-10.0, -67.0, -23.4) | Pos: ( -9.3, -69.0
    , -21.7) | Vel: ( -0.4,  0.3, -1.0) | Acc: ( 0.0,  0.3,  1.9) | Rwd:  7.63
526 [Astar_planner] Segment 6 reached in 18 steps, dist=2.64, axis_error
    =[0.74644375 1.955101  1.6129818 ], reason=running
527 [Astar_planner] Segment 7 starts from actual pos [ -9.254905 -68.
    954094 -21.742352] (planned waypoint was [-10.    -67.    -23.
    352242])
528 [Astar_planner] Segment 7: [ -9.254905 -68.954094 -21.742352] -> [-
    10.    -56.    -24.643232]
529 [Step  24 | Total    0] Target: (-10.0, -56.0, -24.6) | Pos: ( -8.6, -56.8
    , -22.7) | Vel: ( -0.4,  0.4, -1.5) | Acc: ( -1.9,  1.9,  1.1) | Rwd:  7.55
530 [Astar_planner] Segment 7 reached in 24 steps, dist=2.55, axis_error
    =[1.4146652 0.8246498 1.9599113], reason=running
531 [Astar_planner] Segment 8 starts from actual pos [ -8.586547 -56.
    82355 -22.687792] (planned waypoint was [-10.    -56.    -24.
    643232])
532 [Astar_planner] Segment 8: [ -8.586547 -56.82355 -22.687792] -> [ -9
    .    -45.    -24.788965]
533 [Step  12 | Total    0] Target: ( -9.0, -45.0, -24.8) | Pos: ( -9.7, -46.1, -
    22.7) | Vel: ( -0.2,  0.2, -1.5) | Acc: (  1.0, -2.0,  0.8) | Rwd:  6.92
534 [Astar_planner] Segment 8 reached in 12 steps, dist=2.46, axis_error
    =[0.74001694 1.1009979 2.0720406 ], reason=running
535 [Astar_planner] Segment 9 starts from actual pos [ -9.740553 -46.
    100407 -22.721455] (planned waypoint was [ -9.    -45.    -24.
    788965])
536 [Astar_planner] Segment 9: [ -9.740553 -46.100407 -22.721455] -> [ -9
    .    -33.    -21.667706]
537 [Step  12 | Total    0] Target: ( -9.0, -33.0, -21.7) | Pos: (-10.6, -33.3
    , -19.2) | Vel: ( -0.3,  1.0, -1.0) | Acc: ( -0.3, -0.9,  1.8) | Rwd:  5.88
538 [Astar_planner] Segment 9 reached in 12 steps, dist=2.96, axis_error
    =[1.6260786 0.30373383 2.452221 ], reason=running
539 [Astar_planner] Segment 10 starts from actual pos [-10.627119 -33.
    300777 -19.21844 ] (planned waypoint was [ -9.    -33.    -21.
    667706])
540 [Astar_planner] Segment 10: [-10.627119 -33.300777 -19.21844 ] -> [ -
    9.    -21.    -21.941229]
541 [Step  18 | Total    0] Target: ( -9.0, -21.0, -21.9) | Pos: ( -8.7, -22.8, -
    19.8) | Vel: ( -0.6,  0.7, -0.9) | Acc: ( -2.0,  1.7,  0.3) | Rwd:  6.30
542 [Astar_planner] Segment 10 reached in 18 steps, dist=2.85, axis_error
    =[0.26215744 1.8024864 2.1868763 ], reason=running
543 [Astar_planner] Segment 11 starts from actual pos [ -8.741538 -22.
    798458 -19.75993 ] (planned waypoint was [ -9.    -21.    -21.
    941229])
544 [Astar_planner] Segment 11: [ -8.741538 -22.798458 -19.75993 ] -> [ -
    9.    -9.    -21.699133]
545 [Step  17 | Total    0] Target: ( -9.0, -9.0, -21.7) | Pos: ( -9.4, -10.3, -

```

```

545 20.1) | Vel: ( -0.5, 0.2, -0.8) | Acc: ( -0.5, 0.5, 0.7) | Rwd: 606.55
    -> Episode End: ARRIVED!
546
547 [Astar_planner] Segment 11 reached in 17 steps, dist=2.04, axis_error
    =[0.39976406 1.2705421 1.5508022 ], reason=arrived
548 [Astar_planner] Segment 12 starts from actual pos [ -9.401405 -10.
    269858 -20.15079 ] (planned waypoint was [ -9.    -9.    -21.
    699133])
549 [Astar_planner] Segment 12: [ -9.401405 -10.269858 -20.15079 ] -> [ -
    4.    0.    -22.89254]
550 [Step 24 | Total 0] Target: ( -4.0, 0.0, -22.9) | Pos: ( -5.7, -1.7, -
    20.8) | Vel: ( -0.0, 0.2, -1.1) | Acc: ( -1.3, 0.5, 1.2) | Rwd: 4.44
551 [Astar_planner] Segment 12 reached in 24 steps, dist=3.19, axis_error
    =[1.654057 1.7053877 2.1222496], reason=running
552 [Astar_planner] Segment 13 starts from actual pos [ -5.654057 -1.
    7053877 -20.77029 ] (planned waypoint was [ -4.    0.    -22.89254
    ])
553 [Astar_planner] Segment 13: [ -5.654057 -1.7053877 -20.77029
    ] -> [ 7.    1.    -22.11203]
554 [Step 16 | Total 0] Target: ( 7.0, 1.0, -22.1) | Pos: ( 4.1, 0.2, -
    20.1) | Vel: ( 0.5, 0.1, 0.1) | Acc: ( 0.0, -0.0, -0.4) | Rwd: 0.66
555 [Astar_planner] Segment 13 reached in 16 steps, dist=3.66, axis_error
    =[2.9398012 0.7648096 2.0411797], reason=running
556 [Astar_planner] Segment 14 starts from actual pos [ 4.063349 0.
    2358848 -20.070494 ] (planned waypoint was [ 7.    1.    -22.11203
    ])
557 [Astar_planner] Segment 14: [ 4.063349 0.2358848 -20.070494
    ] -> [ 19.    2.    -20.327833]
558 [Step 16 | Total 0] Target: ( 19.0, 2.0, -20.3) | Pos: ( 16.6, 2.1, -
    19.9) | Vel: ( 1.0, -0.2, 0.0) | Acc: ( -0.9, -0.4, 0.7) | Rwd: 1.78
559 [Astar_planner] Segment 14 reached in 16 steps, dist=2.45, axis_error
    =[2.4093533 0.11053133 0.42947388], reason=running
560 [Astar_planner] Segment 15 starts from actual pos [ 16.59646 2.
    1095307 -19.89837 ] (planned waypoint was [ 19.    2.    -20.
    327833])
561 [Astar_planner] Segment 15: [ 16.59646 2.1095307 -19.89837
    ] -> [ 29.    7.    -22.390709]
562 [Step 16 | Total 0] Target: ( 29.0, 7.0, -22.4) | Pos: ( 26.7, 7.0, -
    20.6) | Vel: ( 1.3, -0.9, 0.0) | Acc: ( 0.3, -1.6, 0.3) | Rwd: 1.38
563 [Astar_planner] Segment 15 reached in 16 steps, dist=2.87, axis_error
    =[2.269394 0.04597521 1.7585907 ], reason=running
564 [Astar_planner] Segment 16 starts from actual pos [ 26.734531 6.
    9513335 -20.63212 ] (planned waypoint was [ 29.    7.    -22.
    390709])
565 [Astar_planner] Segment 16: [ 26.734531 6.9513335 -20.63212
    ] -> [ 38.    11.    -24.020601]
566 [Step 129 | Total 0] Target: ( 38.0, 11.0, -24.0) | Pos: ( 38.1, 8.1
    , -22.2) | Vel: ( -0.0, 0.1, -1.2) | Acc: ( -0.1, 0.3, 1.9) | Rwd: 4.05

```

```

567 [Astar_planner] Segment 16 reached in 129 steps, dist=3.41,
    axis_error=[0.09513092 2.8599024 1.8613853 ], reason=running
568 [Astar_planner] Segment 17 starts from actual pos [ 38.095024 8.
    140256 -22.162882] (planned waypoint was [ 38.    11.    -24.
    020601])
569 [Astar_planner] Segment 17: [ 38.095024 8.140256 -22.162882] -> [
    48.    15.    -24.539593]
570 [Step 15 | Total    0] Target: ( 48.0, 15.0, -24.5) | Pos: ( 45.2, 13.0
    , -21.6) | Vel: ( 1.2, -0.1, 0.1) | Acc: ( 0.4, -0.7, 0.1) | Rwd: 0.02
571 [Astar_planner] Segment 17 reached in 15 steps, dist=4.50, axis_error
    =[2.7926674 2.026661 2.8942509], reason=running
572 [Astar_planner] Segment 18 starts from actual pos [ 45.214622 12.
    972616 -21.644823] (planned waypoint was [ 48.    15.    -24.
    539593])
573 [Astar_planner] Segment 18: [ 45.214622 12.972616 -21.644823] -> [
    58. 20. -11.3]
574 [Step 19 | Total    0] Target: ( 58.0, 20.0, -11.3) | Pos: ( 58.2, 18.6
    , -10.3) | Vel: ( 0.2, 0.5, 0.4) | Acc: (-2.0, -1.9, 1.4) | Rwd: 599.34
    -> Episode End: ARRIVED!
575
576 [Astar_planner] Segment 18 reached in 19 steps, dist=1.70, axis_error
    =[0.19961166 1.3657818 0.98796654], reason=arrived
577 Task 8 result: success (Success: Reached destination (19/19
    segments). Shield interventions=42, emergencies=0, recoveries=7,
    recovery_steps=42.)
578
579 --- Task 9/38 ---
580 From [ 23.5 140.08 -4. ] to [ 17.8 -86.92 -8.9 ]
581 [Astar_planner] Top-down occupancy: 350x390, x=[-179.0, 170.0], y=[-
    200.0, 189.0], occupied=72869 cells (53.4%)
582 [Astar_planner] Occupancy base_grid=390x350, occupied=72869 cells
    (53.4%)
583 [Astar_planner] Occupancy A*: grid=390x350, occupied=53.4%,
    safety_margin=1.5
584 Path planned via occupancy (Points: 30, Dist: 376.5m). Local target z/
    alt: 0:-4.01(4.01m), 1:-23.40(23.40m), 2:-23.44(23.44m), 3:-22.04(22.
    04m), 4:-24.44(24.44m), 5:-22.68(22.68m), 6:-21.02(21.02m), 7:-23.22
    (23.22m), 8:-24.96(24.96m), 9:-20.67(20.67m), 10:-22.46(22.46m), 11
    :-22.81(22.81m), 12:-23.09(23.09m), 13:-23.61(23.61m), 14:-20.81(20.
    81m), 15:-24.30(24.30m), 16:-21.40(21.40m), 17:-22.94(22.94m), 18:-
    24.79(24.79m), 19:-24.93(24.93m), 20:-20.18(20.18m), 21:-20.26(20.
    26m), 22:-21.01(21.01m), 23:-23.85(23.85m), 24:-23.38(23.38m), 25:-
    20.02(20.02m), 26:-24.55(24.55m), 27:-24.18(24.18m), 28:-24.68(24.
    68m), 29:-8.90(8.90m)
585 [Astar_planner] Segment 0: [ 24.    140.    -4.0109267] -> [ 16.
    149.    -23.3997]
586 [Astar_planner] Placing first segment start at [ 24.    140.    -4.
    0109267]

```

```

587 [Astar_planner] Initial climb to 20.0m.
588 [Step 30 | Total 0] Target: ( 16.0, 149.0, -23.4) | Pos: ( 18.3, 154.9
, -22.3) | Vel: ( 0.0, 0.0, 0.0) | Acc: ( 0.8, 0.1, -1.7) | Rwd: -903.50
-> Episode End: CRASHED!
589
590 [Astar_planner] Segment 0 ended: collision
591 [Astar_planner] Stop after failed segment 0.
592 Task 9 result: failed (Failed at segment 0: collision (0/29 reached).
Shield interventions=7, emergencies=0, recoveries=2, recovery_steps
=7.)
593
594 --- Task 10/38 ---
595 From [ 17.8 -86.92 -8.9 ] to [-10.5 163.38 -8.94]
596 [Astar_planner] Top-down occupancy: 350x390, x=[-179.0, 170.0], y=[-
200.0, 189.0], occupied=72869 cells (53.4%)
597 [Astar_planner] Occupancy base_grid=390x350, occupied=72869 cells
(53.4%)
598 [Astar_planner] Occupancy A*: grid=390x350, occupied=53.4%,
safety_margin=1.5
599 Path planned via occupancy (Points: 26, Dist: 333.0m). Local target z/
alt: 0:-8.90(8.90m), 1:-24.70(24.70m), 2:-20.31(20.31m), 3:-20.55(20.
55m), 4:-23.26(23.26m), 5:-21.98(21.98m), 6:-22.64(22.64m), 7:-22.55
(22.55m), 8:-22.22(22.22m), 9:-24.98(24.98m), 10:-23.19(23.19m), 11
:-24.99(24.99m), 12:-23.71(23.71m), 13:-24.63(24.63m), 14:-22.30(22.
30m), 15:-24.06(24.06m), 16:-23.99(23.99m), 17:-24.02(24.02m), 18:-
22.48(22.48m), 19:-23.19(23.19m), 20:-24.84(24.84m), 21:-22.71(22.
71m), 22:-23.55(23.55m), 23:-24.25(24.25m), 24:-24.54(24.54m), 25:-
8.94(8.94m)
600 [Astar_planner] Segment 0: [ 18. -87. -8.900034] -> [ 5. -
88. -24.703037]
601 [Astar_planner] Placing first segment start at [ 18. -87. -8.
900034]
602 [Astar_planner] Initial climb to 20.0m.
603 [Step 20 | Total 0] Target: ( 5.0, -88.0, -24.7) | Pos: ( 5.0, -90.6, -
22.3) | Vel: ( -0.8, 0.5, -0.6) | Acc: ( -0.3, 1.7, -1.0) | Rwd: 3.11
604 [Astar_planner] Segment 0 reached in 20 steps, dist=3.50, axis_error
=[0.02667809 2.5578537 2.3881626 ], reason=running
605 [Astar_planner] Segment 1 starts from actual pos [ 5.0241766 -90.
55645 -22.316593 ] (planned waypoint was [ 5. -88. -24.
703037])
606 [Astar_planner] Segment 1: [ 5.0241766 -90.55645 -22.316593
]-> [ -8. -88. -20.307644]
607 [Step 33 | Total 0] Target: ( -8.0, -88.0, -20.3) | Pos: ( -5.1, -88.0, -
19.3) | Vel: ( -0.6, 0.2, -0.0) | Acc: ( -1.7, 1.9, 0.0) | Rwd: 2.18
608 [Astar_planner] Segment 1 reached in 33 steps, dist=3.09, axis_error
=[2.9184866 0.01997375 1.011509 ], reason=running
609 [Astar_planner] Segment 2 starts from actual pos [ -5.0831947 -87.
97933 -19.296186 ] (planned waypoint was [ -8. -88. -20.

```

```

609 307644])
610 [Astar_planner] Segment 2: [ -5.0831947 -87.97933 -19.296186 ] -> [ -
    11.    -76.    -20.551737]
611 [Step 39 | Total    0] Target: (-11.0, -76.0, -20.6) | Pos: (-11.8, -75.9
    , -18.5) | Vel: ( 0.6,  0.8,  0.5) | Acc: ( 0.8,  2.0, -1.1) | Rwd:  5.49
612 [Astar_planner] Segment 2 reached in 39 steps, dist=2.19, axis_error
    =[0.78900814 0.12226868 2.0357285 ], reason=running
613 [Astar_planner] Segment 3 starts from actual pos [-11.787313 -75.
    87543 -18.514515] (planned waypoint was [-11.    -76.    -20.
    551737])
614 [Astar_planner] Segment 3: [-11.787313 -75.87543 -18.514515] -> [ -
    10.    -63.    -23.263937]
615 [Step 49 | Total    0] Target: (-10.0, -63.0, -23.3) | Pos: ( -7.6, -62.3
    , -20.3) | Vel: ( 0.1,  0.2, -0.1) | Acc: ( -0.3, -1.6, -2.0) | Rwd:  4.87
616 [Astar_planner] Segment 3 reached in 49 steps, dist=3.90, axis_error
    =[2.420218 0.6651726 2.9884815], reason=running
617 [Astar_planner] Segment 4 starts from actual pos [ -7.5794096 -62.
    334217 -20.275797 ] (planned waypoint was [-10.    -63.    -23.
    263937])
618 [Astar_planner] Segment 4: [ -7.5794096 -62.334217 -20.275797
    ] -> [ -9.    -51.    -21.977524]
619 [Step 37 | Total    0] Target: ( -9.0, -51.0, -22.0) | Pos: ( -7.6, -53.8, -
    19.7) | Vel: ( -0.0,  0.3, -0.0) | Acc: ( 0.5, -0.4,  1.4) | Rwd: -1.18
620 [Astar_planner] Segment 4 reached in 37 steps, dist=3.84, axis_error
    =[1.3878112 2.7713203 2.261465 ], reason=running
621 [Astar_planner] Segment 5 starts from actual pos [ -7.6122856 -53.
    770527 -19.716179 ] (planned waypoint was [ -9.    -51.    -21.
    977524])
622 [Astar_planner] Segment 5: [ -7.6122856 -53.770527 -19.716179
    ] -> [ -9.    -38.    -22.639086]
623 [Step 45 | Total    0] Target: ( -9.0, -38.0, -22.6) | Pos: ( -9.2, -40.5, -
    19.7) | Vel: ( -0.1,  0.2,  0.6) | Acc: ( -1.3, -0.8, -1.6) | Rwd:  2.56
624 [Astar_planner] Segment 5 reached in 45 steps, dist=3.90, axis_error
    =[0.23034954 2.498787  2.9811935 ], reason=running
625 [Astar_planner] Segment 6 starts from actual pos [ -9.23035 -40.
    498787 -19.657892] (planned waypoint was [ -9.    -38.    -22.
    639086])
626 [Astar_planner] Segment 6: [ -9.23035 -40.498787 -19.657892] -> [ -9
    .    -24.    -22.546743]
627 [Step 54 | Total    0] Target: ( -9.0, -24.0, -22.5) | Pos: (-10.5, -23.1
    , -20.0) | Vel: ( -0.5,  0.3,  0.4) | Acc: ( -1.7,  1.3, -0.5) | Rwd:  3.62
628 [Astar_planner] Segment 6 reached in 54 steps, dist=3.10, axis_error
    =[1.5090437 0.9189453 2.5474224], reason=running
629 [Astar_planner] Segment 7 starts from actual pos [-10.510675 -23.
    080046 -19.998232] (planned waypoint was [ -9.    -24.    -22.
    546743])
630 [Astar_planner] Segment 7: [-10.510675 -23.080046 -19.998232] -> [ -
    9.    -11.    -22.222342]

```

```

631 [Step 39 | Total 0] Target: (-9.0, -11.0, -22.2) | Pos: (-9.7, -11.3, -
    19.3) | Vel: ( 0.3, 0.5, 0.6) | Acc: ( 0.9, 1.3, -0.2) | Rwd: 0.71
632 [Astar_planner] Segment 7 reached in 39 steps, dist=3.05, axis_error
    =[0.70246124 0.3048067 2.9515076 ], reason=running
633 [Astar_planner] Segment 8 starts from actual pos [-9.701644 -11.
    3032465 -19.26916 ] (planned waypoint was [-9. -11. -22.
    222342])
634 [Astar_planner] Segment 8: [-9.701644 -11.3032465 -19.26916 ] -> [-
    14. -1. -24.97672]
635 [Step 19 | Total 0] Target: (-14.0, -1.0, -25.0) | Pos: (-14.4, -1.0, -
    22.5) | Vel: (-0.1, 0.9, -0.9) | Acc: (-0.2, 2.0, 2.0) | Rwd: 7.30
636 [Astar_planner] Segment 8 reached in 19 steps, dist=2.54, axis_error
    =[0.43853092 0.01861322 2.497364 ], reason=running
637 [Astar_planner] Segment 9 starts from actual pos [-14.438684 -1.
    0159675 -22.48189 ] (planned waypoint was [-14. -1. -24.
    97672])
638 [Astar_planner] Segment 9: [-14.438684 -1.0159675 -22.48189 ] -> [-
    26. 3. -23.193481]
639 [Step 34 | Total 0] Target: (-26.0, 3.0, -23.2) | Pos: (-23.4, 2.2, -
    20.7) | Vel: (-0.2, -0.4, 0.6) | Acc: (-0.5, -1.0, -1.0) | Rwd: 1.51
640 [Astar_planner] Segment 9 reached in 34 steps, dist=3.66, axis_error
    =[2.5611954 0.77465177 2.4919453 ], reason=running
641 [Astar_planner] Segment 10 starts from actual pos [-23.439379 2.
    22423 -20.699606] (planned waypoint was [-26. 3. -23.
    193481])
642 [Astar_planner] Segment 10: [-23.439379 2.22423 -20.699606] -> [-
    35. 12. -24.988083]
643 [Step 59 | Total 0] Target: (-35.0, 12.0, -25.0) | Pos: (-31.5, 12.0
    , -20.2) | Vel: (-0.3, 0.9, -0.1) | Acc: (-0.6, 2.0, 1.0) | Rwd: -2.18
644 [Astar_planner] Segment 10 reached in 59 steps, dist=5.90, axis_error
    =[3.5124512 0.02129745 4.7440376 ], reason=
    stuck_near_intermediate
645 [Astar_planner] Segment 11 starts from actual pos [-31.488337 12.
    023886 -20.244263] (planned waypoint was [-35. 12. -24.
    988083])
646 [Astar_planner] Segment 11: [-31.488337 12.023886 -20.244263] -> [-
    44. 22. -23.705223]
647 [Step 19 | Total 0] Target: (-44.0, 22.0, -23.7) | Pos: (-42.7, 21.7
    , -21.8) | Vel: (-0.5, 0.1, -1.2) | Acc: (-0.4, 0.3, 0.0) | Rwd: 606.14
    -> Episode End: ARRIVED!
648
649 [Astar_planner] Segment 11 reached in 19 steps, dist=2.35, axis_error
    =[1.340435 0.31235123 1.9092159 ], reason=arrived
650 [Astar_planner] Segment 12 starts from actual pos [-42.662376 21.
    688509 -21.803366] (planned waypoint was [-44. 22. -23.
    705223])
651 [Astar_planner] Segment 12: [-42.662376 21.688509 -21.803366] -> [-
    50. 32. -24.630688]

```

```

652 [Step 19 | Total 0] Target: (-50.0, 32.0, -24.6) | Pos: (-50.7, 30.4
, -22.7) | Vel: ( -0.1, 0.3, -1.0) | Acc: ( 0.5, -0.9, 1.2) | Rwd: 7.11
653 [Astar_planner] Segment 12 reached in 19 steps, dist=2.60, axis_error
=[0.66487885 1.5887451 1.9538231 ], reason=running
654 [Astar_planner] Segment 13 starts from actual pos [-50.66512 30.
412264 -22.67986 ] (planned waypoint was [-50. 32. -24.
630688])
655 [Astar_planner] Segment 13: [-50.66512 30.412264 -22.67986 ] -> [-
55. 43. -22.296463]
656 [Step 13 | Total 0] Target: (-55.0, 43.0, -22.3) | Pos: (-53.1, 40.2
, -19.8) | Vel: ( -0.5, 0.6, -0.0) | Acc: ( 0.1, -1.0, 0.7) | Rwd: 0.25
657 [Astar_planner] Segment 13 reached in 13 steps, dist=4.19, axis_error
=[1.8561325 2.8038368 2.4945164], reason=running
658 [Astar_planner] Segment 14 starts from actual pos [-53.145466 40.
197884 -19.802063] (planned waypoint was [-55. 43. -22.
296463])
659 [Astar_planner] Segment 14: [-53.145466 40.197884 -19.802063] -> [-
59. 54. -24.063917]
660 [Step 21 | Total 0] Target: (-59.0, 54.0, -24.1) | Pos: (-57.6, 53.4
, -21.5) | Vel: ( -0.4, -0.1, -1.1) | Acc: ( -0.6, -1.3, 1.5) | Rwd: 6.15
661 [Astar_planner] Segment 14 reached in 21 steps, dist=3.02, axis_error
=[1.4037437 0.6310806 2.5957909], reason=running
662 [Astar_planner] Segment 15 starts from actual pos [-57.598686 53.
368572 -21.47481 ] (planned waypoint was [-59. 54. -24.
063917])
663 [Astar_planner] Segment 15: [-57.598686 53.368572 -21.47481 ] -> [-
50. 64. -23.989468]
664 [Step 12 | Total 0] Target: (-50.0, 64.0, -24.0) | Pos: (-52.7, 62.6
, -22.0) | Vel: ( 0.7, 0.6, 0.1) | Acc: ( 0.5, 0.0, -0.7) | Rwd: 1.83
665 [Astar_planner] Segment 15 reached in 12 steps, dist=3.59, axis_error
=[2.6518593 1.4020195 1.971447 ], reason=running
666 [Astar_planner] Segment 16 starts from actual pos [-52.647743 62.
601833 -22.017693] (planned waypoint was [-50. 64. -23.
989468])
667 [Astar_planner] Segment 16: [-52.647743 62.601833 -22.017693] -> [-
41. 71. -24.022715]
668 [Step 87 | Total 0] Target: (-41.0, 71.0, -24.0) | Pos: (-43.7, 71.5
, -22.3) | Vel: ( 1.4, 0.2, -0.1) | Acc: ( 0.9, 0.4, 0.4) | Rwd: 1.87
669 [Astar_planner] Segment 16 reached in 87 steps, dist=3.27, axis_error
=[2.7091217 0.5226593 1.7579765], reason=running
670 [Astar_planner] Segment 17 starts from actual pos [-43.705063 71.
523224 -22.265041] (planned waypoint was [-41. 71. -24.
022715])
671 [Astar_planner] Segment 17: [-43.705063 71.523224 -22.265041] -> [-
40. 84. -22.482668]
672 [Step 37 | Total 0] Target: (-40.0, 84.0, -22.5) | Pos: (-38.2, 81.3
, -19.3) | Vel: ( 0.1, 0.0, 0.1) | Acc: ( 0.1, 0.0, -0.3) | Rwd: -1.48
673 [Astar_planner] Segment 17 reached in 37 steps, dist=4.51, axis_error

```

```

673 =[1.7852135 2.6676025 3.169098 ], reason=stuck_near_intermediate
674 [Astar_planner] Segment 18 starts from actual pos [-38.214516 81.
33248 -19.313334] (planned waypoint was [-40.    84.    -22.
482668])
675 [Astar_planner] Segment 18: [-38.214516 81.33248 -19.313334] -> [-
31.    93.    -23.18574]
676 [Step 24 | Total    0] Target: (-31.0, 93.0, -23.2) | Pos: (-32.9, 91.3
, -21.2) | Vel: ( 0.2, -0.1, -1.8) | Acc: ( -0.2, -0.7, 1.6) | Rwd: 5.29
677 [Astar_planner] Segment 18 reached in 24 steps, dist=3.20, axis_error
=[1.914051 1.6746292 1.9483643], reason=running
678 [Astar_planner] Segment 19 starts from actual pos [-32.913334 91.
32504 -21.242628] (planned waypoint was [-31.    93.    -23.18574
])
679 [Astar_planner] Segment 19: [-32.913334 91.32504 -21.242628] -> [-
28.    105.    -24.838854]
680 [Step 23 | Total    0] Target: (-28.0, 105.0, -24.8) | Pos: (-29.4, 103.
4, -22.0) | Vel: ( -0.3, 0.4, -2.9) | Acc: ( -0.3, -1.0, 1.4) | Rwd: 6.12
681 [Astar_planner] Segment 19 reached in 23 steps, dist=3.58, axis_error
=[1.3927116 1.6346436 2.8609009], reason=running
682 [Astar_planner] Segment 20 starts from actual pos [-29.392712 103.
36536 -21.977953] (planned waypoint was [-28.    105.    -24.
838854])
683 [Astar_planner] Segment 20: [-29.392712 103.36536 -21.977953] -> [-
28.    118.    -22.707104]
684 [Step 67 | Total    0] Target: (-28.0, 118.0, -22.7) | Pos: (-26.5, 118.
2, -20.6) | Vel: ( -0.8, 0.1, -2.7) | Acc: ( 1.7, -0.7, 0.2) | Rwd: 4.68
685 [Astar_planner] Segment 20 reached in 67 steps, dist=2.66, axis_error
=[1.5425224 0.16625214 2.1564865 ], reason=running
686 [Astar_planner] Segment 21 starts from actual pos [-26.459991 118.
166405 -20.558588] (planned waypoint was [-28.    118.    -22.
707104])
687 [Astar_planner] Segment 21: [-26.459991 118.166405 -20.558588
] -> [-25.    130.    -23.551796]
688 [Step 14 | Total    0] Target: (-25.0, 130.0, -23.6) | Pos: (-24.5, 130.
1, -21.3) | Vel: ( -0.1, 2.6, -1.7) | Acc: ( -0.3, 0.5, 2.0) | Rwd: 5.96
689 [Astar_planner] Segment 21 reached in 14 steps, dist=2.32, axis_error
=[0.4994774 0.07147217 2.2600117 ], reason=running
690 [Astar_planner] Segment 22 starts from actual pos [-24.500961 130.
07913 -21.296814] (planned waypoint was [-25.    130.    -23.
551796])
691 [Astar_planner] Segment 22: [-24.500961 130.07913 -21.296814] -> [-
25.    143.    -24.252651]
692 [Step 66 | Total    0] Target: (-25.0, 143.0, -24.3) | Pos: (-24.5, 140.
6, -22.1) | Vel: ( -0.7, 0.1, -2.1) | Acc: ( -0.5, 1.5, 2.0) | Rwd: 3.91
693 [Astar_planner] Segment 22 reached in 66 steps, dist=3.27, axis_error
=[0.5193386 2.4395294 2.1124325], reason=running
694 [Astar_planner] Segment 23 starts from actual pos [-24.482677 140.
56065 -22.146505] (planned waypoint was [-25.    143.    -24.

```



```

694 252651])
695 [Astar_planner] Segment 23: [-24.482677 140.56065 -22.146505] -> [-
20.    154.    -24.540615]
696 [Step 59 | Total    0] Target: (-20.0, 154.0, -24.5) | Pos: (-19.2, 151.
3, -21.7) | Vel: ( -0.3, -0.3, -2.7) | Acc: ( -0.5, -1.1, 1.6) | Rwd: 6.04
697 [Astar_planner] Segment 23 reached in 59 steps, dist=4.04, axis_error
=[0.8060074 2.706421 2.886362 ], reason=running
698 [Astar_planner] Segment 24 starts from actual pos [-19.194857 151.
2926 -21.66228 ] (planned waypoint was [-20.    154.    -24.
540615])
699 [Astar_planner] Segment 24: [-19.194857 151.2926 -21.66228 ] -> [-
10.    163.    -8.939931]
700 [Step 16 | Total    0] Target: (-10.0, 163.0, -8.9) | Pos: (-11.2, 164.0
, -8.8) | Vel: ( 1.2, 0.9, 0.1) | Acc: ( -1.7, -0.7, 1.3) | Rwd: 0.57
701 [Astar_planner] Segment 24 reached in 16 steps, dist=1.56, axis_error
=[1.2167339 0.9773102 0.10683537], reason=running
702 Task 10 result: success (Success: Reached destination (25/25
segments). Shield interventions=168, emergencies=3, recoveries=28,
recovery_steps=168.)
703
704 --- Task 11/38 ---
705 From [-123.9 -90.92 -4. ] to [-117.3 1.48 -9. ]
706 [Astar_planner] Top-down occupancy: 350x390, x=[-179.0, 170.0], y=[-
200.0, 189.0], occupied=72869 cells (53.4%)
707 [Astar_planner] Occupancy base_grid=390x350, occupied=72869 cells
(53.4%)
708 [Astar_planner] Occupancy A*: grid=390x350, occupied=53.4%,
safety_margin=1.5
709 Path planned via occupancy (Points: 9, Dist: 113.1m). Local target z/alt
: 0:-4.01(4.01m), 1:-21.99(21.99m), 2:-24.33(24.33m), 3:-24.55(24.
55m), 4:-20.91(20.91m), 5:-24.31(24.31m), 6:-22.33(22.33m), 7:-21.44
(21.44m), 8:-8.98(8.98m)
710 [Astar_planner] Segment 0: [-124.    -91.    -4.006912] -> [-122
.    -79.    -21.98968]
711 [Astar_planner] Placing first segment start at [-124.    -91.    -4.
006912]
712 [Astar_planner] Initial climb to 20.0m.
713 [Step 43 | Total    0] Target: (-122.0, -79.0, -22.0) | Pos: (-121.2, -82
.5, -18.9) | Vel: ( 0.0, 0.0, 0.0) | Acc: ( 0.0, 0.0, -0.4) | Rwd: -1.31
714 [Astar_planner] Segment 0 reached in 43 steps, dist=4.69, axis_error
=[0.7835922 3.4617004 3.061079 ], reason=stuck_near_intermediate
715 [Astar_planner] Segment 1 starts from actual pos [-121.21636 -82.
46168 -18.928488] (planned waypoint was [-122.    -79.    -21.
98968])
716 [Astar_planner] Segment 1: [-121.21636 -82.46168 -18.928488] -> [-
119.    -68.    -24.329298]
717 [Step 42 | Total    0] Target: (-119.0, -68.0, -24.3) | Pos: (-116.3, -70
.4, -21.3) | Vel: ( 0.1, 0.1, 0.0) | Acc: ( 0.5, 1.3, -0.6) | Rwd: -0.94

```

```

718 [Astar_planner] Segment 1 reached in 42 steps, dist=4.68, axis_error
    =[2.6623306 2.4332733 2.9802608], reason=running
719 [Astar_planner] Segment 2 starts from actual pos [-116.33739 -70.
    43296 -21.348986] (planned waypoint was [-119. -68. -24.
    329298])
720 [Astar_planner] Segment 2: [-116.33739 -70.43296 -21.348986] -> [-
    111. -59. -24.554476]
721 [Step 40 | Total 0] Target: (-111.0, -59.0, -24.6) | Pos: (-113.8, -58
    .8, -21.8) | Vel: ( 0.0, 0.5, -1.2) | Acc: ( -0.6, 1.3, 0.5) | Rwd: 6.66
722 [Astar_planner] Segment 2 reached in 40 steps, dist=3.91, axis_error
    =[2.788391 0.18517303 2.7284107 ], reason=running
723 [Astar_planner] Segment 3 starts from actual pos [-113.78825 -58.
    813396 -21.829693] (planned waypoint was [-111. -59. -24.
    554476])
724 [Astar_planner] Segment 3: [-113.78825 -58.813396 -21.829693
    ] -> [-110. -46. -20.906473]
725 [Step 76 | Total 0] Target: (-110.0, -46.0, -20.9) | Pos: (-114.8, -54
    .9, -20.5) | Vel: ( -0.0, 0.5, -0.1) | Acc: ( -0.5, 2.0, 1.9) | Rwd: -909.
    81 -> Episode End: CRASHED!
726
727 [Astar_planner] Segment 3 ended: collision
728 [Astar_planner] Stop after failed segment 3.
729 Task 11 result: failed (Failed at segment 3: collision (3/8 reached).
    Shield interventions=66, emergencies=7, recoveries=11,
    recovery_steps=66.)
730
731 --- Task 12/38 ---
732 From [-141.4 -191.82 -9. ] to [-125.1 185.28 -4. ]
733 [Astar_planner] Top-down occupancy: 350x390, x=[-179.0, 170.0], y=[-
    200.0, 189.0], occupied=72869 cells (53.4%)
734 [Astar_planner] Occupancy base_grid=390x350, occupied=72869 cells
    (53.4%)
735 [Astar_planner] Occupancy A*: grid=390x350, occupied=53.4%,
    safety_margin=1.5
736 Path planned via occupancy (Points: 33, Dist: 445.5m). Local target z/
    alt: 0:-8.99(8.99m), 1:-22.29(22.29m), 2:-24.87(24.87m), 3:-22.53(22.
    53m), 4:-24.47(24.47m), 5:-22.83(22.83m), 6:-24.49(24.49m), 7:-21.57
    (21.57m), 8:-20.06(20.06m), 9:-22.96(22.96m), 10:-24.25(24.25m), 11
    :-21.08(21.08m), 12:-24.38(24.38m), 13:-21.84(21.84m), 14:-23.74(23.
    74m), 15:-20.73(20.73m), 16:-22.71(22.71m), 17:-23.25(23.25m), 18:-
    20.11(20.11m), 19:-23.51(23.51m), 20:-24.41(24.41m), 21:-24.95(24.
    95m), 22:-20.79(20.79m), 23:-20.73(20.73m), 24:-21.99(21.99m), 25:-
    21.22(21.22m), 26:-23.61(23.61m), 27:-23.98(23.98m), 28:-24.94(24.
    94m), 29:-24.25(24.25m), 30:-24.32(24.32m), 31:-21.30(21.30m), 32:-
    4.00(4.00m)
737 [Astar_planner] Segment 0: [-141. -192. -8.994189] -> [-152
    . -186. -22.293535]
738 [Astar_planner] Placing first segment start at [-141. -192. -8.

```

```

738 994189]
739 [Astar_planner] Initial climb to 20.0m.
740 [Step 94 | Total 0] Target: (-152.0, -186.0, -22.3) | Pos: (-146.9, -
    189.2, -19.8) | Vel: ( 0.0, -0.1, 0.0) | Acc: ( 0.3, -0.8, -0.0) | Rwd: -
    0.86
741 [Astar_planner] Segment 0 reached in 94 steps, dist=6.48, axis_error
    =[5.059433 3.1803284 2.4959927], reason=stuck_near_intermediate
742 [Astar_planner] Segment 1 starts from actual pos [-146.94049 -189.
    18051 -19.797535] (planned waypoint was [-152. -186. -22.
    293535])
743 [Astar_planner] Segment 1: [-146.94049 -189.18051 -19.797535
    ] -> [-166. -186. -24.86938]
744 [Step 81 | Total 0] Target: (-166.0, -186.0, -24.9) | Pos: (-159.5, -
    190.1, -21.7) | Vel: ( 0.1, 0.1, 0.0) | Acc: ( 0.3, 0.8, -0.5) | Rwd: -0
    .52
745 [Astar_planner] Segment 1 reached in 81 steps, dist=8.33, axis_error
    =[6.5056458 4.096695 3.2128716], reason=stuck_near_intermediate
746 [Astar_planner] Segment 2 starts from actual pos [-159.49416 -190.
    09645 -21.656479] (planned waypoint was [-166. -186. -24.
    86938])
747 [Astar_planner] Segment 2: [-159.49416 -190.09645 -21.656479
    ] -> [-173. -176. -22.530539]
748 [Step 180 | Total 0] Target: (-173.0, -176.0, -22.5) | Pos: (-166.6, -
    202.7, -3.4) | Vel: ( 0.2, 0.1, -0.1) | Acc: ( 1.5, 2.0, 2.0) | Rwd: -2.
    42
749 [Astar_planner] Segment 2 timeout, dist=33.45, budget=180
750 [Astar_planner] Stop after failed segment 2.
751 Task 12 result: failed (Failed at segment 2: segment_timeout (2/32
    reached). Shield interventions=0, emergencies=0, recoveries=0,
    recovery_steps=0.)
752
753 --- Task 13/38 ---
754 From [103.4 79.58 -9. ] to [-111.5 118.98 -4. ]
755 [Astar_planner] Top-down occupancy: 350x390, x=[-179.0, 170.0], y=[-
    200.0, 189.0], occupied=72869 cells (53.4%)
756 [Astar_planner] Occupancy base_grid=390x350, occupied=72869 cells
    (53.4%)
757 [Astar_planner] Occupancy A*: grid=390x350, occupied=53.4%,
    safety_margin=1.5
758 Path planned via occupancy (Points: 23, Dist: 317.0m). Local target z/
    alt: 0:-8.99(8.99m), 1:-23.04(23.04m), 2:-22.15(22.15m), 3:-24.33(24.
    33m), 4:-21.00(21.00m), 5:-22.34(22.34m), 6:-23.49(23.49m), 7:-21.90
    (21.90m), 8:-24.96(24.96m), 9:-20.81(20.81m), 10:-24.21(24.21m), 11
    :-24.15(24.15m), 12:-23.68(23.68m), 13:-24.05(24.05m), 14:-21.54(21.
    54m), 15:-20.12(20.12m), 16:-22.50(22.50m), 17:-23.71(23.71m), 18:-
    22.78(22.78m), 19:-24.76(24.76m), 20:-23.89(23.89m), 21:-24.99(24.
    99m), 22:-4.00(4.00m)
759 [Astar_planner] Segment 0: [103. 80. -8.986727] -> [105.

```

```

759 93.    -23.041117]
760 [Astar_planner] Placing first segment start at [103.    80.    -8.
    986727]
761 [Astar_planner] Initial climb to 20.0m.
762 [Step 20 | Total    0] Target: (105.0, 93.0, -23.0) | Pos: (104.3, 93.7
    , -21.1) | Vel: ( 0.3, 2.3, -1.4) | Acc: ( -1.4, 1.7, 2.0) | Rwd: 604.19
    -> Episode End: ARRIVED!
763
764 [Astar_planner] Segment 0 reached in 20 steps, dist=2.19, axis_error
    =[0.67533875 0.7353897 1.9525623 ], reason=arrived
765 [Astar_planner] Segment 1 starts from actual pos [104.325516 93.
    742424 -21.092672] (planned waypoint was [105.    93.    -23.
    041117])
766 [Astar_planner] Segment 1: [104.325516 93.742424 -21.092672] -> [
    105.    107.    -22.151894]
767 [Step 11 | Total    0] Target: (105.0, 107.0, -22.2) | Pos: (102.6, 108.
    6, -19.3) | Vel: ( 0.1, 2.1, -2.6) | Acc: ( -0.1, -0.8, 2.0) | Rwd: 5.36
768 [Astar_planner] Segment 1 reached in 11 steps, dist=4.09, axis_error
    =[2.3865128 1.6272736 2.8936634], reason=running
769 [Astar_planner] Segment 2 starts from actual pos [102.61349 108.
    62727 -19.25823] (planned waypoint was [105.    107.    -22.
    151894])
770 [Astar_planner] Segment 2: [102.61349 108.62727 -19.25823] -> [105
    .    121.    -24.326357]
771 [Step 10 | Total    0] Target: (105.0, 121.0, -24.3) | Pos: (106.8, 118.
    1, -21.9) | Vel: ( 0.6, 0.2, -1.1) | Acc: ( -0.0, -1.9, -0.7) | Rwd: -1.79
772 [Astar_planner] Segment 2 reached in 10 steps, dist=4.19, axis_error
    =[1.7643433 2.889618 2.470562 ], reason=running
773 [Astar_planner] Segment 3 starts from actual pos [106.7662 118.
    111115 -21.85898 ] (planned waypoint was [105.    121.    -24.
    326357])
774 [Astar_planner] Segment 3: [106.7662 118.111115 -21.85898 ] -> [
    101.    133.    -21.000456]
775 [Step 41 | Total    0] Target: (101.0, 133.0, -21.0) | Pos: ( 98.5, 135.
    4, -18.9) | Vel: ( 1.1, -0.9, -1.0) | Acc: ( 0.1, 0.5, -1.6) | Rwd: 1.87
776 [Astar_planner] Segment 3 reached in 41 steps, dist=4.01, axis_error
    =[2.451912 2.3906555 2.0832157], reason=running
777 [Astar_planner] Segment 4 starts from actual pos [ 98.551285 135.
    3879 -18.920221] (planned waypoint was [101.    133.    -21.
    000456])
778 [Astar_planner] Segment 4: [ 98.551285 135.3879 -18.920221] -> [
    88.    135.    -22.344114]
779 [Step 23 | Total    0] Target: ( 88.0, 135.0, -22.3) | Pos: ( 90.9, 133.5
    , -19.7) | Vel: ( -0.3, -0.0, 0.0) | Acc: ( 0.7, -1.1, 0.2) | Rwd: -0.17
780 [Astar_planner] Segment 4 reached in 23 steps, dist=4.23, axis_error
    =[2.9125824 1.466217 2.6928577], reason=running
781 [Astar_planner] Segment 5 starts from actual pos [ 90.91175 133.
    53378 -19.651228] (planned waypoint was [ 88.    135.    -22.

```

```

781 344114])
782 [Astar_planner] Segment 5: [ 90.91175 133.53378 -19.651228] -> [
74.    135.    -23.485046]
783 [Step 27 | Total    0] Target: ( 74.0, 135.0, -23.5) | Pos: ( 75.0, 133.7
, -20.5) | Vel: ( 0.1, 0.9, -3.0) | Acc: ( 1.8, 1.9, 0.4) | Rwd: 5.34
784 [Astar_planner] Segment 5 reached in 27 steps, dist=3.39, axis_error
=[0.98791504 1.3257446 2.963913 ], reason=running
785 [Astar_planner] Segment 6 starts from actual pos [ 74.98835 133.
67703 -20.529984] (planned waypoint was [ 74.    135.    -23.
485046])
786 [Astar_planner] Segment 6: [ 74.98835 133.67703 -20.529984] -> [
60.    136.    -21.89877]
787 [Step 22 | Total    0] Target: ( 60.0, 136.0, -21.9) | Pos: ( 62.5, 136.8
, -20.0) | Vel: ( -1.2, 0.4, 0.3) | Acc: ( 0.6, -1.7, 1.3) | Rwd: -0.11
788 [Astar_planner] Segment 6 reached in 22 steps, dist=3.28, axis_error
=[2.5434494 0.83883667 1.8954563 ], reason=running
789 [Astar_planner] Segment 7 starts from actual pos [ 62.54345 136.
83884 -20.003313] (planned waypoint was [ 60.    136.    -21.89877
])
790 [Astar_planner] Segment 7: [ 62.54345 136.83884 -20.003313] -> [
46.    136.    -24.955326]
791 [Step 43 | Total    0] Target: ( 46.0, 136.0, -25.0) | Pos: ( 48.2, 134.8
, -22.3) | Vel: ( -0.3, 0.1, 0.2) | Acc: ( -0.8, 0.7, -0.4) | Rwd: 3.24
792 [Astar_planner] Segment 7 reached in 43 steps, dist=3.63, axis_error
=[2.2307167 1.1601868 2.6160202], reason=running
793 [Astar_planner] Segment 8 starts from actual pos [ 48.230717 134.
83981 -22.339306] (planned waypoint was [ 46.    136.    -24.
955326])
794 [Astar_planner] Segment 8: [ 48.230717 134.83981 -22.339306] -> [
32.    136.    -20.808414]
795 [Step 21 | Total    0] Target: ( 32.0, 136.0, -20.8) | Pos: ( 32.9, 133.5
, -19.9) | Vel: ( -1.2, 1.5, -0.2) | Acc: ( 0.3, 0.0, 0.0) | Rwd: 1.83
796 [Astar_planner] Segment 8 reached in 21 steps, dist=2.84, axis_error
=[0.9395561 2.5061646 0.95550156], reason=running
797 [Astar_planner] Segment 9 starts from actual pos [ 32.935936 133.
49821 -19.853624] (planned waypoint was [ 32.    136.    -20.
808414])
798 [Astar_planner] Segment 9: [ 32.935936 133.49821 -19.853624] -> [
22.    144.    -24.212358]
799 [Step 21 | Total    0] Target: ( 22.0, 144.0, -24.2) | Pos: ( 24.6, 143.0
, -21.5) | Vel: ( -0.8, 0.6, 0.1) | Acc: ( 0.4, -0.7, 0.1) | Rwd: -0.30
800 [Astar_planner] Segment 9 reached in 21 steps, dist=3.90, axis_error
=[2.560833 1.0455933 2.7480927], reason=running
801 [Astar_planner] Segment 10 starts from actual pos [ 24.55838 142.
95615 -21.463917] (planned waypoint was [ 22.    144.    -24.
212358])
802 [Astar_planner] Segment 10: [ 24.55838 142.95615 -21.463917] -> [
10.    150.    -24.146053]

```

```

803 [Step 17 | Total 0] Target: ( 10.0, 150.0, -24.1) | Pos: ( 11.7, 149.1
, -21.9) | Vel: ( -0.5, 0.5, -2.1) | Acc: ( 1.7, 1.0, 2.0) | Rwd: 5.08
804 [Astar_planner] Segment 10 reached in 17 steps, dist=2.90, axis_error
=[1.6522703 0.91275024 2.2036552 ], reason=running
805 [Astar_planner] Segment 11 starts from actual pos [ 11.650857 149.
0888 -21.948738] (planned waypoint was [ 10. 150. -24.
146053])
806 [Astar_planner] Segment 11: [ 11.650857 149.0888 -21.948738] -> [ -
3. 153. -23.675154]
807 [Step 28 | Total 0] Target: ( -3.0, 153.0, -23.7) | Pos: ( -1.0, 151.6
, -21.9) | Vel: ( -0.4, -0.3, -0.9) | Acc: ( -0.8, -0.0, 2.0) | Rwd: 6.22
808 [Astar_planner] Segment 11 reached in 28 steps, dist=3.07, axis_error
=[2.0360935 1.4473572 1.7864723], reason=running
809 [Astar_planner] Segment 12 starts from actual pos [ -0.96507204 151.
55188 -21.891438 ] (planned waypoint was [ -3. 153. -23.
675154])
810 [Astar_planner] Segment 12: [ -0.96507204 151.55188 -21.891438
] -> [ -16. 153. -24.05106]
811 [Step 16 | Total 0] Target: ( -16.0, 153.0, -24.1) | Pos: ( -13.6, 150.
7, -21.6) | Vel: ( -1.2, 0.1, 0.1) | Acc: ( 0.3, -0.8, -0.0) | Rwd: 0.38
812 [Astar_planner] Segment 12 reached in 16 steps, dist=4.15, axis_error
=[2.416418 2.27417 2.4886818], reason=running
813 [Astar_planner] Segment 13 starts from actual pos [ -13.587155 150.
72624 -21.562214] (planned waypoint was [ -16. 153. -24.05106
])
814 [Astar_planner] Segment 13: [ -13.587155 150.72624 -21.562214] -> [ -
30. 153. -21.544397]
815 [Step 50 | Total 0] Target: ( -30.0, 153.0, -21.5) | Pos: ( -28.9, 151.
1, -19.2) | Vel: ( 0.2, -0.3, 0.4) | Acc: ( 0.1, -0.3, -1.7) | Rwd: 2.83
816 [Astar_planner] Segment 13 reached in 50 steps, dist=3.21, axis_error
=[1.0943031 1.9446564 2.305132 ], reason=running
817 [Astar_planner] Segment 14 starts from actual pos [ -28.905003 151.
05444 -19.23814 ] (planned waypoint was [ -30. 153. -21.
544397])
818 [Astar_planner] Segment 14: [ -28.905003 151.05444 -19.23814 ] -> [ -
43. 157. -20.115852]
819 [Step 40 | Total 0] Target: ( -43.0, 157.0, -20.1) | Pos: ( -41.0, 156.
7, -17.6) | Vel: ( -2.2, 0.1, 0.8) | Acc: ( -1.3, 1.3, 0.0) | Rwd: -1.29
820 [Astar_planner] Segment 14 reached in 40 steps, dist=3.20, axis_error
=[1.9542007 0.30836487 2.509449 ], reason=running
821 [Astar_planner] Segment 15 starts from actual pos [ -41.0458 156.
69164 -17.606403] (planned waypoint was [ -43. 157. -20.
115852])
822 [Astar_planner] Segment 15: [ -41.0458 156.69164 -17.606403] -> [ -
57. 157. -22.496267]
823 [Step 11 | Total 0] Target: ( -57.0, 157.0, -22.5) | Pos: ( -55.2, 157.
0, -19.7) | Vel: ( -2.5, 0.9, -0.0) | Acc: ( 0.4, 0.8, -0.0) | Rwd: 0.52
824 [Astar_planner] Segment 15 reached in 11 steps, dist=3.33, axis_error

```

```

824 =[1.8095398 0.00816345 2.793604 ], reason=running
825 [Astar_planner] Segment 16 starts from actual pos [-55.198017 157.
01086 -19.702671] (planned waypoint was [-57.    157.    -22.
496267])
826 [Astar_planner] Segment 16: [-55.198017 157.01086 -19.702671] -> [-
70.    156.    -23.706083]
827 [Step 25 | Total    0] Target: (-70.0, 156.0, -23.7) | Pos: (-72.0, 158.
1, -21.5) | Vel: ( -3.0, 0.3, -1.8) | Acc: ( 1.1, 0.2, 1.0) | Rwd: 0.92
828 [Astar_planner] Segment 16 reached in 25 steps, dist=3.66, axis_error
=[2.04319 2.0509796 2.2414665], reason=running
829 [Astar_planner] Segment 17 starts from actual pos [-72.05213 158.
05183 -21.46993] (planned waypoint was [-70.    156.    -23.
706083])
830 [Astar_planner] Segment 17: [-72.05213 158.05183 -21.46993] -> [-84
.    156.    -22.77693]
831 [Step 8 | Total    0] Target: (-84.0, 156.0, -22.8) | Pos: (-82.0, 156.3
, -21.3) | Vel: ( -2.0, 0.3, -0.2) | Acc: ( 1.1, 1.7, 0.8) | Rwd: 2.41
832 [Astar_planner] Segment 17 reached in 8 steps, dist=2.52, axis_error
=[1.9907379 0.29418945 1.523037 ], reason=running
833 [Astar_planner] Segment 18 starts from actual pos [-82.01527 156.
29521 -21.254593] (planned waypoint was [-84.    156.    -22.77693
])
834 [Astar_planner] Segment 18: [-82.01527 156.29521 -21.254593] -> [-
97.    154.    -24.76475]
835 [Step 17 | Total    0] Target: (-97.0, 154.0, -24.8) | Pos: (-96.0, 154.
2, -22.6) | Vel: ( -0.5, -0.6, -2.4) | Acc: ( 1.9, -2.0, 2.0) | Rwd: 5.25
836 [Astar_planner] Segment 18 reached in 17 steps, dist=2.39, axis_error
=[0.9507675 0.19447327 2.1791687 ], reason=running
837 [Astar_planner] Segment 19 starts from actual pos [-96.05058 154.
19273 -22.592659] (planned waypoint was [-97.    154.    -24.76475
])
838 [Astar_planner] Segment 19: [-96.05058 154.19273 -22.592659] -> [-
100.    142.    -23.887629]
839 [Step 11 | Total    0] Target: (-100.0, 142.0, -23.9) | Pos: (-98.4, 144
.3, -22.0) | Vel: ( -0.1, -0.7, -0.1) | Acc: ( 0.3, -1.7, 1.2) | Rwd: 1.14

840 [Astar_planner] Segment 19 reached in 11 steps, dist=3.36, axis_error
=[1.553894 2.272766 1.9217625], reason=running
841 [Astar_planner] Segment 20 starts from actual pos [-98.44638 144.
27065 -21.966026] (planned waypoint was [-100.    142.    -23.
887629])
842 [Astar_planner] Segment 20: [-98.44638 144.27065 -21.966026] -> [-
104.    129.    -24.987192]
843 [Step 17 | Total    0] Target: (-104.0, 129.0, -25.0) | Pos: (-102.0,
127.7, -22.9) | Vel: ( -0.1, -0.9, 0.7) | Acc: ( -0.2, -1.8, -1.5) | Rwd:
3.38
844 [Astar_planner] Segment 20 reached in 17 steps, dist=3.20, axis_error
=[1.9976349 1.3030472 2.1299458], reason=running

```

```

845 [Astar_planner] Segment 21 starts from actual pos [-102.002365 127.69695 -22.857246] (planned waypoint was [-104. 129. -24.987192])
846 [Astar_planner] Segment 21: [-102.002365 127.69695 -22.857246] -> [-112. 119. -4.]
847 [Step 27 | Total 0] Target: (-112.0, 119.0, -4.0) | Pos: (-113.5, 119.9, -4.9) | Vel: ( 0.7, 0.7, -0.4) | Acc: ( 0.4, -1.3, -0.7) | Rwd: -1.31
848 [Astar_planner] Segment 21 reached in 27 steps, dist=1.96, axis_error=[1.4532318 0.901062 0.9487572], reason=running
849 Task 13 result: success (Success: Reached destination (22/22 segments). Shield interventions=64, emergencies=1, recoveries=11, recovery_steps=64.)
850
851 --- Task 14/38 ---
852 From [-111.5 118.98 -4. ] to [63.1 4.08 -7.7 ]
853 [Astar_planner] Top-down occupancy: 350x390, x=[-179.0, 170.0], y=[-200.0, 189.0], occupied=72869 cells (53.4%)
854 [Astar_planner] Occupancy base_grid=390x350, occupied=72869 cells (53.4%)
855 [Astar_planner] Occupancy A*: grid=390x350, occupied=53.4%, safety_margin=1.5
856 Path planned via occupancy (Points: 24, Dist: 295.4m). Local target z/alt: 0:-4.01(4.01m), 1:-22.23(22.23m), 2:-20.27(20.27m), 3:-21.88(21.88m), 4:-21.85(21.85m), 5:-22.45(22.45m), 6:-24.14(24.14m), 7:-21.09(21.09m), 8:-22.91(22.91m), 9:-20.82(20.82m), 10:-24.18(24.18m), 11:-20.17(20.17m), 12:-24.47(24.47m), 13:-21.74(21.74m), 14:-20.01(20.01m), 15:-20.29(20.29m), 16:-22.82(22.82m), 17:-22.16(22.16m), 18:-24.68(24.68m), 19:-24.51(24.51m), 20:-22.53(22.53m), 21:-24.62(24.62m), 22:-23.03(23.03m), 23:-7.70(7.70m)
857 [Astar_planner] Segment 0: [-112. 119. -4.008858] -> [-103. 110. -22.228642]
858 [Astar_planner] Placing first segment start at [-112. 119. -4.008858]
859 [Astar_planner] Initial climb to 20.0m.
860 [Step 16 | Total 0] Target: (-103.0, 110.0, -22.2) | Pos: (-105.9, 110.8, -22.2) | Vel: ( 0.5, -0.3, -0.0) | Acc: ( 1.2, -1.9, 0.1) | Rwd: 2.57
861 [Astar_planner] Segment 0 reached in 16 steps, dist=3.04, axis_error=[2.945343 0.7531891 0.01574707], reason=running
862 [Astar_planner] Segment 1 starts from actual pos [-105.9439 110.75238 -22.212934] (planned waypoint was [-103. 110. -22.228642])
863 [Astar_planner] Segment 1: [-105.9439 110.75238 -22.212934] -> [-100. 99. -20.26704]
864 [Step 14 | Total 0] Target: (-100.0, 99.0, -20.3) | Pos: (-102.7, 101.5, -18.7) | Vel: ( 0.3, -0.2, 0.0) | Acc: ( 0.9, -1.3, -0.5) | Rwd: 1.99

```



```

865 [Astar_planner] Segment 1 reached in 14 steps, dist=3.99, axis_error
    =[2.7371826 2.4558716 1.5440998], reason=running
866 [Astar_planner] Segment 2 starts from actual pos [-102.73617 101.
    455215 -18.722889] (planned waypoint was [-100. 99. -20.
    26704])
867 [Astar_planner] Segment 2: [-102.73617 101.455215 -18.722889
    ] -> [-93. 90. -21.884686]
868 [Step 30 | Total 0] Target: (-93.0, 90.0, -21.9) | Pos: (-95.7, 90.0
    , -20.3) | Vel: ( 0.3, -0.3, -0.0) | Acc: ( 0.1, -1.9, 0.2) | Rwd: 1.00
869 [Astar_planner] Segment 2 reached in 30 steps, dist=3.07, axis_error
    =[2.6516953 0.03371429 1.5374737 ], reason=running
870 [Astar_planner] Segment 3 starts from actual pos [-95.65067 90.
    03297 -20.347233] (planned waypoint was [-93. 90. -21.
    884686])
871 [Astar_planner] Segment 3: [-95.65067 90.03297 -20.347233] -> [-81
    . 89. -21.848425]
872 [Step 23 | Total 0] Target: (-81.0, 89.0, -21.8) | Pos: (-81.0, 89.4
    , -20.0) | Vel: ( 2.0, 0.4, -2.0) | Acc: ( 2.0, 0.6, 1.4) | Rwd: 606.72
    -> Episode End: ARRIVED!
873
874 [Astar_planner] Segment 3 reached in 23 steps, dist=1.91, axis_error
    =[0.03457642 0.3860016 1.8706989 ], reason=arrived
875 [Astar_planner] Segment 4 starts from actual pos [-81.02858 89.
    38733 -19.983631] (planned waypoint was [-81. 89. -21.
    848425])
876 [Astar_planner] Segment 4: [-81.02858 89.38733 -19.983631] -> [-68
    . 89. -22.452133]
877 [Step 15 | Total 0] Target: (-68.0, 89.0, -22.5) | Pos: (-65.0, 90.6
    , -20.3) | Vel: ( -0.6, -0.3, 0.1) | Acc: ( -1.3, -0.2, -1.3) | Rwd: 1.74
878 [Astar_planner] Segment 4 reached in 15 steps, dist=4.01, axis_error
    =[2.9636612 1.5793152 2.1952858], reason=running
879 [Astar_planner] Segment 5 starts from actual pos [-65.03634 90.
    579315 -20.256847] (planned waypoint was [-68. 89. -22.
    452133])
880 [Astar_planner] Segment 5: [-65.03634 90.579315 -20.256847] -> [-
    56. 89. -24.142435]
881 [Step 13 | Total 0] Target: (-56.0, 89.0, -24.1) | Pos: (-57.6, 88.7
    , -21.2) | Vel: ( 0.6, 0.1, -0.6) | Acc: ( 0.8, 1.6, 0.8) | Rwd: 4.49
882 [Astar_planner] Segment 5 reached in 13 steps, dist=3.36, axis_error
    =[1.5763664 0.28855133 2.9500237 ], reason=running
883 [Astar_planner] Segment 6 starts from actual pos [-57.57457 88.
    71162 -21.19424] (planned waypoint was [-56. 89. -24.142435
    ])
884 [Astar_planner] Segment 6: [-57.57457 88.71162 -21.19424] -> [-46
    . 82. -21.090273]
885 [Step 17 | Total 0] Target: (-46.0, 82.0, -21.1) | Pos: (-48.9, 82.5
    , -19.6) | Vel: ( 0.3, -0.2, -0.0) | Acc: ( 0.6, -0.4, 0.7) | Rwd: -0.08
886 [Astar_planner] Segment 6 reached in 17 steps, dist=3.31, axis_error

```

```

886 =[2.9070969 0.51065063 1.5038376 ], reason=running
887 [Astar_planner] Segment 7 starts from actual pos [-48.90605 82.
      51006 -19.586555] (planned waypoint was [-46.      82.      -21.
      090273])
888 [Astar_planner] Segment 7: [-48.90605 82.51006 -19.586555] -> [-41
      .      72.      -22.910088]
889 [Step 25 | Total 0] Target: (-41.0, 72.0, -22.9) | Pos: (-39.7, 74.5
      , -21.8) | Vel: ( 1.0, -0.5, -0.1) | Acc: ( 0.5, -0.7, -0.0) | Rwd: 0.30
890 [Astar_planner] Segment 7 reached in 25 steps, dist=3.07, axis_error
      =[1.3128433 2.5461502 1.1138802], reason=running
891 [Astar_planner] Segment 8 starts from actual pos [-39.684113 74.
      544556 -21.796467] (planned waypoint was [-41.      72.      -22.
      910088])
892 [Astar_planner] Segment 8: [-39.684113 74.544556 -21.796467] -> [-
      49.      65.      -20.817326]
893 [Step 11 | Total 0] Target: (-49.0, 65.0, -20.8) | Pos: (-47.5, 67.7
      , -19.6) | Vel: ( -1.0, -0.5, -0.1) | Acc: ( -2.0, -0.7, 0.9) | Rwd: 1.88
894 [Astar_planner] Segment 8 reached in 11 steps, dist=3.33, axis_error
      =[1.526474 2.7213593 1.1753998], reason=running
895 [Astar_planner] Segment 9 starts from actual pos [-47.473526 67.
      72136 -19.641926] (planned waypoint was [-49.      65.      -20.
      817326])
896 [Astar_planner] Segment 9: [-47.473526 67.72136 -19.641926] -> [-
      58.      56.      -24.175215]
897 [Step 34 | Total 0] Target: (-58.0, 56.0, -24.2) | Pos: (-55.3, 58.7
      , -21.8) | Vel: ( -0.5, -0.5, 0.0) | Acc: ( -0.9, -1.8, -0.5) | Rwd: 2.13
898 [Astar_planner] Segment 9 reached in 34 steps, dist=4.54, axis_error
      =[2.7491264 2.7114944 2.3913403], reason=running
899 [Astar_planner] Segment 10 starts from actual pos [-55.252388 58.
      709946 -21.783821] (planned waypoint was [-58.      56.      -24.
      175215])
900 [Astar_planner] Segment 10: [-55.252388 58.709946 -21.783821] -> [-
      57.      45.      -20.166447]
901 [Step 19 | Total 0] Target: (-57.0, 45.0, -20.2) | Pos: (-60.0, 46.0
      , -19.9) | Vel: ( 0.3, -0.1, 0.0) | Acc: ( 1.2, -1.1, -0.8) | Rwd: 2.54
902 [Astar_planner] Segment 10 reached in 19 steps, dist=3.15, axis_error
      =[2.971939 1.0234451 0.25816917], reason=running
903 [Astar_planner] Segment 11 starts from actual pos [-59.97194 46.
      023445 -19.908278] (planned waypoint was [-57.      45.      -20.
      166447])
904 [Astar_planner] Segment 11: [-59.97194 46.023445 -19.908278] -> [-
      50.      36.      -24.471016]
905 [Step 24 | Total 0] Target: (-50.0, 36.0, -24.5) | Pos: (-52.9, 34.5
      , -22.2) | Vel: ( 0.3, -0.3, 0.1) | Acc: ( 0.4, -1.8, -1.4) | Rwd: 0.81
906 [Astar_planner] Segment 11 reached in 24 steps, dist=4.00, axis_error
      =[2.9408798 1.4775352 2.2745705], reason=running
907 [Astar_planner] Segment 12 starts from actual pos [-52.93987 34.
      521706 -22.19629 ] (planned waypoint was [-50.      36.      -24.

```

```

907 471016])
908 [Astar_planner] Segment 12: [-52.93987 34.521706 -22.19629 ] -> [-
47. 25. -21.736248]
909 [Step 13 | Total 0] Target: (-47.0, 25.0, -21.7) | Pos: (-49.6, 24.9
, -20.8) | Vel: ( 0.4, -0.2, -0.0) | Acc: ( 0.9, -1.9, 0.3) | Rwd: 1.73
910 [Astar_planner] Segment 12 reached in 13 steps, dist=2.80, axis_error
=[2.628437 0.05825043 0.97405815], reason=running
911 [Astar_planner] Segment 13 starts from actual pos [-49.62717 24.
941101 -20.762241] (planned waypoint was [-47. 25. -21.
736248])
912 [Astar_planner] Segment 13: [-49.62717 24.941101 -20.762241] -> [-
38. 16. -20.00702]
913 [Step 23 | Total 0] Target: (-38.0, 16.0, -20.0) | Pos: (-40.6, 13.3
, -17.8) | Vel: ( 0.5, -0.1, -0.0) | Acc: ( 1.5, -1.5, 0.1) | Rwd: 0.26
914 [Astar_planner] Segment 13 reached in 23 steps, dist=4.33, axis_error
=[2.5655365 2.7202883 2.1800728], reason=running
915 [Astar_planner] Segment 14 starts from actual pos [-40.564125 13.
27938 -17.827011] (planned waypoint was [-38. 16. -20.00702
])
916 [Astar_planner] Segment 14: [-40.564125 13.27938 -17.827011] -> [-
30. 7. -20.291876]
917 [Step 22 | Total 0] Target: (-30.0, 7.0, -20.3) | Pos: (-32.7, 6.8, -
18.2) | Vel: ( 0.5, 0.1, -0.0) | Acc: ( 0.3, 0.7, 0.1) | Rwd: 0.74
918 [Astar_planner] Segment 14 reached in 22 steps, dist=3.40, axis_error
=[2.6953773 0.20577812 2.0675735 ], reason=running
919 [Astar_planner] Segment 15 starts from actual pos [-32.693813 6.
7944603 -18.224352 ] (planned waypoint was [-30. 7. -20.
291876])
920 [Astar_planner] Segment 15: [-32.693813 6.7944603 -18.224352
] -> [-20. 0. -22.818281]
921 [Step 136 | Total 0] Target: (-20.0, 0.0, -22.8) | Pos: (-22.4, -0.2
, -20.8) | Vel: ( 0.2, 0.5, -0.2) | Acc: ( -1.9, 2.0, 1.3) | Rwd: 4.33
922 [Astar_planner] Segment 15 reached in 136 steps, dist=3.09,
axis_error=[2.3687286 0.17397258 1.9731236 ], reason=running
923 [Astar_planner] Segment 16 starts from actual pos [-22.36826 -0.
17259674 -20.845827 ] (planned waypoint was [-20. 0. -22.
818281])
924 [Astar_planner] Segment 16: [-22.36826 -0.17259674 -20.845827
] -> [-9. -2. -22.161]
925 [Step 44 | Total 0] Target: ( -9.0, -2.0, -22.2) | Pos: (-11.8, -3.2, -
19.3) | Vel: ( 0.5, -0.3, -0.0) | Acc: ( -0.1, -0.4, 0.7) | Rwd: -0.96
926 [Astar_planner] Segment 16 reached in 44 steps, dist=4.20, axis_error
=[2.7932587 1.1933737 2.899374 ], reason=running
927 [Astar_planner] Segment 17 starts from actual pos [-11.791902 -3.
194205 -19.261724] (planned waypoint was [ -9. -2. -22.161])
928 [Astar_planner] Segment 17: [-11.791902 -3.194205 -19.261724] -> [
2. 1. -24.683254]
929 [Step 32 | Total 0] Target: ( 2.0, 1.0, -24.7) | Pos: ( -0.8, -1.1, -

```

```

929 21.8) | Vel: ( 0.3, 0.3, -0.0) | Acc: ( -0.9, 0.6, 0.5) | Rwd: -0.79
930 [Astar_planner] Segment 17 reached in 32 steps, dist=4.53, axis_error
    =[2.7926798 2.1434736 2.858492 ], reason=running
931 [Astar_planner] Segment 18 starts from actual pos [ -0.79267985 -1.
    1434737 -21.824762 ] (planned waypoint was [ 2.      1.      -24.
    683254])
932 [Astar_planner] Segment 18: [ -0.79267985 -1.1434737 -21.824762
    ] -> [ 15.      1.      -24.514248]
933 [Step 30 | Total 0] Target: ( 15.0, 1.0, -24.5) | Pos: ( 12.0, 0.5, -
    22.0) | Vel: ( 0.4, 0.0, 0.2) | Acc: ( -0.5, -0.7, -1.2) | Rwd: 1.07
934 [Astar_planner] Segment 18 reached in 30 steps, dist=3.90, axis_error
    =[2.9627695 0.48669964 2.4876728 ], reason=running
935 [Astar_planner] Segment 19 starts from actual pos [ 12.039626 0.
    5135659 -22.025167 ] (planned waypoint was [ 15.      1.      -24.
    514248])
936 [Astar_planner] Segment 19: [ 12.039626 0.5135659 -22.025167
    ] -> [ 27.      1.      -22.532476]
937 [Step 15 | Total 0] Target: ( 27.0, 1.0, -22.5) | Pos: ( 24.2, -0.0, -
    21.5) | Vel: ( 0.7, 0.5, 0.0) | Acc: ( -0.3, -0.1, -0.0) | Rwd: 1.31
938 [Astar_planner] Segment 19 reached in 15 steps, dist=3.13, axis_error
    =[2.7691307 1.0457325 1.0150795], reason=running
939 [Astar_planner] Segment 20 starts from actual pos [ 24.23286 -0.
    04412692 -21.51731 ] (planned waypoint was [ 27.      1.      -22.
    532476])
940 [Astar_planner] Segment 20: [ 24.23286 -0.04412692 -21.51731
    ] -> [ 39.      1.      -24.623606]
941 [Step 27 | Total 0] Target: ( 39.0, 1.0, -24.6) | Pos: ( 36.3, -0.1, -
    22.5) | Vel: ( 0.8, -0.8, 0.0) | Acc: ( -1.1, -1.4, 0.4) | Rwd: -0.08
942 [Astar_planner] Segment 20 reached in 27 steps, dist=3.59, axis_error
    =[2.670601 1.112436 2.118744], reason=running
943 [Astar_planner] Segment 21 starts from actual pos [ 36.33186 -0.
    11489634 -22.504816 ] (planned waypoint was [ 39.      1.      -24.
    623606])
944 [Astar_planner] Segment 21: [ 36.33186 -0.11489634 -22.504816
    ] -> [ 51.      2.      -23.032238]
945 [Step 12 | Total 0] Target: ( 51.0, 2.0, -23.0) | Pos: ( 48.4, 1.5, -
    22.3) | Vel: ( 1.1, 0.9, -0.0) | Acc: ( -1.8, 1.6, 0.9) | Rwd: 2.15
946 [Astar_planner] Segment 21 reached in 12 steps, dist=2.75, axis_error
    =[2.6050682 0.4623549 0.75079155], reason=running
947 [Astar_planner] Segment 22 starts from actual pos [ 48.39832 1.
    5402793 -22.281572 ] (planned waypoint was [ 51.      2.      -23.
    032238])
948 [Astar_planner] Segment 22: [ 48.39832 1.5402793 -22.281572
    ] -> [63.      4.      -7.6993003]
949 [Step 28 | Total 0] Target: ( 63.0, 4.0, -7.7) | Pos: ( 63.8, 2.5, -
    7.7) | Vel: ( -0.3, 0.3, -0.1) | Acc: ( 0.5, 0.4, 0.4) | Rwd: 1.89
950 [Astar_planner] Segment 22 reached in 28 steps, dist=1.70, axis_error
    =[0.8207855 1.4925959 0.01098824], reason=running

```

```

951 Task 14 result: success (Success: Reached destination (23/23
    segments). Shield interventions=36, emergencies=0, recoveries=6,
    recovery_steps=36.)
952
953 --- Task 15/38 ---
954 From [ 78.3 -148.32 -9.4 ] to [101.7 130.78 -5.3 ]
955 [Astar_planner] Top-down occupancy: 350x390, x=[-179.0, 170.0], y=[-
    200.0, 189.0], occupied=72869 cells (53.4%)
956 [Astar_planner] Occupancy base_grid=390x350, occupied=72869 cells
    (53.4%)
957 [Astar_planner] Endpoint nearest-free cells are disconnected; using
    reachable cells start=(82, 22), goal=(53, 347), search_radius=36.
958 [Astar_planner] Occupancy A*: grid=390x350, occupied=53.4%,
    safety_margin=1.5
959 [Astar_planner] Endpoint nearest-free cells are disconnected; using
    reachable cells start=(82, 22), goal=(53, 347), search_radius=36.
960 [Astar_planner] Occupancy A*: grid=390x350, occupied=53.4%,
    safety_margin=1.0
961 [Astar_planner] Endpoint nearest-free cells are disconnected; using
    reachable cells start=(82, 22), goal=(53, 347), search_radius=36.
962 [Astar_planner] Occupancy A*: grid=390x350, occupied=53.4%,
    safety_margin=0.6
963 [Astar_planner] Endpoint nearest-free cells are disconnected; using
    reachable cells start=(82, 22), goal=(53, 347), search_radius=36.
964 [Astar_planner] Occupancy A*: grid=390x350, occupied=53.4%,
    safety_margin=0.3
965 [Astar_planner] Safe occupancy A* failed; retrying legacy A* without
    clearance penalties.
966 [Astar_planner] Endpoint nearest-free cells are disconnected; using
    reachable cells start=(82, 22), goal=(53, 347), search_radius=36.
967 [Astar_planner] Legacy occupancy A*: grid=390x350, occupied=53.4
    %, safety_margin=1.5
968 [Astar_planner] WARNING: using legacy occupancy fallback; local
    targets may be closer to obstacles.
969 Path planned via occupancy_legacy (Points: 39, Dist: 432.4m). Local
    target z/alt: 0:-8.94(8.94m), 1:-24.62(24.62m), 2:-20.39(20.39m), 3:-20.
    34(20.34m), 4:-24.20(24.20m), 5:-23.99(23.99m), 6:-20.45(20.45m), 7
    :-22.99(22.99m), 8:-23.79(23.79m), 9:-20.17(20.17m), 10:-24.03(24.
    03m), 11:-21.26(21.26m), 12:-21.47(21.47m), 13:-23.53(23.53m), 14:-
    24.57(24.57m), 15:-24.39(24.39m), 16:-20.84(20.84m), 17:-24.60(24.
    60m), 18:-21.09(21.09m), 19:-20.48(20.48m), 20:-22.00(22.00m), 21:-
    24.69(24.69m), 22:-24.48(24.48m), 23:-23.11(23.11m), 24:-23.07(23.
    07m), 25:-21.93(21.93m), 26:-24.61(24.61m), 27:-20.51(20.51m), 28:-
    20.10(20.10m), 29:-22.33(22.33m), 30:-24.79(24.79m), 31:-23.16(23.
    16m), 32:-24.26(24.26m), 33:-20.18(20.18m), 34:-22.03(22.03m), 35:-
    21.50(21.50m), 36:-21.96(21.96m), 37:-20.84(20.84m), 38:-5.30(5.
    30m)
970 [Astar_planner] Segment 0: [ 88.      -178.      -8.942908] -> [ 90

```

```

970 .      -180.      -24.623146]
971 [Astar_planner] Placing first segment start at [ 88.      -178.      -8.
942908]
972 [Astar_planner] Initial climb to 20.0m.
973 [Step 24 | Total    0] Target: ( 90.0, -180.0, -24.6) | Pos: ( 90.5, -
178.1, -22.6) | Vel: ( 0.5, -0.3, -1.3) | Acc: ( 0.5, 0.8, 0.5) | Rwd:
3.84
974 [Astar_planner] Segment 0 reached in 24 steps, dist=2.82, axis_error
=[0.49018097 1.9445953 1.9787273 ], reason=running
975 [Astar_planner] Segment 1 starts from actual pos [ 90.4917 -178.
0564 -22.648321] (planned waypoint was [ 90.      -180.      -24.
623146])
976 [Astar_planner] Segment 1: [ 90.4917 -178.0564 -22.648321
]-> [ 95.      -183.      -20.386494]
977 [Step 5 | Total    0] Target: ( 95.0, -183.0, -20.4) | Pos: ( 93.5, -
180.8, -18.5) | Vel: ( 1.4, -2.0, 1.0) | Acc: ( -1.9, -2.0, -0.2) | Rwd
: 0.22
978 [Astar_planner] Segment 1 reached in 5 steps, dist=3.24, axis_error
=[1.4823074 2.1848297 1.8803883], reason=running
979 [Astar_planner] Segment 2 starts from actual pos [ 93.52194 -180.
82103 -18.503233] (planned waypoint was [ 95.      -183.      -20.
386494])
980 [Astar_planner] Segment 2: [ 93.52194 -180.82103 -18.503233
]-> [ 102.      -186.      -20.343142]
981 [Step 8 | Total    0] Target: (102.0, -186.0, -20.3) | Pos: ( 99.5, -
185.8, -19.8) | Vel: ( 1.5, -0.6, -0.2) | Acc: ( -0.9, -0.3, 0.6) | Rwd
: 2.01
982 [Astar_planner] Segment 2 reached in 8 steps, dist=2.58, axis_error
=[2.519844 0.17710876 0.5213337 ], reason=running
983 [Astar_planner] Segment 3 starts from actual pos [ 99.48453 -185.
82463 -19.822317] (planned waypoint was [ 102.      -186.      -20.
343142])
984 [Astar_planner] Segment 3: [ 99.48453 -185.82463 -19.822317
]-> [ 103.      -186.      -24.19513]
985 [Step 3 | Total    0] Target: (103.0, -186.0, -24.2) | Pos: (101.2, -
187.0, -22.0) | Vel: ( 0.2, -0.5, -1.9) | Acc: ( -1.3, -0.1, 2.0) | Rwd
: 4.50
986 [Astar_planner] Segment 3 reached in 3 steps, dist=3.05, axis_error
=[1.8329086 1.0186768 2.2106571], reason=running
987 [Astar_planner] Segment 4 starts from actual pos [ 101.167725 -187.
02005 -21.99014 ] (planned waypoint was [ 103.      -186.      -24.
19513])
988 [Astar_planner] Segment 4: [ 101.167725 -187.02005 -21.99014
]-> [ 107.      -180.      -23.987352]
989 [Step 26 | Total    0] Target: (107.0, -180.0, -24.0) | Pos: (107.8, -
184.7, -19.3) | Vel: ( 0.1, 0.1, 0.4) | Acc: ( -0.1, -0.4, 0.4) | Rwd
: -2.75
990 [Astar_planner] Segment 4 reached in 26 steps, dist=6.68, axis_error

```

```

990 =[0.8067932 4.7361603 4.639139 ], reason=stuck_near_intermediate
991 [Astar_planner] Segment 5 starts from actual pos [ 107.80712 -184.
73576 -19.347158] (planned waypoint was [ 107. -180. -23.
987352])
992 [Astar_planner] Segment 5: [ 107.80712 -184.73576 -19.347158
]-> [ 111. -173. -20.453316]
993 [Step 15 | Total 0] Target: (111.0, -173.0, -20.5) | Pos: (108.2, -
173.6, -18.5) | Vel: ( -0.4, 0.8, 0.1) | Acc: ( -0.8, 2.0, 1.4) | Rwd:
6.03
994 [Astar_planner] Segment 5 reached in 15 steps, dist=3.43, axis_error
=[2.7743378 0.64183044 1.9111099 ], reason=running
995 [Astar_planner] Segment 6 starts from actual pos [ 108.22437 -173.
63956 -18.54196] (planned waypoint was [ 111. -173. -20.
453316])
996 [Astar_planner] Segment 6: [ 108.22437 -173.63956 -18.54196] -> [
115. -166. -22.99265]
997 [Step 13 | Total 0] Target: (115.0, -166.0, -23.0) | Pos: (112.7, -
164.5, -20.2) | Vel: ( 1.5, 1.6, -0.8) | Acc: ( 0.1, 0.7, 0.5) | Rwd:
1.84
998 [Astar_planner] Segment 6 reached in 13 steps, dist=3.90, axis_error
=[2.2778473 1.507782 2.7882366], reason=running
999 [Astar_planner] Segment 7 starts from actual pos [ 112.726685 -164.
48749 -20.206656] (planned waypoint was [ 115. -166. -22.
99265])
1000 [Astar_planner] Segment 7: [ 112.726685 -164.48749 -20.206656
]-> [ 119. -159. -23.790998]
1001 [Step 7 | Total 0] Target: (119.0, -159.0, -23.8) | Pos: (116.1, -
159.4, -20.9) | Vel: ( 0.5, 1.0, -0.0) | Acc: ( -1.7, 1.3, -0.1) | Rwd
: 0.59
1002 [Astar_planner] Segment 7 reached in 7 steps, dist=4.17, axis_error
=[2.9402237 0.3738861 2.9403572], reason=running
1003 [Astar_planner] Segment 8 starts from actual pos [ 116.06126 -159.
37091 -20.850693] (planned waypoint was [ 119. -159. -23.
790998])
1004 [Astar_planner] Segment 8: [ 116.06126 -159.37091 -20.850693
]-> [ 122. -154. -20.16785]
1005 [Step 9 | Total 0] Target: (122.0, -154.0, -20.2) | Pos: (119.2, -
152.3, -19.1) | Vel: ( 0.7, 1.4, 0.0) | Acc: ( -1.7, 0.6, -0.4) | Rwd:
0.19
1006 [Astar_planner] Segment 8 reached in 9 steps, dist=3.42, axis_error
=[2.767662 1.6744232 1.1113815], reason=running
1007 [Astar_planner] Segment 9 starts from actual pos [ 119.23443 -152.
32137 -19.056467] (planned waypoint was [ 122. -154. -20.
16785])
1008 [Astar_planner] Segment 9: [ 119.23443 -152.32137 -19.056467
]-> [ 122. -144. -24.027672]
1009 [Step 10 | Total 0] Target: (122.0, -144.0, -24.0) | Pos: (121.1, -
145.9, -21.2) | Vel: ( 0.1, 0.5, -1.2) | Acc: ( -1.3, -0.4, -0.1) | Rwd

```

```

1009 : 3.89
1010 [Astar_planner] Segment 9 reached in 10 steps, dist=3.55, axis_error
=[0.87322235 1.9082336 2.8628254 ], reason=running
1011 [Astar_planner] Segment 10 starts from actual pos [ 121.12718 -145.
90665 -21.168436] (planned waypoint was [ 122. -144. -24.
027672])
1012 [Astar_planner] Segment 10: [ 121.12718 -145.90665 -21.168436
]-> [ 122. -137. -21.263199]
1013 [Step 35 | Total 0] Target: (122.0, -137.0, -21.3) | Pos: (124.7, -
138.9, -17.7) | Vel: ( -0.1, -0.2, 0.0) | Acc: ( -0.2, -1.3, -0.3) | Rwd
: -1.59
1014 [Astar_planner] Segment 10 reached in 35 steps, dist=4.88,
axis_error=[2.7325516 1.9009094 3.5645447], reason=
stuck_near_intermediate
1015 [Astar_planner] Segment 11 starts from actual pos [ 124.732315 -138
.90141 -17.698593] (planned waypoint was [ 122. -137. -21.
263199])
1016 [Astar_planner] Segment 11: [ 124.732315 -138.90141 -17.698593
]-> [ 133. -126. -21.467434]
1017 [Step 32 | Total 0] Target: (133.0, -126.0, -21.5) | Pos: (134.1, -
129.0, -18.7) | Vel: ( 0.3, 0.3, -0.1) | Acc: ( -0.3, -0.4, 0.6) | Rwd
: -0.72
1018 [Astar_planner] Segment 11 reached in 32 steps, dist=4.22,
axis_error=[1.0546417 2.9818878 2.7990932], reason=running
1019 [Astar_planner] Segment 12 starts from actual pos [ 134.05542 -128.
98105 -18.668716] (planned waypoint was [ 133. -126. -21.
467434])
1020 [Astar_planner] Segment 12: [ 134.05542 -128.98105 -18.668716
]-> [ 144. -115. -23.527931]
1021 [Step 25 | Total 0] Target: (144.0, -115.0, -23.5) | Pos: (142.7, -
117.9, -20.8) | Vel: ( 0.9, 0.4, -0.3) | Acc: ( 0.2, -0.4, 0.0) | Rwd:
0.39
1022 [Astar_planner] Segment 12 reached in 25 steps, dist=4.22,
axis_error=[1.3326263 2.9008713 2.7556343], reason=running
1023 [Astar_planner] Segment 13 starts from actual pos [ 142.66737 -117.
90087 -20.772297] (planned waypoint was [ 144. -115. -23.
527931])
1024 [Astar_planner] Segment 13: [ 142.66737 -117.90087 -20.772297
]-> [ 155. -104. -24.5655]
1025 [Step 22 | Total 0] Target: (155.0, -104.0, -24.6) | Pos: (152.6, -
106.1, -22.7) | Vel: ( 1.6, 0.1, -0.4) | Acc: ( 0.4, -0.8, 0.2) | Rwd:
1.76
1026 [Astar_planner] Segment 13 reached in 22 steps, dist=3.68,
axis_error=[2.4212952 2.0510406 1.8634167], reason=running
1027 [Astar_planner] Segment 14 starts from actual pos [ 152.58339 -106.
05089 -22.703217] (planned waypoint was [ 155. -104. -24.
5655])
1028 [Astar_planner] Segment 14: [ 152.58339 -106.05089 -22.703217

```



```

1028 ] -> [160.    -99.    -24.388771]
1029 [Step 13 | Total    0] Target: (160.0, -99.0, -24.4) | Pos: (160.3, -
    101.6, -21.5) | Vel: ( 1.3,  1.0,  0.5) | Acc: (-1.4,  0.6,  2.0) | Rwd: -
    0.62
1030 [Astar_planner] Segment 14 reached in 13 steps, dist=3.89,
    axis_error=[0.27772522 2.60643  2.8729  ], reason=running
1031 [Astar_planner] Segment 15 starts from actual pos [ 160.28172 -101.
    603424 -21.514353] (planned waypoint was [160.    -99.    -24.
    388771])
1032 [Astar_planner] Segment 15: [ 160.28172 -101.603424 -21.514353
    ] -> [160.    -84.    -20.83767]
1033 [Step 200 | Total    0] Target: (160.0, -84.0, -20.8) | Pos: (157.1, -
    100.0,  2.9) | Vel: (-0.4,  0.2, -0.2) | Acc: (-1.9,  2.0,  1.7) | Rwd: -
    252.62 -> Episode End: TIMEOUT!
1034
1035 [Astar_planner] Segment 15 ended: timeout
1036 [Astar_planner] Stop after failed segment 15.
1037 Task 15 result: failed (Failed at segment 15: timeout (15/38 reached
    ). Shield interventions=40, emergencies=1, recoveries=7,
    recovery_steps=40.)
1038
1039 --- Task 16/38 ---
1040 From [150.3  37.28 -10.4 ] to [ 72.7 -198.02 -13.3 ]
1041 [Astar_planner] Top-down occupancy: 350x390, x=[-179.0, 170.0], y
    =[-200.0, 189.0], occupied=72869 cells (53.4%)
1042 [Astar_planner] Occupancy base_grid=390x350, occupied=72869
    cells (53.4%)
1043 [Astar_planner] Endpoint nearest-free cells are disconnected; using
    reachable cells start=(20, 237), goal=(97, 2), search_radius=36.
1044 [Astar_planner] Occupancy A*: grid=390x350, occupied=53.4%,
    safety_margin=1.5
1045 [Astar_planner] Endpoint nearest-free cells are disconnected; using
    reachable cells start=(20, 237), goal=(97, 2), search_radius=36.
1046 [Astar_planner] Occupancy A*: grid=390x350, occupied=53.4%,
    safety_margin=1.0
1047 [Astar_planner] Endpoint nearest-free cells are disconnected; using
    reachable cells start=(20, 237), goal=(97, 2), search_radius=36.
1048 [Astar_planner] Occupancy A*: grid=390x350, occupied=53.4%,
    safety_margin=0.6
1049 [Astar_planner] Endpoint nearest-free cells are disconnected; using
    reachable cells start=(20, 237), goal=(97, 2), search_radius=36.
1050 [Astar_planner] Occupancy A*: grid=390x350, occupied=53.4%,
    safety_margin=0.3
1051 [Astar_planner] Safe occupancy A* failed; retrying legacy A* without
    clearance penalties.
1052 [Astar_planner] Legacy occupancy A*: grid=390x350, occupied=53.4
    %, safety_margin=1.5
1053 [Astar_planner] WARNING: using legacy occupancy fallback; local

```

```

1053 targets may be closer to obstacles.
1054 Path planned via occupancy_legacy (Points: 30, Dist: 288.1m). Local
target z/alt: 0:-10.40(10.40m), 1:-24.27(24.27m), 2:-22.13(22.13m), 3
:-22.68(22.68m), 4:-21.84(21.84m), 5:-22.50(22.50m), 6:-23.04(23.
04m), 7:-20.38(20.38m), 8:-21.86(21.86m), 9:-24.47(24.47m), 10:-24.
24(24.24m), 11:-22.28(22.28m), 12:-24.66(24.66m), 13:-21.57(21.
57m), 14:-20.28(20.28m), 15:-20.58(20.58m), 16:-21.16(21.16m), 17
:-21.60(21.60m), 18:-23.25(23.25m), 19:-20.78(20.78m), 20:-23.50(
23.50m), 21:-20.33(20.33m), 22:-24.37(24.37m), 23:-24.80(24.80m),
24:-22.24(22.24m), 25:-21.34(21.34m), 26:-22.33(22.33m), 27:-23.34
(23.34m), 28:-23.71(23.71m), 29:-13.30(13.30m)
1055 [Astar_planner] Segment 0: [150.    37.    -10.404803] -> [150
.    22.    -24.274466]
1056 [Astar_planner] Placing first segment start at [150.    37.    -10.
404803]
1057 [Astar_planner] Initial climb to 20.0m.
1058 [Step 17 | Total    0] Target: (150.0, 22.0, -24.3) | Pos: (150.4, 24.
5, -23.8) | Vel: ( 1.2, -1.4, -0.4) | Acc: ( 0.4, -1.1, 0.8) | Rwd: 2.
17
1059 [Astar_planner] Segment 0 reached in 17 steps, dist=2.61, axis_error
=[0.42633057 2.5281734 0.46928406], reason=running
1060 [Astar_planner] Segment 1 starts from actual pos [150.42633 24.
528173 -23.805182] (planned waypoint was [150.    22.    -24.
274466])
1061 [Astar_planner] Segment 1: [150.42633 24.528173 -23.805182] -> [
150.    7.    -22.125542]
1062 [Step 21 | Total    0] Target: (150.0, 7.0, -22.1) | Pos: (152.4, 9.1
, -21.9) | Vel: (-0.5, -1.1, -0.0) | Acc: (-0.2, -1.6, -0.1) | Rwd: 2.55
1063 [Astar_planner] Segment 1 reached in 21 steps, dist=3.21, axis_error
=[2.3804169 2.143197 0.20088768], reason=running
1064 [Astar_planner] Segment 2 starts from actual pos [152.38042 9.
143197 -21.924654] (planned waypoint was [150.    7.    -22.
125542])
1065 [Astar_planner] Segment 2: [152.38042 9.143197 -21.924654] -> [
150.    -8.    -22.67786]
1066 [Step 18 | Total    0] Target: (150.0, -8.0, -22.7) | Pos: (152.1, -5.
7, -21.3) | Vel: (-0.2, -1.0, -0.0) | Acc: (-0.2, 1.3, -0.8) | Rwd: 1.
45
1067 [Astar_planner] Segment 2 reached in 18 steps, dist=3.47, axis_error
=[2.1497803 2.3362012 1.3924274], reason=running
1068 [Astar_planner] Segment 3 starts from actual pos [152.1493 -5.
6666894 -21.285439] (planned waypoint was [150.    -8.    -22.
67786])
1069 [Astar_planner] Segment 3: [152.1493 -5.6666894 -21.285439
] -> [150.    -23.    -21.83886]
1070 [Step 24 | Total    0] Target: (150.0, -23.0, -21.8) | Pos: (152.7, -21
.0, -21.4) | Vel: (-0.4, -0.7, 0.0) | Acc: (-0.1, 1.7, -0.9) | Rwd: 2.
85

```

```

1071 [Astar_planner] Segment 3 reached in 24 steps, dist=3.37, axis_error
    =[2.6571198 2.0226612 0.48713303], reason=running
1072 [Astar_planner] Segment 4 starts from actual pos [152.6558 -20.
    979572 -21.35161 ] (planned waypoint was [150. -23. -21.
    83886])
1073 [Astar_planner] Segment 4: [152.6558 -20.979572 -21.35161 ] -> [
    149. -30. -22.500357]
1074 [Step 10 | Total 0] Target: (149.0, -30.0, -22.5) | Pos: (151.2, -27
    .0, -21.8) | Vel: ( -0.1, -1.2, -0.7) | Acc: ( 0.3, 1.7, -0.8) | Rwd: 2.
    79
1075 [Astar_planner] Segment 4 reached in 10 steps, dist=3.76, axis_error
    =[2.1532898 2.9912128 0.7248459], reason=running
1076 [Astar_planner] Segment 5 starts from actual pos [151.15312 -27.
    012396 -21.777637] (planned waypoint was [149. -30. -22.
    500357])
1077 [Astar_planner] Segment 5: [151.15312 -27.012396 -21.777637] -> [
    146. -37. -23.041708]
1078 [Step 10 | Total 0] Target: (146.0, -37.0, -23.0) | Pos: (148.7, -34
    .5, -22.1) | Vel: ( -0.3, -1.1, -0.0) | Acc: ( 0.3, 1.6, -1.3) | Rwd: 2.
    55
1079 [Astar_planner] Segment 5 reached in 10 steps, dist=3.82, axis_error
    =[2.7006836 2.52557 0.9473133], reason=running
1080 [Astar_planner] Segment 6 starts from actual pos [148.69978 -34.
    477848 -22.094505] (planned waypoint was [146. -37. -23.
    041708])
1081 [Astar_planner] Segment 6: [148.69978 -34.477848 -22.094505] -> [
    143. -44. -20.377842]
1082 [Step 8 | Total 0] Target: (143.0, -44.0, -20.4) | Pos: (145.6, -41.
    4, -19.6) | Vel: ( -0.6, -1.9, 0.0) | Acc: ( 0.0, -1.7, -0.7) | Rwd: 3.
    06
1083 [Astar_planner] Segment 6 reached in 8 steps, dist=3.77, axis_error
    =[2.645996 2.5522156 0.8151207], reason=running
1084 [Astar_planner] Segment 7 starts from actual pos [145.64427 -41.
    45339 -19.562675] (planned waypoint was [143. -44. -20.
    377842])
1085 [Astar_planner] Segment 7: [145.64427 -41.45339 -19.562675] -> [
    141. -59. -21.86095]
1086 [Step 15 | Total 0] Target: (141.0, -59.0, -21.9) | Pos: (143.1, -56
    .3, -21.1) | Vel: ( -0.0, -1.3, 0.2) | Acc: ( 0.9, 0.2, 1.1) | Rwd: 1.
    21
1087 [Astar_planner] Segment 7 reached in 15 steps, dist=3.51, axis_error
    =[2.0687103 2.739666 0.7149601], reason=running
1088 [Astar_planner] Segment 8 starts from actual pos [143.06865 -56.
    264362 -21.145456] (planned waypoint was [141. -59. -21.
    86095])
1089 [Astar_planner] Segment 8: [143.06865 -56.264362 -21.145456] -> [
    141. -74. -24.471073]
1090 [Step 28 | Total 0] Target: (141.0, -74.0, -24.5) | Pos: (143.8, -73

```

```

1090 .6, -22.6) | Vel: ( -0.8, -1.0, 0.1) | Acc: ( -0.1, -1.7, 1.8) | Rwd: -0.
44
1091 [Astar_planner] Segment 8 reached in 28 steps, dist=3.43, axis_error
=[2.8213654 0.42958832 1.8942451 ], reason=running
1092 [Astar_planner] Segment 9 starts from actual pos [143.81892 -73.
573555 -22.576574] (planned waypoint was [141. -74. -24.
471073])
1093 [Astar_planner] Segment 9: [143.81892 -73.573555 -22.576574] -> [
139. -89. -24.237253]
1094 [Step 19 | Total 0] Target: (139.0, -89.0, -24.2) | Pos: (141.9, -89
.6, -24.4) | Vel: ( -0.5, -1.6, 0.0) | Acc: ( -0.7, 1.4, -0.0) | Rwd: -0.
04
1095 [Astar_planner] Segment 9 reached in 19 steps, dist=2.95, axis_error
=[2.8837128 0.5888138 0.11799431], reason=running
1096 [Astar_planner] Segment 10 starts from actual pos [141.88208 -89.
59347 -24.355244] (planned waypoint was [139. -89. -24.
237253])
1097 [Astar_planner] Segment 10: [141.88208 -89.59347 -24.355244
] -> [ 139. -104. -22.2768]
1098 [Step 12 | Total 0] Target: (139.0, -104.0, -22.3) | Pos: (138.6, -
101.0, -20.0) | Vel: ( -0.2, -1.7, -0.3) | Acc: ( 0.6, 0.4, -0.1) | Rwd
: 1.45
1099 [Astar_planner] Segment 10 reached in 12 steps, dist=3.74,
axis_error=[0.3829956 2.9839554 2.2287598], reason=running
1100 [Astar_planner] Segment 11 starts from actual pos [ 138.6163 -101.
02119 -20.048815] (planned waypoint was [ 139. -104. -22.
2768])
1101 [Astar_planner] Segment 11: [ 138.6163 -101.02119 -20.048815
] -> [ 139. -115. -24.658474]
1102 [Step 11 | Total 0] Target: (139.0, -115.0, -24.7) | Pos: (141.8, -
112.4, -24.6) | Vel: ( -0.1, -1.5, -0.2) | Acc: ( 0.0, -0.8, 0.0) | Rwd
: 0.84
1103 [Astar_planner] Segment 11 reached in 11 steps, dist=3.82,
axis_error=[2.7809753 2.6227493 0.03852463], reason=running
1104 [Astar_planner] Segment 12 starts from actual pos [ 141.78073 -112.
38188 -24.620403] (planned waypoint was [ 139. -115. -24.
658474])
1105 [Astar_planner] Segment 12: [ 141.78073 -112.38188 -24.620403
] -> [ 135. -121. -21.566948]
1106 [Step 10 | Total 0] Target: (135.0, -121.0, -21.6) | Pos: (137.5, -
119.1, -19.3) | Vel: ( -1.0, -1.5, -0.3) | Acc: ( -0.5, -0.6, -1.6) | Rwd
: 0.73
1107 [Astar_planner] Segment 12 reached in 10 steps, dist=3.88,
axis_error=[2.5191193 1.8709946 2.2837029], reason=running
1108 [Astar_planner] Segment 13 starts from actual pos [ 137.51338 -119.
137825 -19.285227] (planned waypoint was [ 135. -121. -21.
566948])
1109 [Astar_planner] Segment 13: [ 137.51338 -119.137825 -19.285227

```

```

1109 ]-> [ 129.    -127.    -20.284515]
1110 [Step 13 | Total    0] Target: (129.0, -127.0, -20.3) | Pos: (128.3, -
    124.4, -19.1) | Vel: ( -0.6, -1.1,  0.8) | Acc: ( 1.0,  0.5,  1.8) | Rwd
    : -0.53
1111 [Astar_planner] Segment 13 reached in 13 steps, dist=2.94,
    axis_error=[0.6590271 2.6280212 1.1450195], reason=running
1112 [Astar_planner] Segment 14 starts from actual pos [ 128.33925 -124.
    37533 -19.137072] (planned waypoint was [ 129.    -127.    -20.
    284515])
1113 [Astar_planner] Segment 14: [ 128.33925 -124.37533 -19.137072
    ]-> [ 122.    -134.    -20.578924]
1114 [Step 13 | Total    0] Target: (122.0, -134.0, -20.6) | Pos: (120.4, -
    131.5, -18.7) | Vel: ( -0.6, -0.9,  0.6) | Acc: ( -1.0,  1.1,  1.4) | Rwd
    : -1.16
1115 [Astar_planner] Segment 14 reached in 13 steps, dist=3.48,
    axis_error=[1.6242523 2.4674377 1.8387527], reason=running
1116 [Astar_planner] Segment 15 starts from actual pos [ 120.37401 -131.
    53539 -18.738468] (planned waypoint was [ 122.    -134.    -20.
    578924])
1117 [Astar_planner] Segment 15: [ 120.37401 -131.53539 -18.738468
    ]-> [ 122.    -149.    -21.155663]
1118 [Step 16 | Total    0] Target: (122.0, -149.0, -21.2) | Pos: (124.2, -
    146.6, -20.4) | Vel: ( -0.1, -1.2,  0.3) | Acc: ( -0.2,  1.0,  0.1) | Rwd
    :  0.29
1119 [Astar_planner] Segment 15 reached in 16 steps, dist=3.31,
    axis_error=[2.2035294 2.3611603 0.7282448], reason=running
1120 [Astar_planner] Segment 16 starts from actual pos [ 124.20309 -146.
    64236 -20.426525] (planned waypoint was [ 122.    -149.    -21.
    155663])
1121 [Astar_planner] Segment 16: [ 124.20309 -146.64236 -20.426525
    ]-> [ 122.    -154.    -21.600414]
1122 [Step  7 | Total    0] Target: (122.0, -154.0, -21.6) | Pos: (123.6, -
    151.1, -18.9) | Vel: ( -0.4, -1.4, -0.0) | Acc: ( -0.6,  0.4, -0.4) | Rwd
    : -0.32
1123 [Astar_planner] Segment 16 reached in 7 steps, dist=4.23, axis_error
    =[1.5572586 2.9031372 2.6553497], reason=running
1124 [Astar_planner] Segment 17 starts from actual pos [ 123.55726 -151.
    09686 -18.945065] (planned waypoint was [ 122.    -154.    -21.
    600414])
1125 [Astar_planner] Segment 17: [ 123.55726 -151.09686 -18.945065
    ]-> [ 121.    -156.    -23.248472]
1126 [Step  6 | Total    0] Target: (121.0, -156.0, -23.2) | Pos: (122.8, -
    154.9, -20.5) | Vel: ( -0.2, -1.0, -1.9) | Acc: (  0.7, -0.2, -0.6) | Rwd
    :  1.42
1127 [Astar_planner] Segment 17 reached in 6 steps, dist=3.45, axis_error
    =[1.806282 1.1495209 2.708376 ], reason=running
1128 [Astar_planner] Segment 18 starts from actual pos [ 122.80581 -154.
    8535 -20.545902] (planned waypoint was [ 121.    -156.    -23.

```

```

1128 248472])
1129 [Astar_planner] Segment 18: [ 122.80581 -154.8535 -20.545902
]-> [ 117. -162. -20.781279]
1130 [Step 9 | Total 0] Target: (117.0, -162.0, -20.8) | Pos: (117.8, -
159.5, -19.6) | Vel: ( -1.5, -1.2, -0.1) | Acc: ( -1.7, -1.7, -0.5) | Rwd
: 2.56
1131 [Astar_planner] Segment 18 reached in 9 steps, dist=2.92, axis_error
=[0.81559753 2.5385132 1.1840916 ], reason=running
1132 [Astar_planner] Segment 19 starts from actual pos [ 117.81114 -159.
46518 -19.59752] (planned waypoint was [ 117. -162. -20.
781279])
1133 [Astar_planner] Segment 19: [ 117.81114 -159.46518 -19.59752
]-> [ 114. -168. -23.498795]
1134 [Step 34 | Total 0] Target: (114.0, -168.0, -23.5) | Pos: (116.8, -
169.4, -21.4) | Vel: ( -0.5, -0.4, 0.0) | Acc: ( 0.9, 0.2, -0.5) | Rwd
: -0.24
1135 [Astar_planner] Segment 19 reached in 34 steps, dist=3.73,
axis_error=[2.756668 1.3561707 2.120306 ], reason=running
1136 [Astar_planner] Segment 20 starts from actual pos [ 116.75521 -169.
3575 -21.378422] (planned waypoint was [ 114. -168. -23.
498795])
1137 [Astar_planner] Segment 20: [ 116.75521 -169.3575 -21.378422
]-> [ 113. -169. -20.334587]
1138 [Step 1 | Total 0] Target: (113.0, -169.0, -20.3) | Pos: (116.0, -
169.6, -20.4) | Vel: ( -0.6, 0.2, -0.3) | Acc: ( -2.0, 0.9, 0.0) | Rwd
: 3.07
1139 [Astar_planner] Segment 20 reached in 1 steps, dist=3.05, axis_error
=[2.994873 0.5803375 0.03152275], reason=running
1140 [Astar_planner] Segment 21 starts from actual pos [ 115.993 -169.
57968 -20.366938] (planned waypoint was [ 113. -169. -20.
334587])
1141 [Astar_planner] Segment 21: [ 115.993 -169.57968 -20.366938
]-> [ 110. -175. -24.367676]
1142 [Step 42 | Total 0] Target: (110.0, -175.0, -24.4) | Pos: (112.2, -
176.1, -22.0) | Vel: ( -0.1, -0.1, 1.0) | Acc: ( -0.1, -1.6, 0.2) | Rwd
: 2.44
1143 [Astar_planner] Segment 21 reached in 42 steps, dist=3.40,
axis_error=[2.2173462 1.0636292 2.3455048], reason=running
1144 [Astar_planner] Segment 22 starts from actual pos [ 112.216995 -176
.06393 -22.019293] (planned waypoint was [ 110. -175. -24.
367676])
1145 [Astar_planner] Segment 22: [ 112.216995 -176.06393 -22.019293
]-> [ 109. -176. -24.795383]
1146 [Step 9 | Total 0] Target: (109.0, -176.0, -24.8) | Pos: (112.0, -
178.8, -21.9) | Vel: ( -0.2, -0.3, -0.5) | Acc: ( 0.4, -0.2, 0.7) | Rwd
: 0.07
1147 [Astar_planner] Segment 22 reached in 9 steps, dist=5.01, axis_error
=[2.9709625 2.8207703 2.8880062], reason=running

```

```

1148 [Astar_planner] Segment 23 starts from actual pos [ 111.9703 -178.82164 -21.908934] (planned waypoint was [ 109. -176. -24.795383])
1149 [Astar_planner] Segment 23: [ 111.9703 -178.82164 -21.908934 ] -> [ 106. -182. -22.238907]
1150 [Step 10 | Total 0] Target: (106.0, -182.0, -22.2) | Pos: (108.6, -182.0, -19.4) | Vel: ( -0.8, -0.8, 0.2) | Acc: ( 1.6, -0.4, -0.1) | Rwd : -1.19
1151 [Astar_planner] Segment 23 reached in 10 steps, dist=3.83, axis_error=[2.5726852e+00 7.9345703e-04 2.8339367e+00], reason=running
1152 [Astar_planner] Segment 24 starts from actual pos [ 108.570274 -182.0032 -19.4043 ] (planned waypoint was [ 106. -182. -22.238907])
1153 [Astar_planner] Segment 24: [ 108.570274 -182.0032 -19.4043 ] -> [ 104. -184. -21.335363]
1154 [Step 5 | Total 0] Target: (104.0, -184.0, -21.3) | Pos: (106.8, -184.5, -18.9) | Vel: ( -0.4, -0.8, 0.3) | Acc: ( 1.2, -0.0, 0.3) | Rwd : -1.59
1155 [Astar_planner] Segment 24 reached in 5 steps, dist=3.74, axis_error=[2.800499 0.51091003 2.4261951 ], reason=running
1156 [Astar_planner] Segment 25 starts from actual pos [ 106.799286 -184.5133 -18.908386] (planned waypoint was [ 104. -184. -21.335363])
1157 [Astar_planner] Segment 25: [ 106.799286 -184.5133 -18.908386 ] -> [ 103. -186. -22.328724]
1158 [Step 13 | Total 0] Target: (103.0, -186.0, -22.3) | Pos: (105.1, -188.0, -20.4) | Vel: ( -0.3, -0.5, -2.0) | Acc: ( 0.6, 0.2, 1.4) | Rwd : 3.19
1159 [Astar_planner] Segment 25 reached in 13 steps, dist=3.45, axis_error=[2.0568237 2.0056305 1.9087868], reason=running
1160 [Astar_planner] Segment 26 starts from actual pos [ 105.05607 -188.00705 -20.425838] (planned waypoint was [ 103. -186. -22.328724])
1161 [Astar_planner] Segment 26: [ 105.05607 -188.00705 -20.425838 ] -> [ 97. -187. -23.34459]
1162 [Step 64 | Total 0] Target: ( 97.0, -187.0, -23.3) | Pos: ( 97.9, -189.8, -20.4) | Vel: ( 0.1, -0.0, -1.7) | Acc: ( 0.7, -0.4, -0.9) | Rwd : 1.20
1163 [Astar_planner] Segment 26 reached in 64 steps, dist=4.13, axis_error=[0.8837509 2.801834 2.9014091], reason=running
1164 [Astar_planner] Segment 27 starts from actual pos [ 97.88421 -189.80208 -20.453325] (planned waypoint was [ 97. -187. -23.34459])
1165 [Astar_planner] Segment 27: [ 97.88421 -189.80208 -20.453325 ] -> [ 86. -198. -23.714596]
1166 [Step 13 | Total 0] Target: ( 86.0, -198.0, -23.7) | Pos: ( 88.9, -196.2, -21.5) | Vel: ( -1.0, -1.3, -0.2) | Acc: ( -0.5, -1.0, -1.0) | Rwd

```

```

1166 : 1.28
1167 [Astar_planner] Segment 27 reached in 13 steps, dist=4.09,
axis_error=[2.8895416 1.8054962 2.2562084], reason=running
1168 [Astar_planner] Segment 28 starts from actual pos [ 88.88954 -196.
1945 -21.458387] (planned waypoint was [ 86. -198. -23.
714596])
1169 [Astar_planner] Segment 28: [ 88.88954 -196.1945 -21.458387
]-> [ 73. -198. -13.29868]
1170 [Step 36 | Total 0] Target: ( 73.0, -198.0, -13.3) | Pos: ( 71.9, -
197.3, -13.2) | Vel: ( 2.1, 0.7, -1.1) | Acc: ( 2.0, 1.5, -1.0) | Rwd:
602.52 -> Episode End: ARRIVED!
1171
1172 [Astar_planner] Segment 28 reached in 36 steps, dist=1.32,
axis_error=[1.0853806 0.7370148 0.09878922], reason=arrived
1173 Task 16 result: success (Success: Reached destination (29/29
segments). Shield interventions=0, emergencies=0, recoveries=0,
recovery_steps=0.)
1174
1175 --- Task 17/38 ---
1176 From [ 84.8 -145.42 -4. ] to [ -64.4 -146.72 -4. ]
1177 [Astar_planner] Top-down occupancy: 350x390, x=[-179.0, 170.0], y
=[-200.0, 189.0], occupied=72869 cells (53.4%)
1178 [Astar_planner] Occupancy base_grid=390x350, occupied=72869
cells (53.4%)
1179 [Astar_planner] Occupancy A*: grid=390x350, occupied=53.4%,
safety_margin=1.5
1180 Path planned via occupancy (Points: 14, Dist: 181.2m). Local target z
/alt: 0:-4.00(4.00m), 1:-20.07(20.07m), 2:-23.82(23.82m), 3:-23.90(23
.90m), 4:-22.38(22.38m), 5:-23.14(23.14m), 6:-23.11(23.11m), 7:-24.
05(24.05m), 8:-21.72(21.72m), 9:-23.46(23.46m), 10:-20.59(20.59m
), 11:-23.87(23.87m), 12:-22.61(22.61m), 13:-4.00(4.00m)
1181 [Astar_planner] Segment 0: [ 85. -145. -4.] -> [ 77. -154. -20.
06826]
1182 [Astar_planner] Placing first segment start at [ 85. -145. -4.]
1183 [Astar_planner] Initial climb to 20.0m.
1184 [Step 1 | Total 0] Target: ( 77.0, -154.0, -20.1) | Pos: ( 0.1, 0.0
, -0.2) | Vel: ( 0.0, 0.0, 0.1) | Acc: ( -0.0, -0.0, -1.2) | Rwd: -909.
40 -> Episode End: CRASHED!
1185
1186 [Astar_planner] Segment 0 ended: collision
1187 [Astar_planner] Stop after failed segment 0.
1188 Task 17 result: failed (Failed at segment 0: collision (0/13 reached).
Shield interventions=1, emergencies=0, recoveries=1,
recovery_steps=1.)
1189
1190 --- Task 18/38 ---
1191 From [-94.8 92.08 -4. ] to [118.1 76.88 -4. ]
1192 [Astar_planner] Top-down occupancy: 350x390, x=[-179.0, 170.0], y

```



```

1192 =[ -200.0, 189.0], occupied=72869 cells (53.4%)
1193 [Astar_planner] Occupancy base_grid=390x350, occupied=72869
      cells (53.4%)
1194 [Astar_planner] Occupancy A*: grid=390x350, occupied=53.4%,
      safety_margin=1.5
1195 Path planned via occupancy (Points: 25, Dist: 333.1m). Local target z
      /alt: 0:-4.00(4.00m), 1:-20.77(20.77m), 2:-21.62(21.62m), 3:-20.30(20
      .30m), 4:-21.86(21.86m), 5:-21.61(21.61m), 6:-20.80(20.80m), 7:-20.
      84(20.84m), 8:-23.76(23.76m), 9:-23.78(23.78m), 10:-22.31(22.31m
      ), 11:-20.63(20.63m), 12:-20.13(20.13m), 13:-20.86(20.86m), 14:-22.
      51(22.51m), 15:-24.67(24.67m), 16:-23.42(23.42m), 17:-22.85(22.
      85m), 18:-24.21(24.21m), 19:-22.48(22.48m), 20:-21.32(21.32m), 21
      :-20.02(20.02m), 22:-21.32(21.32m), 23:-22.01(22.01m), 24:-4.00(4.
      00m)
1196 [Astar_planner] Segment 0: [-95.  92.  -4.] -> [-83.    89.   -20.
      771948]
1197 [Astar_planner] Placing first segment start at [-95.  92.  -4.]
1198 [Astar_planner] Initial climb to 20.0m.
1199 [Step  15 | Total    0] Target: (-83.0,  89.0, -20.8) | Pos: (-84.6,  89.
      4, -18.5) | Vel: (  2.2,  0.5, -0.1) | Acc: (  2.0,  0.3,  0.0) | Rwd:  2.38

1200 [Astar_planner] Segment 0 reached in 15 steps, dist=2.79, axis_error
      =[1.5628662  0.40210724  2.280199 ], reason=running
1201 [Astar_planner] Segment 1 starts from actual pos [-84.55632  89.
      403595 -18.492126] (planned waypoint was [-83.    89.   -20.
      771948])
1202 [Astar_planner] Segment 1: [-84.55632  89.403595 -18.492126] -> [-
      69.    89.   -21.615353]
1203 [Step  15 | Total    0] Target: (-69.0,  89.0, -21.6) | Pos: (-67.5,  91.
      8, -20.0) | Vel: ( -0.2, -0.2,  0.2) | Acc: ( -0.7, -0.1, -1.8) | Rwd:  1.
      63
1204 [Astar_planner] Segment 1 reached in 15 steps, dist=3.55, axis_error
      =[1.5326843  2.7643967  1.6208401], reason=running
1205 [Astar_planner] Segment 2 starts from actual pos [-67.467804  91.
      763824 -19.994026] (planned waypoint was [-69.    89.   -21.
      615353])
1206 [Astar_planner] Segment 2: [-67.467804  91.763824 -19.994026] -> [-
      56.    89.   -20.304657]
1207 [Step  17 | Total    0] Target: (-56.0,  89.0, -20.3) | Pos: (-55.8,  88.
      3, -17.7) | Vel: (  0.7, -0.2, -0.3) | Acc: (  0.8, -1.2, -0.8) | Rwd:  2.
      19
1208 [Astar_planner] Segment 2 reached in 17 steps, dist=2.72, axis_error
      =[0.20022202  0.70001984  2.6220646 ], reason=running
1209 [Astar_planner] Segment 3 starts from actual pos [-55.79776  88.
      29928 -17.68336] (planned waypoint was [-56.    89.   -20.
      304657])
1210 [Astar_planner] Segment 3: [-55.79776  88.29928 -17.68336] -> [-42
      .    89.   -21.858234]

```

```

1211 [Step 21 | Total 0] Target: (-42.0, 89.0, -21.9) | Pos: (-41.9, 91.
    3, -19.3) | Vel: ( 1.6, 0.4, -1.2) | Acc: ( -1.6, 0.2, -0.8) | Rwd: 2.
    07
1212 [Astar_planner] Segment 3 reached in 21 steps, dist=3.45, axis_error
    =[0.11568451 2.2768707 2.5908146 ], reason=running
1213 [Astar_planner] Segment 4 starts from actual pos [-41.874844 91.
    27947 -19.274343] (planned waypoint was [-42. 89. -21.
    858234])
1214 [Astar_planner] Segment 4: [-41.874844 91.27947 -19.274343] -> [-
    30. 94. -21.607262]
1215 [Step 33 | Total 0] Target: (-30.0, 94.0, -21.6) | Pos: (-32.9, 93.
    2, -20.1) | Vel: ( 1.3, -0.2, 0.0) | Acc: ( 0.8, -0.3, 0.4) | Rwd: 1.38
1216 [Astar_planner] Segment 4 reached in 33 steps, dist=3.38, axis_error
    =[2.9190521 0.8455963 1.4712124], reason=running
1217 [Astar_planner] Segment 5 starts from actual pos [-32.915016 93.
    15367 -20.13598 ] (planned waypoint was [-30. 94. -21.
    607262])
1218 [Astar_planner] Segment 5: [-32.915016 93.15367 -20.13598 ] -> [-
    25. 105. -20.797647]
1219 [Step 14 | Total 0] Target: (-25.0, 105.0, -20.8) | Pos: (-27.7, 106
    .5, -18.6) | Vel: ( 0.7, 1.6, -0.4) | Acc: ( -1.6, 1.2, 0.4) | Rwd: 0.
    25
1220 [Astar_planner] Segment 5 reached in 14 steps, dist=3.76, axis_error
    =[2.7005672 1.4698792 2.1564274], reason=running
1221 [Astar_planner] Segment 6 starts from actual pos [-27.698505 106.
    474655 -18.64232 ] (planned waypoint was [-25. 105. -20.
    797647])
1222 [Astar_planner] Segment 6: [-27.698505 106.474655 -18.64232 ] -> [-
    25. 118. -20.838034]
1223 [Step 13 | Total 0] Target: (-25.0, 118.0, -20.8) | Pos: (-25.0, 115
    .5, -18.6) | Vel: ( 0.0, 1.9, -0.6) | Acc: ( -1.6, 0.4, -0.1) | Rwd: 2.
    42
1224 [Astar_planner] Segment 6 reached in 13 steps, dist=3.39, axis_error
    =[0.02089691 2.5444412 2.246046 ], reason=running
1225 [Astar_planner] Segment 7 starts from actual pos [-24.979103 115.
    45556 -18.591988] (planned waypoint was [-25. 118. -20.
    838034])
1226 [Astar_planner] Segment 7: [-24.979103 115.45556 -18.591988] -> [-
    25. 132. -23.764637]
1227 [Step 14 | Total 0] Target: (-25.0, 132.0, -23.8) | Pos: (-23.6, 132
    .8, -21.8) | Vel: ( -0.4, 1.7, -1.7) | Acc: ( 0.5, -0.4, 2.0) | Rwd: 4.
    10
1228 [Astar_planner] Segment 7 reached in 14 steps, dist=2.54, axis_error
    =[1.4138966 0.8452606 1.9311562], reason=running
1229 [Astar_planner] Segment 8 starts from actual pos [-23.587387 132.
    85034 -21.83843 ] (planned waypoint was [-25. 132. -23.
    764637])

```

```

1230 [Astar_planner] Segment 8: [-23.587387 132.85034 -21.83843 ] -> [-
25.    145.    -23.777246]
1231 [Step  65 | Total    0] Target: (-25.0, 145.0, -23.8) | Pos: (-23.3, 143
.4, -21.7) | Vel: ( -0.4, -0.3, -1.6) | Acc: ( 0.0, -1.3,  1.9) | Rwd: 7.
45
1232 [Astar_planner] Segment 8 reached in 65 steps, dist=3.12, axis_error
=[1.6890278 1.6383209 2.0535374], reason=running
1233 [Astar_planner] Segment 9 starts from actual pos [-23.313328 143.
36005 -21.733175] (planned waypoint was [-25.    145.    -23.
777246])
1234 [Astar_planner] Segment 9: [-23.313328 143.36005 -21.733175] -> [-
17.    153.    -22.308111]
1235 [Step  19 | Total    0] Target: (-17.0, 153.0, -22.3) | Pos: (-18.5, 150
.3, -19.4) | Vel: ( 0.9,  0.6,  0.1) | Acc: ( 1.5, -0.9, -0.3) | Rwd: -0.
31
1236 [Astar_planner] Segment 9 reached in 19 steps, dist=4.26, axis_error
=[1.4507561 2.7135468 2.9432335], reason=running
1237 [Astar_planner] Segment 10 starts from actual pos [-18.44538 150.
29002 -19.3645 ] (planned waypoint was [-17.    153.    -22.
308111])
1238 [Astar_planner] Segment 10: [-18.44538 150.29002 -19.3645 ] -> [-3
.    153.    -20.634693]
1239 [Step  16 | Total    0] Target: ( -3.0, 153.0, -20.6) | Pos: ( -5.7, 155.
3, -18.7) | Vel: ( 1.4, -0.8,  0.0) | Acc: ( 0.8, -0.2, -0.1) | Rwd: 2.
27
1240 [Astar_planner] Segment 10 reached in 16 steps, dist=3.99,
axis_error=[2.676087 2.281952 1.8950233], reason=running
1241 [Astar_planner] Segment 11 starts from actual pos [ -5.6719446 155.
27956 -18.739525 ] (planned waypoint was [ -3.    153.    -20.
634693])
1242 [Astar_planner] Segment 11: [ -5.6719446 155.27956 -18.739525
] -> [ 10.    151.    -20.127441]
1243 [Step  12 | Total    0] Target: ( 10.0, 151.0, -20.1) | Pos: (  8.6, 150.
6, -17.6) | Vel: ( 2.1, -1.0, -1.4) | Acc: ( 1.1,  1.2, -0.7) | Rwd: 4.
38
1244 [Astar_planner] Segment 11 reached in 12 steps, dist=2.92,
axis_error=[1.372222 0.35266113 2.5543842 ], reason=running
1245 [Astar_planner] Segment 12 starts from actual pos [  8.633928 150.
64438 -17.577374] (planned waypoint was [ 10.    151.    -20.
127441])
1246 [Astar_planner] Segment 12: [  8.633928 150.64438 -17.577374
] -> [ 21.    145.    -20.857985]
1247 [Step   8 | Total    0] Target: ( 21.0, 145.0, -20.9) | Pos: ( 18.1, 146.
2, -19.7) | Vel: ( 2.3, -1.2, -0.7) | Acc: ( 1.4, -0.8, -1.3) | Rwd: 4.
98
1248 [Astar_planner] Segment 12 reached in 8 steps, dist=3.36, axis_error
=[2.883751 1.2483063 1.1908035], reason=running
1249 [Astar_planner] Segment 13 starts from actual pos [ 18.129995 146.

```

```

1249 24138 -19.671162] (planned waypoint was [ 21.    145.   -20.
      857985])
1250 [Astar_planner] Segment 13: [ 18.129995 146.24138 -19.671162
    ] -> [ 31.    137.   -22.505169]
1251 [Step 18 | Total    0] Target: ( 31.0, 137.0, -22.5) | Pos: ( 28.7, 134.
    0, -21.0) | Vel: ( 1.1, -0.8, 0.0) | Acc: ( -1.5, -1.8, 0.2) | Rwd: -0.
    35
1252 [Astar_planner] Segment 13 reached in 18 steps, dist=4.07,
    axis_error=[2.313816 2.977005 1.545105], reason=running
1253 [Astar_planner] Segment 14 starts from actual pos [ 28.689573 134.
    02065 -20.960022] (planned waypoint was [ 31.    137.   -22.
    505169])
1254 [Astar_planner] Segment 14: [ 28.689573 134.02065 -20.960022
    ] -> [ 44.    137.   -24.665487]
1255 [Step 14 | Total    0] Target: ( 44.0, 137.0, -24.7) | Pos: ( 44.5, 139.
    6, -22.0) | Vel: ( 1.2, -0.7, -0.0) | Acc: ( -1.8, 1.1, -0.4) | Rwd: 0.
    62
1256 [Astar_planner] Segment 14 reached in 14 steps, dist=3.73,
    axis_error=[0.48537064 2.582428 2.651701 ], reason=running
1257 [Astar_planner] Segment 15 starts from actual pos [ 44.489113 139.
    58037 -22.013906] (planned waypoint was [ 44.    137.   -24.
    665487])
1258 [Astar_planner] Segment 15: [ 44.489113 139.58037 -22.013906
    ] -> [ 58.    137.   -23.42028]
1259 [Step 9 | Total    0] Target: ( 58.0, 137.0, -23.4) | Pos: ( 55.1, 136.
    4, -21.6) | Vel: ( 2.6, 0.1, 0.1) | Acc: ( -0.0, 1.2, -0.5) | Rwd: 1.61
1260 [Astar_planner] Segment 15 reached in 9 steps, dist=3.49, axis_error
    =[2.8887672 0.5664215 1.8655663], reason=running
1261 [Astar_planner] Segment 16 starts from actual pos [ 55.119114 136.
    43375 -21.554436] (planned waypoint was [ 58.    137.   -23.
    42028])
1262 [Astar_planner] Segment 16: [ 55.119114 136.43375 -21.554436
    ] -> [ 71.    135.   -22.84989]
1263 [Step 10 | Total    0] Target: ( 71.0, 135.0, -22.8) | Pos: ( 68.5, 133.
    8, -21.4) | Vel: ( 2.7, 0.7, -0.4) | Acc: ( -2.0, 1.6, -0.2) | Rwd: 3.
    91
1264 [Astar_planner] Segment 16 reached in 10 steps, dist=3.15,
    axis_error=[2.524086 1.1681976 1.4778805], reason=running
1265 [Astar_planner] Segment 17 starts from actual pos [ 68.48393 133.
    83386 -21.373236] (planned waypoint was [ 71.    135.   -22.
    84989])
1266 [Astar_planner] Segment 17: [ 68.48393 133.83386 -21.373236
    ] -> [ 84.    135.   -24.209085]
1267 [Step 22 | Total    0] Target: ( 84.0, 135.0, -24.2) | Pos: ( 81.3, 134.
    5, -22.2) | Vel: ( 1.6, -0.7, 0.5) | Acc: ( -0.2, -0.4, 1.0) | Rwd: -0.
    25
1268 [Astar_planner] Segment 17 reached in 22 steps, dist=3.39,

```

```

1268 axis_error=[2.6615677 0.46354675 2.0445461 ], reason=running
1269 [Astar_planner] Segment 18 starts from actual pos [ 81.34334 134.
5345 -22.163149] (planned waypoint was [ 84. 135. -24.
209085])
1270 [Astar_planner] Segment 18: [ 81.34334 134.5345 -22.163149] -> [
98. 135. -22.478079]
1271 [Step 28 | Total 0] Target: ( 98.0, 135.0, -22.5) | Pos: ( 95.8, 134.
8, -19.9) | Vel: ( 0.2, 1.1, -2.2) | Acc: ( -1.3, 1.5, 2.0) | Rwd: 2.41

1272 [Astar_planner] Segment 18 reached in 28 steps, dist=3.41,
axis_error=[2.1731262 0.19880676 2.62286 ], reason=running
1273 [Astar_planner] Segment 19 starts from actual pos [ 95.82738 134.
80435 -19.861717] (planned waypoint was [ 98. 135. -22.
478079])
1274 [Astar_planner] Segment 19: [ 95.82738 134.80435 -19.861717] -> [
104. 125. -21.319597]
1275 [Step 14 | Total 0] Target: (104.0, 125.0, -21.3) | Pos: (103.0,
127.9, -20.3) | Vel: ( 0.8, -1.7, -0.0) | Acc: ( 0.1, -2.0, 1.1) | Rwd
: 2.58
1276 [Astar_planner] Segment 19 reached in 14 steps, dist=3.23,
axis_error=[1.0145798 2.8807907 1.0551052], reason=running
1277 [Astar_planner] Segment 20 starts from actual pos [102.98542 127.
88079 -20.264492] (planned waypoint was [104. 125. -21.
319597])
1278 [Astar_planner] Segment 20: [102.98542 127.88079 -20.264492
]-> [104. 111. -20.023855]
1279 [Step 30 | Total 0] Target: (104.0, 111.0, -20.0) | Pos: (106.5,
108.4, -18.2) | Vel: ( 0.7, 1.0, -0.5) | Acc: ( -0.8, -0.8, -0.7) | Rwd
: -3.13
1280 [Astar_planner] Segment 20 reached in 30 steps, dist=4.07,
axis_error=[2.5158386 2.6473389 1.7984428], reason=running
1281 [Astar_planner] Segment 21 starts from actual pos [106.517975 108.
35576 -18.226828] (planned waypoint was [104. 111. -20.
023855])
1282 [Astar_planner] Segment 21: [106.517975 108.35576 -18.226828
]-> [104. 98. -21.315033]
1283 [Step 10 | Total 0] Target: (104.0, 98.0, -21.3) | Pos: (107.6, 106
.1, -21.9) | Vel: ( 0.1, 0.1, -1.0) | Acc: ( 0.2, 1.8, -0.7) | Rwd: -922
.82 -> Episode End: CRASHED!

1284
1285 [Astar_planner] Segment 21 ended: collision
1286 [Astar_planner] Stop after failed segment 21.
1287 Task 18 result: failed (Failed at segment 21: collision (21/24 reached
). Shield interventions=36, emergencies=5, recoveries=7,
recovery_steps=36.)
1288
1289 --- Task 19/38 ---
1290 From [107. -82.52 -4. ] to [ -63.8 -136.72 -4. ]

```

```

1291 [Astar_planner] Top-down occupancy: 350x390, x=[-179.0, 170.0], y
    =[-200.0, 189.0], occupied=72869 cells (53.4%)
1292 [Astar_planner] Occupancy base_grid=390x350, occupied=72869
    cells (53.4%)
1293 [Astar_planner] Occupancy A*: grid=390x350, occupied=53.4%,
    safety_margin=1.5
1294 Path planned via occupancy (Points: 20, Dist: 263.7m). Local target z
    /alt: 0:-4.00(4.00m), 1:-24.41(24.41m), 2:-20.77(20.77m), 3:-21.68(21
    .68m), 4:-24.32(24.32m), 5:-22.98(22.98m), 6:-23.98(23.98m), 7:-24.
    39(24.39m), 8:-24.78(24.78m), 9:-20.33(20.33m), 10:-24.54(24.54m
    ), 11:-24.06(24.06m), 12:-23.93(23.93m), 13:-21.07(21.07m), 14:-24.
    37(24.37m), 15:-21.94(21.94m), 16:-22.37(22.37m), 17:-21.26(21.
    26m), 18:-23.14(23.14m), 19:-4.00(4.00m)
1295 [Astar_planner] Segment 0: [107. -83. -4.] -> [100.    -93.    -24.
    405222]
1296 [Astar_planner] Placing first segment start at [107. -83. -4.]
1297 [Astar_planner] Initial climb to 20.0m.
1298 [Step 13 | Total    0] Target: (100.0, -93.0, -24.4) | Pos: (102.4, -90
    .9, -23.4) | Vel: ( -0.6, -1.1,  0.0) | Acc: (  0.2,  1.5, -0.8) | Rwd: 2.
    82
1299 [Astar_planner] Segment 0 reached in 13 steps, dist=3.35, axis_error
    =[2.3746185 2.126503  1.0202904], reason=running
1300 [Astar_planner] Segment 1 starts from actual pos [102.3727 -90.
    876816 -23.384912] (planned waypoint was [100.    -93.    -24.
    405222])
1301 [Astar_planner] Segment 1: [102.3727 -90.876816 -23.384912] -> [
    100.    -106.    -20.770947]
1302 [Step 26 | Total    0] Target: (100.0, -106.0, -20.8) | Pos: (100.7, -
    103.6, -19.6) | Vel: ( -0.4, -0.9, -0.0) | Acc: (  1.8,  1.6, -0.4) | Rwd
    :  1.60
1303 [Astar_planner] Segment 1 reached in 26 steps, dist=2.74, axis_error
    =[0.66672516 2.403534  1.1417694 ], reason=running
1304 [Astar_planner] Segment 2 starts from actual pos [ 100.66567 -103.
    59905 -19.629227] (planned waypoint was [ 100.    -106.    -20.
    770947])
1305 [Astar_planner] Segment 2: [ 100.66567 -103.59905 -19.629227
    ] -> [  94.    -117.    -21.678215]
1306 [Step 16 | Total    0] Target: ( 94.0, -117.0, -21.7) | Pos: ( 93.4, -
    114.4, -21.2) | Vel: ( -0.4, -0.9, -0.1) | Acc: (  0.7,  1.9, -0.1) | Rwd
    :  1.59
1307 [Astar_planner] Segment 2 reached in 16 steps, dist=2.71, axis_error
    =[0.5990982 2.607483  0.45931816], reason=running
1308 [Astar_planner] Segment 3 starts from actual pos [ 93.39984 -114.
    395134 -21.219053] (planned waypoint was [  94.    -117.    -21.
    678215])
1309 [Astar_planner] Segment 3: [ 93.39984 -114.395134 -21.219053
    ] -> [  85.    -126.    -24.324219]
1310 [Step 12 | Total    0] Target: ( 85.0, -126.0, -24.3) | Pos: ( 86.8, -

```

```

1310 123.0, -23.6) | Vel: ( -0.4, -1.0, 0.0) | Acc: ( 0.0, -0.1, -0.3) | Rwd
      : 1.47
1311 [Astar_planner] Segment 3 reached in 12 steps, dist=3.57, axis_error
      =[1.7993774 2.9939194 0.7405987], reason=running
1312 [Astar_planner] Segment 4 starts from actual pos [ 86.79938 -123.
      00608 -23.58362] (planned waypoint was [ 85.    -126.    -24.
      324219])
1313 [Astar_planner] Segment 4: [ 86.79938 -123.00608 -23.58362] -> [
      82.    -137.    -22.978104]
1314 [Step 23 | Total    0] Target: ( 82.0, -137.0, -23.0) | Pos: ( 84.9, -
      135.4, -22.0) | Vel: ( -0.8, -0.6, 0.0) | Acc: ( 0.0, 0.5, 1.3) | Rwd:
      0.51
1315 [Astar_planner] Segment 4 reached in 23 steps, dist=3.44, axis_error
      =[2.85672 1.6443634 0.9980335], reason=running
1316 [Astar_planner] Segment 5 starts from actual pos [ 84.85197 -135.
      35928 -21.97991] (planned waypoint was [ 82.    -137.    -22.
      978104])
1317 [Astar_planner] Segment 5: [ 84.85197 -135.35928 -21.97991] -> [
      80.    -150.    -23.982594]
1318 [Step 20 | Total    0] Target: ( 80.0, -150.0, -24.0) | Pos: ( 82.8, -
      147.4, -23.5) | Vel: ( -0.5, -0.8, -0.0) | Acc: ( 0.3, 0.7, -0.3) | Rwd
      : 1.89
1319 [Astar_planner] Segment 5 reached in 20 steps, dist=3.86, axis_error
      =[2.7821274 2.6313019 0.5098896], reason=running
1320 [Astar_planner] Segment 6 starts from actual pos [ 82.78213 -147.
      3687    -23.472704] (planned waypoint was [ 80.    -150.    -23.
      982594])
1321 [Astar_planner] Segment 6: [ 82.78213 -147.3687    -23.472704
      ] -> [ 71.    -156.    -24.386908]
1322 [Step 12 | Total    0] Target: ( 71.0, -156.0, -24.4) | Pos: ( 73.9, -
      154.8, -22.9) | Vel: ( -1.1, -0.9, 0.1) | Acc: ( 0.4, -1.3, -0.3) | Rwd
      : 1.76
1323 [Astar_planner] Segment 6 reached in 12 steps, dist=3.49, axis_error
      =[2.9374695 1.196579 1.4595833], reason=running
1324 [Astar_planner] Segment 7 starts from actual pos [ 73.9343 -154.
      80608 -22.927149] (planned waypoint was [ 71.    -156.    -24.
      386908])
1325 [Astar_planner] Segment 7: [ 73.9343 -154.80608 -22.927149
      ] -> [ 58.    -156.    -24.78418]
1326 [Step 30 | Total    0] Target: ( 58.0, -156.0, -24.8) | Pos: ( 58.5, -
      159.0, -23.6) | Vel: ( -0.3, 0.5, -0.0) | Acc: ( -1.0, 1.1, 0.2) | Rwd
      : 3.08
1327 [Astar_planner] Segment 7 reached in 30 steps, dist=3.23, axis_error
      =[0.481575 2.964737 1.1850529], reason=running
1328 [Astar_planner] Segment 8 starts from actual pos [ 58.481575 -158.
      96474 -23.599127] (planned waypoint was [ 58.    -156.    -24.
      78418])
1329 [Astar_planner] Segment 8: [ 58.481575 -158.96474 -23.599127

```

```

1329 ]-> [ 45.    -156.    -20.330814]
1330 [Step 16 | Total    0] Target: ( 45.0, -156.0, -20.3) | Pos: ( 47.8, -
    156.0, -19.3) | Vel: ( -0.5,  0.1, -0.0) | Acc: ( -1.8,  1.4,  0.2) | Rwd
    : 2.23
1331 [Astar_planner] Segment 8 reached in 16 steps, dist=2.97, axis_error
    =[2.796547  0.02285767 1.0019627 ], reason=running
1332 [Astar_planner] Segment 9 starts from actual pos [ 47.795124 -156.
    02254 -19.32891 ] (planned waypoint was [ 45.    -156.    -20.
    330814])
1333 [Astar_planner] Segment 9: [ 47.795124 -156.02254 -19.32891
    ]-> [ 33.    -160.    -24.53882]
1334 [Step 27 | Total    0] Target: ( 33.0, -160.0, -24.5) | Pos: ( 30.8, -
    161.3, -22.4) | Vel: ( -0.0,  0.3,  1.2) | Acc: (  0.1,  1.4, -0.2) | Rwd:
    3.86
1335 [Astar_planner] Segment 9 reached in 27 steps, dist=3.32, axis_error
    =[2.1871853 1.2793732 2.1441593], reason=running
1336 [Astar_planner] Segment 10 starts from actual pos [ 30.812807 -161.
    27835 -22.391094] (planned waypoint was [ 33.    -160.    -24.
    53882])
1337 [Astar_planner] Segment 10: [ 30.812807 -161.27835 -22.391094
    ]-> [ 20.    -160.    -24.05602]
1338 [Step 36 | Total    0] Target: ( 20.0, -160.0, -24.1) | Pos: ( 21.0, -
    162.4, -21.8) | Vel: ( -0.2, -0.4, -1.8) | Acc: (  1.7, -0.7, -0.7) | Rwd
    : 5.66
1339 [Astar_planner] Segment 10 reached in 36 steps, dist=3.46,
    axis_error=[1.0466442 2.3961487 2.2660255], reason=running
1340 [Astar_planner] Segment 11 starts from actual pos [ 21.046028 -162.
    39731 -21.795498] (planned waypoint was [ 20.    -160.    -24.
    05602])
1341 [Astar_planner] Segment 11: [ 21.046028 -162.39731 -21.795498
    ]-> [  8.    -161.    -23.927105]
1342 [Step 69 | Total    0] Target: (  8.0, -161.0, -23.9) | Pos: ( 10.7, -
    163.4, -22.8) | Vel: ( -0.6,  0.0,  0.1) | Acc: (  0.6,  0.1, -0.7) | Rwd:
    1.17
1343 [Astar_planner] Segment 11 reached in 69 steps, dist=3.81,
    axis_error=[2.7072487 2.4424133 1.1203499], reason=running
1344 [Astar_planner] Segment 12 starts from actual pos [ 10.705377 -163.
    4423 -22.806526] (planned waypoint was [  8.    -161.    -23.
    927105])
1345 [Astar_planner] Segment 12: [ 10.705377 -163.4423 -22.806526
    ]-> [ -5.    -161.    -21.074966]
1346 [Step 199 | Total    0] Target: ( -5.0, -161.0, -21.1) | Pos: (  6.3, -
    169.7, -25.0) | Vel: (  0.3, -0.1,  0.0) | Acc: (  0.8, -0.0, -0.0) | Rwd: -
    906.48 -> Episode End: CRASHED!
1347
1348 [Astar_planner] Segment 12 ended: collision
1349 [Astar_planner] Stop after failed segment 12.
1350 Task 19 result: failed (Failed at segment 12: collision (12/19 reached

```



```

1350 ). Shield interventions=251, emergencies=4, recoveries=43,
      recovery_steps=251.)
1351
1352 --- Task 20/38 ---
1353 From [-75.1 -8.22 -16. ] to [101.4 29.68 -6.6 ]
1354 [Astar_planner] Top-down occupancy: 350x390, x=[-179.0, 170.0], y
      =[-200.0, 189.0], occupied=72869 cells (53.4%)
1355 [Astar_planner] Occupancy base_grid=390x350, occupied=72869
      cells (53.4%)
1356 [Astar_planner] Endpoint nearest-free cells are disconnected; using
      reachable cells start=(245, 192), goal=(69, 230), search_radius=36.
1357 [Astar_planner] Occupancy A*: grid=390x350, occupied=53.4%,
      safety_margin=1.5
1358 [Astar_planner] Endpoint nearest-free cells are disconnected; using
      reachable cells start=(245, 192), goal=(69, 230), search_radius=36.
1359 [Astar_planner] Occupancy A*: grid=390x350, occupied=53.4%,
      safety_margin=1.0
1360 Traceback (most recent call last):
1361   File "E:\Github\AirSim_RL\Src\batch_test_script.py", line 110, in <
      module>
1362     run_batch_from_csv(args.csv)
1363   File "E:\Github\AirSim_RL\Src\batch_test_script.py", line 86, in
      run_batch_from_csv
1364     summary = pipeline.run_batch_test(task_list, status_callback=print
      )
1365   File "E:\Github\AirSim_RL\Src\nav_pipeline.py", line 173, in
      run_batch_test
1366     waypoints, msg = self.plan_path(start, end)
1367   File "E:\Github\AirSim_RL\Src\nav_pipeline.py", line 104, in
      plan_path
1368     plan = self.planner.plan(start_pos, end_pos)
1369   File "E:\Github\AirSim_RL\Network\Astar_planner\
      occupancy_planner.py", line 315, in plan
1370     clearance_cost = self._build_clearance_cost(grid)
1371   File "E:\Github\AirSim_RL\Network\Astar_planner\
      occupancy_planner.py", line 808, in _build_clearance_cost
1372     if not grid.is_free(near_cell):
1373   File "E:\Github\AirSim_RL\Network\Astar_planner\
      occupancy_planner.py", line 232, in is_free
1374     return not bool(self.occupied[cell[0], cell[1]])
1375 KeyboardInterrupt
1376
1377 进程已结束 退出代码为-1073741510 (0xC000013A: interrupted by Ctrl+C
      )
1378

```